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College of Information Technology Education

A Data-Driven Funeral Service Management System for Planning and Optimization

In partial fulfillment for the requirements in Capstone 1 and Research Subject

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CHAPTER 1: INTRODUCTION

Introduction

This chapter introduces the overview of the study called ―A Data-Driven Funeral Management System for Planning and Optimization.‖ The discussion on the study was elaborated comprehensively through the identification of the fundamental elements of the study, including its basic foundation. This section comprises the following: background of the study, statement of the problem, objectives of the study, significance of the study, scope and delimitation of the study, and definition of terms. All of these elements contribute to forming the direction and main objective of the study.

The background of the study presents the information regarding the existing problems that have been experienced when managing funerals in terms of record management, scheduling, inventory management, transaction, and reporting purposes. In addition to these, the relevance of computerization and the use of data in funeral management are provided in this section.

The statement of the problem discusses the general and specific problems and concerns that this study attempts to resolve and which serve as the basis for developing the proposed system.

The objectives of the study enumerate both the general and specific objectives that researchers intend to achieve through the study using the developed system. It includes the purposes of carrying out the proposed system.

The significance of the study explains why researchers undertake this study and the advantages that the proposed system will have not only to funeral service providers and managers but also to their staff members and clients as well as future researchers.

The scope and delimitations indicate what areas are covered by the study and what are not. This section describes what functions of the proposed system are included in the study and which are not.

Finally, the definition of terms includes definitions of some key terms used in this study for greater clarity, ensuring that technical and operational concepts are understood within the context of this study.

This chapter forms the basis of the study by providing basic information about the study including its direction and important concepts involved in it. It serves as the roadmap for the entire development process and research work.

1.1 Background of the Study

The funeral home plays a crucial role in the community by providing services that assist families during one of the most difficult times in their lives – the loss of their loved one. These services include the retrieval and preparation of the body, wake arrangements, coordination of the burial or cremation services, and memorial

services. Despite the importance of these services to the community, some funeral homes still rely on manual processes such as paper-based records, handwritten schedules, and unstructured communication between staff and clients, for managing their operation which often results in inefficiencies in the management of their services[1].

Laudon and Laudon states that organizations that continue to depend on manual information processing often experience delays, inaccurate reporting, poor coordination, and inefficient management of operational activities [2]. In an industry where timing, coordination, and emotional sensitivity are critical, such inefficiencies can significantly impact both service providers and grieving families.

Another major challenge in funeral service management is the unpredictability of demand.[3] Death occurrences are naturally uncertain, making it difficult for funeral homes to forecast service needs, allocate resources effectively, and maintain optimal inventory levels. Unlike other service industries where demand follows seasonal or daily patterns, funeral service demand is subject to demographic trends, health events, accidents, and natural causes which are difficult to predict with manual estimation. Without proper data analysis tools, funeral homes struggle to anticipate peak periods, resulting in either over-preparation – which waste resources and increase cost - or under-preparation – which leads to shortages, poor service, and missed opportunities.

Furthermore, many funeral homes lack centralized databases that store client information, service records, and financial transactions. This absence of structured data limits their ability to analyze trends, improve decision-making, and enhance service quality. In today‘s digital age, where modern businesses are increasingly adopting computerized systems and databases technologies in order to manage large amount of data effectively , funeral services remain relatively underdeveloped in terms of technological integration and data utilization [4].

In today‘s business environment, the use of data to support decision-making has become an essential strategy for improving operational efficiency and service quality. A data-driven system utilizes collected and stored data to identify trends, monitor performance, forecast future demands, and support managerial decisionmaking [5]. Through the analysis of historical and current operational data, organizations can make informed decisions regarding resource allocation, inventory management, scheduling, and service planning.

In the context of funeral management, a data-driven system can provide administrators with valuable insights into service demand patterns, inventory utilization, operational performance, and customer service requirements. By analyzing historical records and operational data, funeral homes can improve planning, optimize resource allocation, minimize operational inefficiencies, and enhance the overall quality of service provided to clients [3], [6]. The effective use of data enables funeral service providers to contribute to organizational efficiency, customer satisfaction, and sustainable business operations.

The emergence of data analytics, web-based systems, and cloud computing present a significant opportunity to modernize funeral service management. These technologies enable organizations to collect, store, process, and analyze operational data efficiently through a centralized platform [3]. Through automation and digital record management, funeral home can minimize human errors, reduce paperwork, improve accessibility of information, and increase operational efficiency. Additionally, the set of data analytics allows administrators to transform raw operational data into meaningful insights that support strategic planning and decision-making [5], [6]. For example, analyzing historical data can reveal which services are most in demand during certain months, which resources are most frequently used, and how revenue is distributed across different service packages.

This study proposed the development of a **Data-Driven Funeral Management System** for Alfonso Somo Funeral Homes, a local service provider serving the community of Maasin, Iloilo and surrounding areas. The proposed system seeks to improve funeral service management through automation, centralized information management, scheduling functions, resource monitoring, data analytics, and planning optimization. Specifically, the system intends to provide an efficient and organized platform for managing funeral operations while supporting data-driven decision-making processes. Through a computerized management system, funeral homes can improve efficiency, reduce operational errors, enhance customer service and support better planning and decision-making for sustainable business growth.

1.2 Statement of the Problem

[7] Funeral homes play a vital role in providing essential services to families during times of grief and loss. Their responsibilities extend beyond funeral arrangements and include the management of client information, scheduling of services, inventory control, resource allocation, financial transactions, and coordination among personnel. Because funeral services involve numerous interconnected activities that must be performed within specific timeframes, effective management is necessary to ensure that operations run smoothly and that clients receive quality service.

Despite the importance of efficient management, many funeral homes including Alfonso Somo still relies on traditional and manual methods in handling daily operations. Information is often recorded in logbooks, paper forms, and/or handwritten records. Scheduling activities are coordinated manually, while inventory monitoring and financial recording are often conducted without the assistance of integrated information systems [7]. Although these methods may have been sufficient in the past, they become increasingly ineffective as service demands grow and organizational operations become more complex.

The absence of a centralized and computerized management system creates several operational challenges. Manual record-keeping increases risk of data loss, duplication, and inaccuracies. Retrieving information may become time-consuming, particularly when multiple funeral services are being handles simultaneously. Furthermore, the lack of integrated data management makes it difficult for administrators to analyze operational performance, monitor resource utilization, and make informed decisions regarding future planning [7] [8].

Another significant concern faced by Alfonso Somo Funeral Homes is the unpredictability of service demand. Funeral service providers often experience fluctuations in the number of clients they serve. These variations make it difficult to determine the amount of resources, personnel, equipment, and supplies required at any given time. Without proper forecasting mechanisms and historical data analysis, administrators may encounter shortages during peak periods or excessive idle resources during slow periods [3].

In addition, resource monitoring and service tracking remain challenging in organizations that rely on manual procedures. Funeral homes like Alfonso Somo must manage various resources including hearses, embalming equipment, caskets, flowers, staff assignments/schedules, and transportation services. The absence of real-time monitoring systems may result in scheduling conflicts, inefficient resource allocation, delays in service delivery, and customer dissatisfaction [3] [7].

The growing importance of information technology and data analytics presents opportunities for funeral service providers to modernize their operations [5] [6] [8]. Through the implementation of a data-driven funeral management system, organizations can centralize information, automate operational processes, improve communication, monitor resource effectively, and utilize historical data to support planning and decisionmaking activities.

Considering these challenges, there is a need to develop a Data-Driven Funeral Management System for Alfonso Somo Funeral Homes that improves operational efficiency, enhance service quality, optimize resource utilization, and provide reliable information to support planning and organizational decision-making.

1.2.1 General Problem Statement

Alfonso Somo Funeral Homes experiences operational challenges arising from the use of manual procedures in managing client records, funeral scheduling, inventory monitoring, resource allocation, financial transactions, and reporting activities. The absence of a centralized and data-driven management system limits the organization‘s ability to efficiently store, retrieve, analyze, and utilize operational information. As a result, administrative processes become time-consuming, error-prone, and difficult to monitor.

Furthermore, the unpredictability of funeral service demand creates difficulties in planning resources and scheduling operations. Without appropriate forecasting and analytical tools, management encounters challenges in determining future service requirements and ensuring the availability of necessary resources. The lack of integrated information management also restricts the organization‘s ability to generate meaningful insights that could improve operational efficiency and support strategic decisionmaking.

Therefore, the general problem addressed by this study is the lack of a comprehensive, centralized, and data-driven funeral management system that can effectively support planning, optimization, resource management, and decision-making processes in Alfonso Somo Funeral Homes.

1.2.2 Specific Problems

1. Uses of manual process that can result to errors.

In Alfonso Somo Funeral Homes, most of the current procedures, including the recording of client details, deceased information and booking of funeral services, are conducted using manual systems where notes are

recorded on paper, logbooks and filing system. Although a manual system can be efficient, it could also cause problems like loss of records and mistakes, which might have detrimental effects on the performance of the organization, especially while organizing a funeral event. Finding the record of clients can also become challenging for employees when handling several funeral events simultaneously, leading to delays in service delivery.

Manual systems can similarly cause problems such as lack of accuracy [7] [8], in terms of billing, payments, and records keeping. Calculations are mostly done by hand, leading to potential overcharging or undercharging, and records are often inconsistent between different books. These inaccuracies may lower the level of services offered and erode the trust of the community. To address this, the development of a web-based system is proposed to enhance performance and efficiency within Alfonso Somo funeral homes.

2. **Unpredictable demand for funeral services**

The demand for services at Alfonso Somo Funeral Homes is always uncertain. Sometimes several funeral arrangements happen at once, but during other times, very few services take place. The uncertainty of demand makes it difficult to determine the resources needed during operations. If this problem is left unattended, there may be cases where there will not be sufficient resources – such as hearses, embalming supplies, or staff members - available to complete the task. At other times, the resources may be in excess, meaning that capital and assets are sitting idle, increasing operation costs.

Scheduling issues may arise due to poor planning, lack of visibility, and manual coordination, thus making the work processes less efficient [3]. Conflict in wake schedules, transportation assignments, and staff shifts are common. It is necessary to have a better system of planning in order to solve the issue. This study seeks to provide a solution by formulating a Funeral Management System that will help Alfonso Somo Funeral Homes to forecast demand patterns based on historical data and make plans accordingly.

3. Unsatisfactory tracking of funeral services and resources

Alfonso Somo Funeral Homes do not have a structured system to track the necessary resources such as coffins, hearses, and scheduling of staff. Inventory is checked visually or through rough estimates, and there is no realtime update of stock levels or availability, As a result, there might be delays and problems in the process of service delivery due to the lack of an efficient method of monitoring these resources. For example, promising a specific type of casket to a family only to find out it is out of stock causes significant distress and dissatisfaction.

Moreover, coordinating funeral events becomes increasingly difficult since it is difficult to communicate effectively when the information is not centrally stored with each other when the information is not centrally stored [7] [8]. Staff members work in silos, and updates are passed verbally, leading to miscommunication and errors. This calls for an efficient way to keep track of the necessary resources. Such a system will enable Alfonso Somo Funeral Homes to know exactly what resources are available at any particular time, their condition, and their scheduled use.

4. Unavailability of an organized database to aid planning and optimization

There is no central place where data is stored at Alfonso Somo Funeral Homes. Records are kept in different locations and formats – some in logbooks, some in folders, and some in individual memories. This means that

the process of collecting and accessing records becomes very tedious and time-consuming. It is also difficult for the employees to conduct analysis on some data, especially about the commonly required services, seasonal trends, or which resources should be restocked.

Decision-making at the organization is limited because of the lack of a data management system that helps one gain insight into the company's performance and service demand [8]. This project will seek to come up with a data-driven funeral management system that uses a central database to collect, store, and analyze information, turning raw data into useful knowledge for better planning and optimization.

1.2.3 Research Questions

To address the identified problems, this study seeks to answer the following research questions:

General Research Question

How can a Data-Driven Funeral Management System be developed for Alfonso Somo Funeral Homes to improve planning, resource management, operational efficiency, and decision-making processes?

Specific Research Questions

1. How can the current manual processes used in Alfonso Somo Funeral Homes be analyzed and improved through the implementation of a computerized management system?
2. How can a data-driven funeral management system assist in forecasting funeral service demand using historical operational data?
3. How can data analytics and reporting functionalities support planning, optimization, and decision-making within the funeral home?

1.3 Objectives of the Study

The study objectives are the detailed set of goals the researcher intends to accomplish through the proposed system. They are the targets of the research which the researcher intends to fulfill by developing a system that is useful in solving the problems identified with the current operation process in Alfonso Somo Funeral Homes, especially concerning record keeping, scheduling, resource management, and decision making.

These objectives are divided into two parts:

1.3.1 General Objective

The main purpose of this study is to develop a **Data-Driven Funeral Management System for Planning and Optimization** in Alfonso Somo Funeral Homes aimed at increasing efficiency of its operation, monitoring, and optimizing processes via means of data analysis and management of information. In particular, this system helps the company to organize the process of managing clients' records, monitor available resources, organize schedule, and analyze data on their performance.

Moreover, the development of this proposed data-driven system to funeral operations makes the funeral house work more effectively and technologically. In turn, it helps Alfonso Somo Funeral Homes to organize

its services in a much better way, reduced operational costs, and provide improved quality services to customers.

1.3.2 Specific Objectives

To meet the general objective of the research, the specific objectives are outlined, which are aimed at addressing the analysis, design, development, implementation, and evaluation of the proposed system. The specific objectives have been formulated with the intention to help solve the problems associated with funeral homes regarding record keeping, scheduling, resource management, planning, and decision making. These particular objectives form the foundation on which an efficient data-based funeral home management system can be created. From these objectives, the research is guided to ensure that an efficient system that is capable of solving the existing problems is formulated.

- **Analyze the current situation and requirements of the funeral home and client.**

  It is important for the study to examine and evaluate the process through which the current operations are undertaken, management practices, and the challenges faced both by Alfonso Somo and by their clients. The process through which funeral arrangements are done, scheduling is conducted, inventory control is managed, transactions are completed, and information recorded are some of the areas that were examined. An assessment of the current situation is important in establishing areas where the current process is inefficient, challenging and necessitates the use of technology to ease management.

  In addition, an evaluation of the current situation is essential in determining the requirements and the needs of users concerning the proposed system. An assessment of the current work process facilitates the development of a better understanding of the needs of the funeral home which enables the development of a system suited to meet those needs. Lastly, the examination of requirements of clients is critical in identifying features and functionalities required from a computerized information system. This objective facilitates the development of a system which not only meets the operational objectives of the funeral home but also meets the needs of their clients.

- **Design and develop a web-based funeral management system.**

  In this study, the objective is to design and develop a funeral management system using a webbased interface to facilitate the operation processes of Alfonso Somo Funeral Homes. The creation of the funeral management system offers a digital environment where all management operations of funeral homes were effectively done on one platform. Using the web-based funeral management system, it enables users to interact with data and perform tasks related to the management processes via internet-accessible devices such as computers, tablets, or smartphones.

  Compared to manual systems, a web-based system offers better accessibility of data and faster performance due to the internet connection. The system incorporates a user-friendly interfaces and proper functionalities that facilitate effective utilization by users from various groups such as

administrators, funeral staff, and clients. Some of the main functionalities offered by the system include management of funeral service records, scheduling, inventory control, and reporting processes.

Furthermore, during the development process of the system, it is necessary to apply appropriate software development practices along with database management principles to increase the efficiency, safety, and reliability of the system.

- **Develop a centralized platform for storing client information, schedules, service records, and financial transactions.**

The study aims to create a centralized database and information management platform that facilitates the ease with which the funeral home can store and organize all types of information. It is very important to adopt centralized data management since this involves the collection of all information and storage of it in a single information management system and not in several disconnected systems, including information stored manually.

The adoption of the centralized data management system allows funeral home operators to manage their information more easily since they will find it easy to update their information, retrieve records instantly, and share data securely. Centralized database management is critical since it helps in ensuring data security, organization, and management. This means that the probability of having lost or duplicate information will be minimal. In this way, management will always have easy access to all information.

Lastly, a centralized database is instrumental in ensuring effective communication within a funeral home business setting, ensuring that everyone works form the same accurate set of data.

- **Create a data-driven system that can help families plan according to their preferences and budget.**

Another goal of the study is to design a data-driven funeral management system, which assist clients in arranging funeral services depending on their needs and possibilities. For many families, funeral planning can be a hard thing emotionally and financially, particularly at such a tragic period in life. That is why the developed system aims at assisting clients in planning the funeral services logically and clearly. By using the system, funeral homes were able to present all available service packages, their pricing, as well as other details and resources. By analyzing data and using information management methods, the system can assist in identifying services suitable for a client depending on their budgetary constraints and preferences.

This goal also implies creating transparency in funeral services by giving clients all the required information in terms of costs and included services as well as tracking payment. This way, families feel more confident in planning the funeral without having any confusion about finances and available services. Besides, the system may also help managers analyze preferences and trends among customers to increase service variety and efficiency.

- **Create an admin dashboard for tracking sales, schedule, and resources usage.**

This objective includes the creation of a comprehensive dashboard for efficient management of the operations performed by funeral homes. Through the analysis performed with the use of the dashboard, the

manager can control available resources, monitor current services being rendered, analyze sales performance, and make conclusions about overall trends in operations [6].

In addition, by monitoring operation processes and making analyses of information obtained during this process, it is possible to detect inefficiencies in service provision, identify bottlenecks, and take action aimed at increasing the efficiency of operations.

As a result, development of the admin dashboard is expected to contribute significantly to efficient management of funeral home operations by providing real-time visibility and intelligence.

- **Evaluate the system's data forecasting reliability and usability based on historical trend averaging**

In order to assess the reliability and utility of the generated forecasts according to historical trend averages, the researchers examine the accuracy of the insights produced as well as their usefulness to the management of Alfonso Somo Funeral Homes. Rather than using mostly artificial intelligence techniques, which depend on a large dataset for their operation, this research utilizes historical trend averages in determining the seasonal demand trends, usage trends, and preferences for packages offered by the business through its recorded transactions. Reliability of this objective are proven numerically by comparing the forecasted trends produced by the system to the outcomes that were actually observed within the tracking period, thereby determining the accuracy of the model.

At the same time, the usability component are determined via the process of evaluating the product usage by funeral home administrators and determining whether it is convenient to interpret the provided data presented in the form of the analytical dashboards and graphs, as well as forecasts of demand. Upon reaching the stated aim, one can assume that the developed solution not become just another experiment but a reliable, highly interpretable, and practical decision-making instrument.

1.4 Scope and Delimitation

The scope and delimitation of the study serve to indicate the parameters of the study and define what is included and excluded in the creation of the proposed data-driven system for funeral management of Alfonso Somo. In essence, this part will help to determine the extent of coverage of the Data-Driven Funeral Management System in terms of its capabilities, intended users, and functionality. This is to ensure that the focus of the study is narrowed down to the specified parameters and development remains feasible.

1.4.1 Scope of the Study

This research highlights the development of a Data-Driven Funeral Management System for Alfonso Somo Funeral Homes in a web-based format for the enhancement of efficiency in funeral service management. The system helped centralize the data records, organize procedures for data-driven decisionmaking, and make funeral service delivery more efficient.

It also helps reduce the efforts in handling daily transactions as well as minimize mistakes that may result from keeping manual records. This will go a long way in enhancing the efficiency and effectiveness of the funeral homes.

The proposed system will include the main functionalities such as:

- **User, admin and staff registration and login.**

  The system provides a secure authentication module that allows users, administrators, and staff members to create accounts and log in. This ensures that only authorized individuals can access specific features based on their assigned roles, improving system security and data privacy. Different permission levels ensure that sensitive financial data is only accessible to management, while operational data is accessible to relevant staff.

- **Admin, staff and user dashboard.**

  Each user – admin, staff, and client, has its own dashboard interface designed in accordance with the roles and duties assigned to them. The dashboard acts as the main interface from where users can monitor various actions, access system features, and obtain all necessary information. The admin dashboard focuses on analytics and control, the staff dashboard focuses on daily tasks and schedules, and the client/user dashboard focuses on their specific service details and payments.

- **Funeral scheduling and appointment management.**

  The system enables the client and staff members to book the funeral services as well as their associated activities in a more efficient manner. This ensures that there is no overlapping in booking schedules and also ensures better time management skills. It covers scheduling of wakes, transportation, and service personnel.

- **Data analysis and visualization dashboard.**

  The system includes some functions of analytics, where funeral information is presented in graphs and other visuals, including chart summaries, and trend lines. This helps managers to analyze trends, identify peaks seasons, determine best-selling packages, and make decisions according to collected data.

- **Storage of client information, deceased details, and service records and management**

  The system also include a centralized database that contains everything, including details about the clients, those who have passed away, services offered, and the transactions made. This ensures a complete history of every service provided

- **Inventory and resource monitoring**

  The system provides a module, which keeps track of funeral resources and equipment such as caskets, flowers, vehicles, embalming fluids and chapel amenities. It helps to ensure availability of items when required and avoid under or over-provisioning of items.

- **Online payment integration**

  Digital modes of payment are integrated into the system, providing customers the opportunity to make payments using online means. The benefits include increased convenience, ease of transactions, and improved security.

- **Mapping and location services**

  Location services are offered as an added advantage to help in locating various destinations such as places where the service is required to be delivered, cemeteries, and the funeral home itself, assisting in logistics and coordination.

- **Messaging and communication feature**

  The system also includes the provision for internal messages and notifications through which communication can be achieved among clients, administrators, and staff. It serves as a log of instructions and updates regarding a specific service.

- **Database management**

  There is the use of a properly structured database management system where all the information within the system is stored, retrieved, and managed effectively. It ensures data integrity, backup, and security.

1.4.2 Delimitation of the Study

The study is limited to specific boundaries to make sure that the research remains focused, feasible, and manageable for the development of the system. The delimitations established the parameters of inclusion and exclusion within the study.

The following delimitations are identified in the study:

- **The system is limited to web-based implementation only.**

  The proposed solution is a browser-based and has no independent programs or offline solutions. Users must access the system via a web browser.

- **Mobile application functionality is not included.**

  The system does not cover the creation of any Android or iOS applications. Only the browser-based approach to usage will be considered.

- **Limited predictive analysis (basic forecasting only).**

  Though the system has data analytic capabilities, it has only limited capabilities for forecast generation and trend analysis, with no advanced predictive modeling options or artificial intelligence algorithms. It relies on historical trend averaging.

- **Advanced artificial intelligence features are not included**

  No complicated AI functions such as deep learning, intelligent decisions, or fully automated recommendations are employed in this system. All recommendations and decision remain with the management.

- **Internet connectivity is required to access the system online.**

  As this is a web-based application, internet connectivity is required for accessing all its functionalities. There is no offline mode or local data storage capability built into the user interface.

1.5 Significance of the Study

The development of the **Data-Driven Funeral Service Management System** is expected to provide several benefits to funeral service providers, staff members, clients, researchers, and the academic community. Through the integration of information system, automation, and data analytics, the study seeks to contribute to the modernization and improvement of funeral service management.

- **Alfonso Somo Funeral Home Administrators/Owners**

  The proposed system provides funeral home administrators and owners with an organized platform for managing operational processes more efficiently. Through centralized information management, administrators may easily monitor schedules, customer records, inventory, transactions, and operational activities from a single location.

  The system also helps administrators improve planning and decision-making through analytical reports and operational monitoring tools. By utilizing data analytics, management may identify operational trends, monitor service demand, analyze resource utilization, and evaluate business performance. This leads to better financial management and strategic growth.

- **Alfonso Somo Funeral Home Employees and Staff**

The proposed system assists Alfonso Somo Funeral Homes employees and staff members in performing their responsibilities more effectively and efficiently. Through automated scheduling, organized record management, and centralized communication, employees coordinate funeral service more easily and reduce the time spent on administrative paperwork.

The system also reduces the workload associated with manual documentation and repetitive administrative tasks. Staff members see records and information quickly, and communicate with other personnel through the system, reducing errors and confusion.

- **Clients and Bereaved Families**

The proposed system provides clients and grieving families with improved convenience, accessibility, and transparency in funeral service management. Through digital functionalities, clients access service information, schedules, communication features, and payment details more efficiently, reducing the stress of coordination.

The system also helps families make informed decisions regarding funeral arrangements by providing organized information about available services, schedules, and pricing. Improved communication and coordination contributes to better customer experience and satisfaction.

- **Researchers and Future Developers**

This study serves as a reference for future researchers and system developers who intend to conduct studies related to funeral management systems, information systems, analytics, and service management technologies. It provides a framework for how technology can be applied to service industries.

The concepts presented in this study can contribute to future innovation and improvements within the field of information technology and service management.

- **Academic Community**

This research contributed to the field of Information Technology and Business Analytics by showing how data analytics could be applied in service management, specifically in the funeral industry which is often overlooked in technology studies. It adds to the body of knowledge regarding industry-specific system development.

- **Local Community**

  The proposed system also benefits the local community by improving the efficiency and quality of funeral services provided by Alfonso Somo Funeral Homes. Through better planning, resource allocation, and service coordination, families can receive more organized, timely, and reliable assistance during difficult times. The system‘s ability to streamline operations and reduce service delays can enhance customer satisfaction and strengthen public trust in funeral service providers. Furthermore, the continuous improvement of funeral management practices can contribute to the availability of dependable and accessible funeral services within the community.

Overall, this study contributed to the modernization of funeral service operations through the implementation of a data-driven management system that promotes efficiency, accuracy, effective planning, and improved service delivery for the benefit of both the organization and the community it serves.

1.6 Definition of Terms

The terms below are provided relative to their use in the study. The purpose of defining these terms is to provide a better understanding of certain aspects relating to the creation and assessment of the proposed system.

1. **Agile Development Model –** is an iterative and incremental software development approach that enables developers to deliver software in short cycles while adapting to changing user requirements through continuous collaboration and feedback [9].
   - *In this study,* Agile Development Model refers to the approach used in developing the proposed funeral management system through repeated planning, testing, feedback gathering, and system enhancement.
2. **Analytics -** the process of examining raw data using mathematical, statistical, and computational methods to discover meaningful patterns, extract actionable insights, and inform decision-making [10]
   - *In this study,* analytics refers to the analysis and interpretation of data from the funeral service, including its records and transactions, for better service management and organizational efficiency.
3. **Automation –** the technique of making an apparatus, a process, or a system operate automatically [11]. **-**
   - *In this study,* automation refers to the computerization of various activities performed by the funeral homes including scheduling, inventory tracking, report generation, and record keeping.
4. **Client-Server Architecture -** is a computing model in which client devices request resources or services from centralized servers over a network [12].

client-server architecture refers to the design or model of the suggested funeral website application, whereby the users interact with the server and the database using client devices.

5. **Data Analytics** - refers to the process of examining raw data to identify patterns, gain meaningful insights, and support better decision--making [10].
    - *In this study,* data analytics is the process of gathering, organizing, and analyzing data on funeral services in order to make decisions and increase efficiency in funeral homes' operations.
6. **Data Flow Diagram (DFD) –** a graphical representation of the flow of data within a system and the process involved in handling information [14].
    - *In this study,* Data Flow Diagram refers to the graphical presentation method employed in showing the flow of information within the proposed funeral management system among the user, process, and database.
7. **Data Visualization -** is the graphical representation of information and data using visual elements such as charts and graphs [15].
    - *In this study,* data visualization refers to the use of dashboards, graphs, and visual illustrations to communicate the information about the funeral services in a way that makes it easier for administrators, and employees to understand.
8. **Database -** is an organized collection of data stored and accessed electronically [12].
    - *In this study,* the database refers to a system that can store records of client data, data related to the deceased, scheduling information, and transactions, allowing for effective storage and organization of data concerning the funeral services.
9. **Database Management System (DBMS) –** is a software designed to store, organized, retrieve, and manage data efficiently in a database [16]
    - *In this study,* Database Management System refers to the application used to manage, organize, and protect records contained within the database of the funeral management system.
10. **Data-Driven – r**efers to decisions or processes based on the analysis and interpretation of data rather than intuition [12].
    - *In this study,* data-driven refers to the concept that involves utilizing collected funeral service data to make decisions and manage workflows in a more effective manner.
11. **Decision-making -** the action or process of making decisions, especially important ones [20].
    - *In this study,* decision-making means choosing appropriate decisions to take regarding the running of the funeral home based on analysis of reports.
12. **Entity-Relationship Diagram (ERD) –** a visual representation of entities, attributes, and relationship within a database system [17].

- In this study, Entity-Relationship Diagram refers to the diagram developed by the researchers for designing and organizing the database structure of the funeral management system.

13. **Funeral Home –** is a business that provides services related to funerals, including preparation of the body and arrangements for burial or cremation [12].
    - *In this study,* funeral home refers to a place where there are facilities for preparing bodies, arranging funerals, making schedules, and coordinating memorials that are made optimal by the developed system.
14. **Funeral Management System –** aligns with the concept of a management system, which is a structured framework for managing operations and processes [12].
    - *In this study,* funeral management system refers to the system that is developed for managing funeral homes through the internet and includes functionalities such as managing customer information, scheduling, resource allocation, and other activities.
15. **Information System –** is an organized system for collecting, processing, storing, and distributing information to support organizational operations and decision-making [2].
    - *In this study,* information system means the computerized system that is used to manage the operations related to the funeral service information, processes, and transactions.
16. **Inventory Management –** the process of ordering, storing, tracking, and controlling inventory or stock within an organization [18].
    - *In this study,* inventory management refers to the process of controlling the funeral supplies and equipment through the suggested funeral management system.
17. **Manual Process –** a task performed by hand or without the use of automated systems [12].
    - *In this study,* manual process refers to the traditional practices adopted by funeral homes whereby all procedures are conducted through pen and paper without the use of any computerized methods.
18. **MySQL –** an open-source relational database management system used for storing and managing structured data (Oracle Corporation,n.d).
    - *In this study,* MySQL is defined as the application utilized to manage, organize, and process records and transactions related to the funerals.
19. **Optimization -** is the act of making something as effective, perfect, or functional as possible [15]. **-** *In this study,* optimization refers to improving the operations of funeral services through better scheduling, proper allocation of resources, and effective decision making through data analysis
20. **Planning - i**s the process of making decisions about what will be done in the future [11].
    - *In this study,* planning refers to the process whereby funeral services are systematically planned for, taking into account activities such as scheduling and resource allocation.
21. **Prototype –** is the original model, a sample on which to base future designs [19].

- In this study, prototype means the first model created as an attempt to implement the suggested funeral management system for its evaluation and improvement before deployment.

22. **Scheduling –** setting an order and time for planned events [19]
    - *In this study,* scheduling refers to the management of the appointments, memorial ceremonies, and assignment of employees within the system.
23. **User Interface -** a program that controls a display for the user (usually on a computer monitor) and that allows the user to interact with the system [19].
    - *In this study,* user interface is defined as the way the funeral management system looks, is designed, and allows to operate and use its features.
24. **Web-based System -** is a system that utilizes web services and software agents to efficiently select services and integrate business processes over the Internet [21].
    - *In this study,* the web-based system is refers to the web-based Data-Driven Funeral Management System that allows authorized users to operate the funeral services.

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CHAPTER 2: REVIEW OF RELATED LITERATURE AND STUDIES

Introduction

This chapter gives a comprehensive review of the literature and studies that supported the development and implementation of a Data-Driven Funeral Management System. This chapter provides detailed information taken from journals, articles, thesis, and previous studies involving the topics of funeral management systems, information systems, data analysis, database management, optimization of services and business process automation. In this chapter, information about local and international literature and studies that pertain to the present study and give details on how these previous works contribute in formulating the proposed system are provided. Through the review of literature and studies, the researchers were able to come up with an analysis of the technology and techniques used in previous works as well as the weaknesses and strengths of the systems implemented in the past.

This chapter is divided into three main sections: Foreign Studies, Local Studies, and Related Literature. Following these sections, a Synthesis of the Review is provided to summarize the findings and highlight the relevance to the study.

Foreign studies emphasized the growing use of information systems, automation, and data-driven technologies in improving funeral service operations and related service industries. Local studies showed the gradual adoption of information systems and digital technologies in funeral homes and related service industries in the Philippines, reflecting the unique cultural and operational context of the country.

2.1 Foreign Studies

Foreign studies provide a global perspective on how funeral management is approached and how technology is integrated in different cultural and economic context. These works highlight universal challenges such as the need for efficiency, transparency, and data utilization, which are applicable to the local setting of Alfonso Somo Funeral Homes. They serve as evidence of the effectiveness of technology in this specific industry.

The following foreign studies are reviewed in relation to the proposed study entitled ―A Data-Driven Funeral Management System for Planning and Optimization.‖ These studies discuss the use of information systems in funeral service management, digital record-keeping, funeral planning, resource allocation, and data-driven decision-making. Their findings provide a foundation for developing a system that can improve the operational efficiency of funeral service providers.

Information System in the Organization of Business Processes of Funeral Companies: Features of Efficiency Assessment – Elena Biryukova (2023)

Biryukova [1], in the study ―Information Systems in the Organization of Business Processes of Funeral Companies: Features of Efficiency Assessment,‖ examined the application of information technology to funeral company operations, particularly in document management, workflow automation, and administrative processes.

The study emphasized that automation can reduce the time spent on routine tasks, minimize errors, improve service delivery, and strengthen the monitoring of business processes. It also identified the limited digitalization of funeral-related activities as a challenge, particularly in the registration of deaths and the coordination of funeral arrangements.

The researcher used business process modeling, data collection and analysis, comparison, classification, and economic efficiency assessment to examine existing funeral service processes. Through the ―AS IS‖ model, the study assessed existing practices and identified areas that required technological support. It also considered a proposed information system that could support communication and coordination among the organizations involved in funeral activities. The study concluded that digitalization could establish a more unified information environment, improve the transparency of administrative procedures, and make funeral-related processes more efficient.

This study is relevant to the proposed Data-Driven Funeral Management System for Planning and Optimization because it establishes the importance of automating funeral home operations. Its findings support the integration of client records, service scheduling, administrative transactions, and document management into a centralized system. Furthermore, the use of business process analysis provides a basis for identifying operational problems and determining how an information system can address them. For the proposed system, these principles can be applied to improve the management of funeral services and provide organized data for monitoring performance and supporting operational decisions.

Study on the Strategies of Regional Cooperative Funeral Facilities: A Case Study of New Taipei City, Taiwan – Meng Ju Lin, Kuang Hui Peng, and Lih Yau Song (2019)

Lin, Peng, and Song [3] conducted a study entitled ―Study on the Strategies of Regional Cooperative Funeral Facilities: A Case Study of New Taipei City, Taiwan.‖ The researchers investigated the distribution, utilization, and management of funeral facilities in New Taipei City and its neighboring cities. They identified problems associated with uneven facility distribution, insufficient coordination, and imbalances between supply and demand. These conditions resulted in some residents using funeral facilities located outside their respective cities.

To examine these concerns, the researchers analyzed the service radius and distribution of funeral facilities and used a time-series model to estimate the supply and demand for public cemeteries and columbarium facilities. Their findings revealed that facility utilization varied across locations and that certain areas could experience supply-and-demand imbalances. The study proposed strategies involving regional planning, facility improvement, management systems, and cooperation among neighboring local governments.

Although the study focused primarily on public funeral facilities rather than the internal operations of private funeral homes, its findings demonstrate the importance of using data to support planning and resource allocation. Estimating future demand can help decision-makers identify potential shortages and determine where resources or facilities may be needed. The study also illustrates how information about service locations, utilization patterns, and demand can contribute to more effective management.

For the proposed Data-Driven Funeral Management System for Planning and Optimization, the study provides a foundation for incorporating basic forecasting and analytical dashboards. Historical funeral service records may be analyzed to identify service demand patterns and help management plan the availability of staff, vehicles, funeral packages, and other operational resources. While the proposed system focuses on a funeral home's operations rather than regional government planning, the principle of using historical data to support planning and optimize resource utilization remains applicable.

Development of Mobile App: Funeral Services System (FSS) – Researchers at Universiti Teknologi MARA (2023)

The study conducted by researchers at Universiti Teknologi MARA in Malaysia [4] was aimed at designing a mobile funeral services app aimed at making funeral arrangements easier and more convenient. Malaysia, being a neighboring country with similar cultural and religious diversity regarding death rites, offers valuable insights into user requirements and system features suitable for the Southeast Asia context.

According to the study, planning funerals is always a complicated process that involves numerous activities which might be stressful for bereaved people in an emotional situation. The complexity arises from the need to coordinate between the family, the hospital, the cemetery, and the service provider. Families are often overwhelmed by decisions regarding caskets, flowers, transportation, permits, and scheduling. In order to make funeral arrangements simpler and reduce the cognitive load on grieving families, the researchers designed a mobile application called the Funeral Services System (FSS) [4].

The system developed included attributes such as user registration, online reservation, automatic assignment of the funeral agents according to their locations, the system for the approval of bookings, and data management systems. The researchers highlighted the significance of automation and manipulation of data in order to make the transactions quick and improve the service efficiency. One of the key findings was the speed and clarity of information was the most valued features by clients. They also found that the ability to view available options and pricing clearly reduced the anxiety associated with planning.

Furthermore, security issues and data management issues related to the funeral system was explained by the researchers. Issues related to authentication, password security, and proper management of the databases was discussed by them, emphasizing that records related to the deceased are sensitive and require high-level privacy protection and strict access controls.

The researchers concluded that their development of a mobile application proved to be useful for improving the coordination of funerals as well as making funeral services more accessible. They noted that digitization improves the relationship between the service provider and the client by establishing a channel for continuous communication and transparency.

This article is relevant to the current study since it is about the use of information technology, databases, and automation in improving the management of funeral services. The current study proposes an IT-based management system that aims at planning and optimizing the funeral service process. Just like the current article, the selected article

focuses on the significance of digital solutions, effective data management, and automation of services. While the proposed system in this research is web-based rather than mobile, the features regarding scheduling, client interaction, transparency, and security are directly applicable and validated.

Design, Development and Implementation of a Digital Burial Record-keeping and Management System in Bangladesh – Md Hafizur Rahman et al. (2025)

Rahman et al. [22] presented the study ―Design, Development and Implementation of a Digital Burial Record-Keeping and Management System in Bangladesh,‖ published in the Journal of Global Health in 2025. The researchers developed and evaluated a digital system for recording and managing burial-related information in six graveyards under the Dhaka North City Corporation. The study addressed the limitations of traditional paper-based record-keeping, particularly the difficulty of maintaining accurate records and retrieving information efficiently.

The researchers first examined existing documentation procedures through on-site observations, interviews, and consultations with relevant stakeholders. The information gathered was used to guide the design of the digital record-keeping system. The researchers subsequently evaluated the system using quantitative and qualitative approaches involving 200 application and website users and 14 graveyard record-keepers and managers

The findings indicated that the digital system was generally well received by application and website users, who reported favorable usability and technology acceptance. However, the study also identified differences in acceptance among record-keepers. Some supervisors experienced difficulties using the system, while younger and more technologically experienced operators were generally more receptive to digital record management. Training, technical familiarity, and the availability of appropriate infrastructure were important considerations in implementation [22].

The study demonstrates the potential benefits of replacing paper-based documentation with an organized digital database. Centralized digital records can make information easier to retrieve, improve documentation practices, and provide data that can be analyzed for planning and monitoring purposes. However, the findings also emphasize that system usability and user readiness must be considered when introducing digital solutions.

This study is closely related to the proposed funeral management system because both address the need for accurate, accessible, and organized records concerning funeral or burial activities. Its findings support the development of centralized client records, deceased information, service histories, and transaction records. In addition, the study highlights the importance of designing a user-friendly system that can be used by administrators and staff with varying levels of technological experience. The proposed system extends this record-management approach by incorporating funeral scheduling, inventory monitoring, service tracking, and data visualization to support operational planning and optimization.

Online Presence of the Funeral Industry: The Example of the Quebec Federation of Funeral Cooperatives - Elisabeth Beaunoyer, Alexandre Guitton, and Matthieu J. Guitton (2022)

The research written by Elisabeth Beaunoyer, Alexandre Guitton, and Matthieu J. Guitton [23], focuses on the transformation caused by digital technologies on the way of conducting business and interacting with clients in the funeral industry. The study highlights the importance of digital technologies for the funeral industry as they offer alternative ways for funeral institutions to interact with the community, offer their services to plan a funeral ceremony, help people overcome grief, and improve the practice of memorialization.

Funeral institutions have traditionally played an important role in providing necessary assistance to communities and dealing with their grief. However, with the development of communication technologies and web resources, the methods of working with funeral ceremonies have changed. In particular, technologies such as websites, online memorials, cyber funerals, and grief support platforms become increasingly relevant [23].

According to the authors, the need for digital funeral services became more urgent due to the onset of the COVID-19 pandemic, which made interactions much less common. Funeral establishments were forced to strengthen their online presence as a way to remain relevant in helping mourning families through the difficult process [23]. As a result, funeral companies had started to implement websites, social networks, and other digital communication channels into their practices.

The research focused on the Quebec Federation of Funeral Cooperatives located in Canada as an example of a funeral institution undergoing digitalization. This organization comprised several funeral cooperatives that were spread across Quebec. According to the authors, this union had created a rather impressive digital footprint using websites, social networks, online grief support, and digital funeral planning [23].

For conducting the study, the authors adopted a qualitative virtual anthropology methodology. The study involved an analysis of 21 cooperatives' websites, 1 federation site, and Facebook pages, forums, and other sources related to funeral services. In doing so, the authors looked into how online platforms were affecting the delivery of funeral services, communications between the organizations and their clients, and the way clients viewed funeral organizations.

It was found out that digital technologies enhanced the funeral-planning process through increased availability of the information needed by the clients. The funeral organizations made use of websites in providing information about the services, contact details, virtual memorials, funeral planning, and bereavement assistance services. The researchers stated that more and more people are using the Internet in searching for funeral information than contacting organizations directly.

Another significant feature mentioned in the research was the Family Space platform known as ―Espace Famille‖. This was a digital platform where families could plan funerals and communicate with funeral homes using online platforms. The researchers pointed out that the introduction of this online space for collaboration between the family and funeral homes helped to integrate funeral information into one single digital place.

Moreover, the study emphasized the importance of digital grief support systems. The federation offered online grief counseling facilities as well as discussion forums where bereaved people could share their experiences and get emotionally supported. According to the researchers, digital grief support systems extended the responsibilities of funeral institutions from just arranging funerals to supporting families during mourning.

A further significant result from this study was the role played by website design and user interface. In the study, it is noted that the funeral cooperative websites utilized comforting graphics, effective designs, and navigations systems in order to enhance user experience. Additionally, the funeral companies tailored their web page content to fit the culture of the locality.

It is noted that the digital transformation helped to increase the visibility of funeral services. Information about the funerals, such as prices of the services offered, were provided on the web pages, which allowed customers to engage in funerals digitally.

However, some issues associated with digital inequality were identified by the study. Some older participants and people with low technology skills had problems with utilizing online technologies. It was suggested that funeral companies needed to ensure that they considered accessibility issues and find other means of service provision to those who are not tech-savvy [23].

It is concluded that digital technology made the whole process of service provision much more effective. Communication and even memorials became much easier than before.

This paper is very relevant to the suggested capstone research paper on ―A Data-Driven Funeral Management System for Planning and Optimization,‖ as it shows the ability of digital technologies to enhance funeral management, communication, customer service, and planning processes. The conclusions drawn in the paper will help to develop central funeral management systems which will combine various aspects of online communication, planning, customer services, and information management.

Operational System Exclusive Funeral Muslim in Indonesia "Darul Muqamah Memorial Park" - Adi Hardono et al. (2021)

The research conducted by Adi Hardono et al. [24] discussed the process of developing and analyzing the operational system of management for an exclusive Muslim funeral service in Indonesia. The research stated the importance of having organized and professionally managed funeral services as a result of increasing urbanization, high population rate, and limited burial land in urban areas like Jakarta. The researchers argued that the management of funerals has now become an important problem since an increase in the population also leads to an increase in burial requirements.

According to the study, Indonesia belongs to those developing nations which have large population size, and due to the increase in population size, many problems emerge in cities, such as insufficient residential and burial grounds. It was mentioned by the researchers that, statistically, many deaths occur in Jakarta every year, especially Muslims. With the increasing number of deaths and less cemetery space, researchers highlighted the requirement for a modernized

funerary management system that would help enhance land utility and operational efficiency, among others. Researchers pointed out many shortcomings in the cemetery management of Jakarta, like overcrowding of the cemetery, poor maintenance, lack of infrastructure, disorganized burial services, and inadequate parking facilities[24].

Moreover, the study revealed that upper-middle-class families were more inclined towards private and professionally managed cemeteries since the public burial grounds were congested and lacked proper facilities. After conducting surveys and interviews among the Muslim respondents who belonged to middle- and upper-class societies, the researchers found that the preferred cemetery would be one that was clean, tidy, secured, serene, professionally managed, and had Islamic values. Consequently, the researchers suggested the development of a funeral service business known as Darul Muqamah Memorial Park [24].

Darul Muqamah Memorial Park refers to an exclusively designed funeral service and cemetery management system that offers full-scale services right from body picking up until finalization of burial and maintenance. According to the researchers, the reason behind developing such a system was not only providing the needed space for burial but also offering better funeral services through professional operational management and modern facilities. The researchers explained that the funeral park comprised landscaped cemeteries, organized grave blocks, mosques, parking lots, food courts, fish ponds, gardens, and sitting places among others to comfort the mourners [24].

One of the most significant findings of the study included the introduction of technology in funeral service delivery. According to the study, the Darul Muqamah Memorial Park used online registration and booking for funerals and burials. Customers were allowed to access their funeral services online by registering and booking funeral services on the company's website where they were allowed to book burial plots, enter funeral details, and request funeral services online [24].

According to the study, online registration helped make funeral service preparation easier and faster since the management would have adequate time to arrange for burial grounds, funeral rooms, transport, and other required personnel for the services long before receiving the corpse. This helped in avoiding unnecessary delay of funeral service preparation procedures. The study indicated that such arrangements were crucial due to the Islamic funeral culture.

The digital database management system was another notable aspect of the operational system. The research states that the identity and burial places of the deceased people were recorded in the database of the cemetery management. By means of the digital database system, it would be easier for the family members to locate the grave sites using computers and the database.

The study also highlighted the use of CCTV cameras and security systems are another aspect that the researchers pointed out. The use of security systems increased the overall security of the area and enabled the families to monitor the cemetery grounds from remote places. Furthermore, the use of the digital security systems facilitated the operation management of the memorial park.

For their analysis on funeral operations, the authors adopted a SIPOC (Supplier, Input, Process, Output, and Customer) model. Through the application of the SIPOC model, the authors were able to get the whole picture of all processes involved in funerals and thus come up with ways to enhance operational efficiency in funeral management operations. The funeral operations were divided into major processes, which include funeral registration, body collection, body washing, shrouding, funeral prayers, grave digging, burial, transportation of the dead to the place of burial, and placement of graves.

According to the authors, funerals can be registered manually or via the Internet. In the course of the registration process, clients select burial places and make payments while at the same time giving some information regarding funerals. After the registration process, ambulances take the dead to the memorial park. Body washing and shrouding is then done by qualified personnel following the Islamic culture.

The study highlighted that through the operational system, coordination between funeral employees and family members was achieved appropriately during the entire operation. The team responsible for grave digging prepared burial pits using Islamic standards, whereas the team involved in the funeral was charged with transportation and arranging for burial. The study highlighted that organized operational systems contributed to funeral efficiency despite being emotional tasks.

Moreover, the study pointed out the importance of visual management systems during funerals. The study indicated that visual systems such as maps, direction markers, flow charts, information boards about services, and guidelines contributed to effective communication and helped avoid information shortages. According to the study, an organized visual system helped reduce inquiries and made it easier for visitors to move within the cemetery.

Other important contributions from the study were the emphasis on the operational management and development of infrastructure in the research. In this regard, the research ensured proper planning of the cemetery using architecture experts who developed the plans based on capacity for singles, doubles, and family burials. Planning of infrastructure in the memorial park involved roads, parking, prayer facilities, landscaping, among others.

Operational management, according to the study, is very important in ensuring the effective use of the resources available for performing funeral processes. These resources include manpower, equipment, transportation means, facilities, and burial ground.

The results of the study demonstrated that the use of structured operation system positively affected the effectiveness of funeral services, customer satisfaction, and cemetery management. The authors claimed that technological-based funeral systems made funeral arrangements much faster, better organized, and more accessible for families who have lost their relatives. The use of the SIPOC operation model positively impacted the work of the staff and the processes of funerals organization.

Moreover, according to the study, modern funeral companies should use technologies to solve the problems caused by an increase in urban population and requirements for burials. Technological solutions help manage data,

plan operations, and provide better customer services. The researchers suggested further development of technological-based systems of funeral arrangements to ensure cemetery management in the future.

2.2 Local Studies

Local studies are conducted within the Philippines context. They are highly relevant as they address the specific cultural, regulatory, and operational environment in which Alfonso Somo Funeral Homes operates. These studies highlight the unique challenges faced by Filipino funeral businesses, including the complexity of Filipino funeral traditions, regulatory requirements, and the impact of modernization on traditional practices

Funenaria and Modern Funeral Homes: Change, Modernity, and Sustainability – Derogongan, Agosto, & Bariñan (2019)

A local study by Rohane M. Derogongan, Arnann Jay M. Agosto, and Russel M. Bariñan during the Third International Conference on Sustainable Innovation, [25] looked at the history, modernization, and sustainability of funeral homes in Iligan City through technological innovation and business competition.

This study highlighted that the funeral homes in the Philippines have changed from traditional and familybased ―funenarial into technologically advanced places that offer systematic funeral and burial services. As per the researchers, the funeral homes do not operate just as business ventures, but as social institutions which help in comforting the mourning families, employment generation, public sanitation, and economic development of the region.

In the past, Philippine funeral ceremonies were heavily influenced by the native traditions and belief system. Prior to modern times, funeral preparations were basically organized by the community through manual systems [25]. Nonetheless, the Americans have contributed much to the development of funeral practices in terms of embalming, cremations, and the systematic process of funerals which later gave rise to today‘s modern funeral industry. The institution of civil registration and embalming regulations in the 1950s was an indication of the professionalism of the industry.

The researchers narrated the story of the creation of the first funeral homes established in Iligan City like the Funenaria Halili, Mansueto Funeral Homes, and Gamalinda Funeral Homes. These funeral homes started out using manual methods in embalming, making of caskets, funeral schedule, among others. Even without the use of advanced technology back then, these companies played an important role in society by providing inexpensive funeral services for the poor.

As the process of modernization continued, funeral homes had to adjust to the growing needs of consumers and increasing competition in the industry [25]. Corporate funeral homes like St. Peter Life Plan and Cosmopolitan Memorial Chapels took their place in the funeral homes sector and started implementing modern technologies, funeral services covered by insurance programs, organizational management, online assistance, and technical customer service. Modernization greatly impacted funeral service management, according to the research results.

The study provided a detailed description of how technology changed funeral management practices. Corporate funeral homes adopted websites, online memorials, e-burol (virtual wakes), and other modern means of communication and assistance to allow families living overseas to take part in the wake and burial ceremony – a crucial feature in Philippines where many family members work abroad. In addition, websites have become an effective medium for advertising funeral services, contacting customers, and delivering customer service.

Another critical issue brought out in the study was the difficulty traditional funeral homes have faced trying to compete effectively. A number of the local funeral homes found themselves facing numerous issues such as financial viability, lack of automation, family disputes, aging management, and the failure to adapt to technological advancement. It was observed that the gap between large chains and small independent homes is widening because of technology. It was seen to be the need for adopting systematic and technological means to ensure that they can survive in a very competitive market [25].

The study also explained how the funeral homes were responsible for playing a major role in social organization and public health. This was done through effective management of funerals, keeping records, and proper sanitation practices by the funeral homes. In this regard, the funeral homes played an important role in ensuring the maintenance of social equilibrium.

The importance of this research to the proposed funeral system is in how it addresses modernization and the application of technology in funeral homes. The research findings suggest that there are many funeral business establishments that are using technological tools in order to operate efficiently and effectively. In this regard, the proposed system ―Data Driven Funeral Management System for Planning and Optimization‖ will definitely meet some of their needs.

More specifically, this research pointed out that there are a number of operational problems like scheduling, coordination, communication with customers, and management of funerals that could be overcome by implementing an integrated funeral management system such as the one recommended above that will involve digitizing of funeral records, scheduling of funerals, organization of customer records. Alfonso Somo Funeral Homes is one of this type of independent businesses that needs to modernize to compete and serve the community better.

Moreover, the study shows that digitization is an important aspect in enhancing customer satisfaction and the availability of services. In turn, the proposed capstone system can provide additional benefits through the use of analytical capabilities, planning, optimization, and information management, which would enable more efficient operations without compromising on quality of service provided.

Thus, the study conducted by Derogongan et al. (2019) can be considered as a good basis for the proposed research as it illustrates how modernization and effective management systems have become important aspects of sustainable funeral homes‘ operation in the Philippines [25].

Everplan: A Multiplatform Funeral Arrangement and Management System for Funeral Services – Pangasina State University (2025)

Another was carried out by students and teachers from Pangasinan State University in 2025 and is aimed at simplifying funeral planning and funeral service management. [26]

The study highlighted the fact that funeral planning can be quite distressing and challenging for families. The standard process of funeral arrangement involves personal visits to the funerary house, dealing with lots of paperwork, making manual comparisons between various service packages, and seeking recommendations from friends and relatives. Such process turns into an even greater challenge if families do not have enough information about prices, service types, and timing.

Some problems were noted in regard to both families and funeral service providers. Specifically, families faced problems with locating affordable and diverse funeral services, while funeral service providers found it difficult to communicate with clients, schedule appointments, promote their businesses, and respond to false allegations made by third parties.

To deal with these problems, the research team came up with the idea of creating a new solution called Everplan using the RAD (Rapid Application Development) model. It was developed on the basis of an e-commerce platform where users had the ability to review and compare various packages of funeral services offered by providers [26].

There were several key features integrated in the system such as browsing and reviewing packages, form filling, sending requests for funeral services, notifying about the status of the transaction, making a favorites list or adding items to cart, and editing funeral services details on the internet.

In addition, this study also highlighted the importance of the technology used to enhance transparency and efficiency in managing funerals. Through the digitalization of funeral services transaction processes, the reliance on referrals was minimized and the family was better equipped to make the right choices. The software also ensured easy communication between the service provider and the client through automated messages and management of the service.

To determine whether the system worked well, usability and acceptability tests were performed using ISO 9126-1, which is an internationally recognized software quality evaluation method. This gave a framework for determining and analyzing both the functional and non-functional software attributes that are needed to satisfy the requirements of the end-user [26]. Thirty families and four funeral service providers were involved in the evaluation. Findings showed very high acceptability in all the six qualities of the software: functionality, reliability, usability, efficiency, maintainability, and portability.

The authors' findings revealed that Everplan was able to simplify the funeral process for its users and improve communication between the funeral home and its customers. The authors suggested several enhancements such as messaging capabilities, payment facilities, language options, maps, and alerts [26].

The current study is very relevant to the proposed system because both systems address the same issue of improving the management of funerals with the help of technological innovations. Everplan shows how much funeral services can gain by utilizing such systems [26].

However, as much as Everplan focused on managing the process of funerals and dealing with customers, the new Data-driven Funeral Management System aims at offering more functionalities based on data analysis and management. Some of the features that could be included in the proposed funeral management system are predictive scheduling, inventory analysis, service demand analysis, performance analysis, and planning optimization.

The Everplan research project offers the feasibility and viability of implementing funeral management systems in the Philippines. Moreover, the study shows that both families and the funeral services providers would readily embrace the technology of funeral management if they were offered efficiency, transparency, and convenience.

Information Systems Development Plan for Sta. Teresa Funeral Homes – Davao Del Norte State College (2021)

The local research which was done by students of Davao Del Norte State College in 2021, aims at improving the operations of Sta. Teresa Funeral Homes by introducing information systems and IT systems. [7]

The research stated that although there had been increased technological developments, funeral homes were yet relying heavily on manual operations. The researchers cited several problems being faced by Sta. Teresa Funeral Homes in terms of attending management, payroll computations, production tracking, and inventory management. They pointed out that manual work slowed down processes, produced errors, and made it difficult to monitor performances and production levels [7].

The study included an observation of all the activities carried out in the funeral home, which included assistance services to the customers, burial process, deceased body retrieval, production of the casket, and billings.

Since the funeral home operates round the clock, it was becoming increasingly challenging to manage employee schedules manually.

In order to address these challenges, the researchers introduced two information systems: the Production Monitoring System (PMS) and the Biometric Time and Attendance System (BTAS) [7]. The former aimed at controlling the caskets‘ stocks, avoiding the lack of stock and overproduction, stabilizing production, and better resource management. On the other hand, the BTAS introduced automation for managing the staff attendance and salaries.

By focusing on the use of information systems in funeral homes, the researchers stressed the value of this aspect in increasing the productivity and efficiency of businesses in this area. With the help of automated collection and monitoring of the data, managers could make more accurate decisions about inventory, personnel, and production planning.

Another important contribution made by the study is that it highlighted the need for real-time monitoring and data-based management. As stated in the study, the production monitoring systems ensure paperless production and help businesses make comparisons between intended and actual production outputs. They also enhance scheduling and resource allocation.

The other contribution of this study is its focus on the benefits that can be realized from the use of biometric technology to increase attendance accuracy and accountability. With attendance integrated with the payroll and database system, it becomes easier for funeral homes to manage their employees.

Additionally, the researchers recommended an IT infrastructure structure comprising desktop computers, biometric systems, operating systems, and enterprise software applications to facilitate the implementation of the systems. The researchers also carried out economic feasibility analysis to determine the costs involved in implementing the systems and the feasibility of implementing such systems in funeral homes.

This research is applicable to the capstone project proposal because it is focused on the application of information systems in managing funeral homes. The proposed Data-driven Funeral Management System has similar objectives as the above systems.

While the Sta. Teresa study emphasizes the use of automated processes in production monitoring and attendance control, the intended capstone project aims to create an even more comprehensive management information system with planning, scheduling, customers management, analysis, and optimization. On the other hand, the current study highlights the need for automated processes in funeral homes in order to enhance productivity and accuracy.

Moreover, it proves that the management of operational data is an important aspect in funeral homes. This helps in managing the resources used and the performance of services provided. The mentioned aspects will greatly help in achieving the purpose of the intended capstone project.

Graveyard Management with Geographical Information System and Operational Business Analytics for Negros Occidental – Gonzaga, R. et al. (2023)

Another local study by Roslyn Gonzaga, Jezza Claire Salvideo, Edward John Octava, Geramae Baquiano, and Carl Mathew Velez [27], they developed a Geographical Information System (GIS) integrated with operational business analytics for cemetery and memorial garden management in Negros Occidental, Philippines. The study aimed to address the inefficiencies of traditional manual cemetery operations, including inaccurate burial records, difficulty in locating gravesites, delayed transactions, and poor maintenance monitoring. The system was designed to digitalize cemetery management through automated record keeping, GIS-based mapping, analytics, and maintenance scheduling.

The studies also highlighted the usefulness of GIS and operational analytics in improving cemetery and funeral resource management. The developed system utilized GIS technology to provide a visual representation of cemetery layouts and burial plots, allowing administrators and visitors to easily locate gravesites. The integration of operational

business analytics allows cemetery personnel to analyze operational data, improve decision-making, optimize resources allocation, and generate reports efficiently. The system also included features such as billing automation, ownership transfer tracking, visitor notifications, and gravesite maintenance scheduling. These functionalities improved service delivery, reduced administrative workload, and ensured accurate and updated cemetery records.

Furthermore, Gonzaga et al. (2023) emphasized the importance of digital transformation in cemetery and funeral-related services. The researchers demonstrated how GIS and analytics technologies can improve operational efficiency, customer service, and long-term planning within cemetery management. The findings revealed that the developed system was reliable, efficient, and user-friendly based on evaluations conducted by IT experts and beneficiaries [27].

This study is relevant to the proposed Data-Driven Funeral Management System for Planning and Optimization because both studies focus on the use of data analytics and information systems to improve funeral and cemetery management operations. The study of Gonzaga et al. (2023) provides valuable insights regarding GIS integration, automated record management, and operational analytics, which can support the development of a more efficient, organized, and data-driven funeral management platform [27].

2.3 Related Studies

Management Information System

According to the study by Yousef A. Baker El-Ebiary et al.[8], Management Information System (MIS) plays a significant role in improving organizational decision-making processes. The study explained that MIS is a combination of technology, people, procedures, and data that work together to collect, process, store, and distribute information needed by organizations. Through MIS, managers are provided with accurate, timely, and relevant information that supports planning, monitoring, controlling, and strategic decision-making [8].

The researchers emphasized that effective MIS implementation helps organizations improve operational efficiency and maintain competitiveness in the market. The study identified several factors that contribute to the effectiveness of MIS, including proper system design, high-quality data, user training, system integration, and organizational support. The researchers also noted that some MIS implementations fail due to poor requirement analysis, inadequate training, and ineffective system management. Therefore, organizations must ensure that their information systems are properly developed and aligned with organizational objectives to maximize their benefits in decision-making processes.

Furthermore, the study highlighted that MIS enhances data-driven decision-making by transforming raw data into meaningful information that can guide managers in solving problems and evaluating alternatives. The integration of information systems within organizations allows faster communication, improved coordination, and more reliable reporting processes. These capabilities help organizations reduce errors, optimize resources, and improve service delivery.

This literature is relevant to the study because it supports the importance of implementing an effective Management Information System in improving operational processes and organizational decision-making. The concepts presented

by El-Ebiary et al. (2023) provide a strong foundation of developing a funeral management system that utilizes data-driven technologies to improve planning, record management, resource allocation, and service efficiency [8].

Data-Driven Decision Making

According to Jakob Mokander et al.[28], in their study entitled *Ethics-Based Auditing of Automated Decision-Making Systems: Nature, Scope, and Limitations,* a data-driven decision-making system play an important role in modern organizations by improving efficiency, accuracy, and consistency in decision-making processes. The study explained that automated decision-making system utilize data collection, analysis, and algorithmic processes to generate insights that support organizational operations and strategic planning. Through these systems, organizations can make informed decisions based on reliable and organized data rather than relying solely on manual judgment and traditional methods.

The researchers emphasized that data-driven systems provide organizations with the capability to process large amount of information efficiently while improving transparency and accountability in operations. The study also highlighted the importance of ensuring fairness, reliability, and proper governance in automated systems to maintain the quality and integrity of decision-making processes. Furthermore, the researchers discussed how data analytics and automation can support operational management by identifying patterns, predicting outcomes, and optimizing resources within organizations.

This literature is relevant to the proposed system because it supports the significance of utilizing datadriven technologies in improving operational efficiency and decision-making. The concepts presented by Mokander et.al. (2021) provide a theoretical foundation for the development of a funeral management system that can organize records, analyze operational data, improve planning processes, and support effective management decisions through reliable information systems [28].

Data Analytics and Visualization

Another study explored the role of data analytics in improving organizational performance and enhancing strategic decision-making processes. [6] The researchers discussed how businesses increasingly rely on data-driven approaches to analyze operational activities, customer behavior, and market trends in order to achieve better outcomes and improve efficiency. The study emphasized that organizations generate large amounts of data daily, and without proper analytical tools and techniques, valuable information may remain unused. Through the application of data analytics, companies are able to transform raw and unstructured data into meaningful insights that support planning, forecasting, and optimization.

The study further explained that data analytics enables organizations to identify patterns, trends, and relationships within datasets that may not be immediately visible through manual analysis. By using analytical models and visualization techniques, decision-makers can better understand business performance and predict future outcomes [6]. The researchers highlighted that visualization tools such as graphs, charts, dashboards, and reports play an important role in simplifying complex data and presenting information in a more understandable format. These

visualization techniques improve communication among managers and stakeholders by allowing them to interpret data quickly and make informed decisions efficiently.

Moreover, the study examined how predictive analytics and optimization methods contribute to sales performance improvement. Historical sales records and customer data were analyzed to forecast demand, evaluate business trends, and determine effective strategies for increasing operational productivity [6]. The findings revealed that organizations using data-driven decision-making approaches were more capable of responding to market changes, minimizing inefficiencies, and improving resource allocation. The study concluded that integrating analytics and visualization into management systems significantly enhances organizational planning and operational effectiveness.

This study is highly relevant to the proposed research entitled ―A Data-driven Funeral Management System for Planning and Optimization‖ because it demonstrates the importance of utilizing analytics and visualization tools in managing organizational data. Similar to sales optimization systems, the proposed funeral management system aims to collect, process, and analyze operational information to support better decision-making and planning. Through the use of data analytics, the proposed system may help funeral homes monitor service records, forecast service demand, manage inventory, and optimize scheduling processes. In addition, visualization features such as dashboards and graphical reports can assist administrators in understanding operational trends and improving management efficiency. The study supports the idea that integrating data-driven technologies into management systems can enhance accuracy, productivity, and overall organizational performance [6].

Database Management System

[29] Database management systems play a crucial role in handling and organizing large volumes of structured data in information systems. In the study published in the International Journal of Database Management Systems (IJDMS), the authors discussed the importance of DBMS in ensuring efficient data storage, retrieval, and management while addressing common challenges such as data consistency, security, synchronization, and system performance.

The study emphasized that database systems must be properly designed to support reliable and efficient information processing, especially in environments that handle complex and large-scale datasets. In relation to the proposed study, ―A Data-driven Funeral Management System for Planning and Optimization,‖ a database management system is essential for ensuring the accurate and efficient handling of funeral service records, scheduling data, inventory tracking, and transaction processing.

By applying DBMS principles, the proposed system can improve data integrity, reduce redundancy, and support efficient retrieval of information needed for planning and operational decision-making.

2.4 Synthesis of the Review Literature

The foreign and local studies show that funeral service operations can be significantly improved through the use of data-driven systems and digital technologies. They highlight that traditional processes are often inefficient and risky, while structured systems, automation, and real-time data management help improve planning, coordination, and

service delivery. Overall, integrating technology leads to better decision-making, reduced errors, and a more efficient and user-friendly funeral management processes.

2.4.1 Synthesis of Foreign Studies

It was found from the reviewed foreign research studies that the relevance of information systems, automation, data analysis, digitalization of platforms, and strategic planning in today‘s funeral management is growing. Despite the focus of each individual research paper being slightly varied, they highlighted similar problems faced by all funeral firms including inefficiency of manual procedures, poor coordination and absence of unified databases, scheduling issues, and ineffectiveness in resource management. It becomes apparent that through the application of technologies and data analysis, funeral services can enhance their effectiveness and efficiency as well as their customer service, planning, and decision making. The results of these foreign research studies strongly support the capstone study proposal titled ―A Data-Driven Funeral Management System for Planning and Optimization.‖

According to Elena Biryukova (2023)‘s study named ―Information Systems in the Organization of Business Processes of Funeral Companies: Features of Efficiency Assessment,‖ the major attention was paid to the usage of information systems and automation of business processes in the field of funeral services. It is mentioned in the study that funeral organizations still use traditional approaches for their work, such as writing down necessary documents, using telephones, making direct contacts with other employees and customers, as well as the lack of systematization of their activities. These aspects lead to different problems, including inefficiency and higher costs of their work [1].

One of the significant findings from the Biryukova article is the analysis on the importance of integrated information systems that could handle all funeral-related procedures in one central location. It is indicated that funeral companies have various related processes such as notifications of death, planning of funerals and burials, coordination with customers, transport management, document handling, and cemeteries management. In case the above procedures are carried out without integrated systems, the chances of inefficiencies and miscommunication will be much higher [1].

According to the article, there were various critical operation-related difficulties in the funeral organization. Some of the main challenges include no automation processes, fragmented data bases, inadequate document handling systems, no transparent operations, and no technology for conducting financial transactions. All of these difficulties are likely to result in low efficiency and productivity and poor services offered to the clients. For the purpose of overcoming these challenges, the researcher applied AS-IS and TO-BE models in order to assess current funeral organization procedures and recommend more efficient technology-based solutions [1].

Another important part of the study was related to the description of specialized funeral information systems like AeroGIS ―Kladbishche,‖ GIS ―Pamyat,‖ and 1C: Ritual Services Management. These software applications have shown how modern funeral agencies could use technologies to improve the processes of creating electronic records, managing databases, tracking burial services, processing various documents, managing transport services,

and keeping operational reports [1]. As for the results of the study, it should be stated that information systems can significantly contribute to the improvement of funeral business performance.

Results of the conducted foreign research are quite valuable in terms of their application to the suggested capstone project. Both studies pursue a common aim associated with the improvement of funeral operations through the implementation of advanced technological solutions. Thus, the suggested system of planning and optimization of funerals will be similar to the mentioned funeral information systems as it will include a number of customer management, scheduling, operational reporting, database, and analytics functions. The mentioned foreign research proves the relevance of developing a unified database-based system.

In addition, this study highlighted the importance of data-driven management in enhancing organizational efficiency. Organizations can increase their efficiencies by acquiring funeral related data in an electronic manner and conducting analysis on the acquired data. Data analytics is very important in the proposed capstone and its implementation will be of great importance to organizations involved in offering funeral services.

The second important foreign study done by Meng Ju Lin, Kuang Hui Peng, and Lih Yau Song (2019), and its title is ―Study on the Strategies of Regional Cooperative Funeral Facilities: A Case Study of New Taipei City, Taiwan.‖ The key focus of the study was on the significance of strategic planning, cooperation among the funeral facilities and usage of data analytics in optimizing the management of funeral services. According to the research team, numerous issues arise from the poor planning, unbalance between the supply and demand for funeral services, ineffective allocation of resources, and inequality of their distribution [3].

In order to estimate the effectiveness of funeral facilities' operation, various types of analytical methods were used, including service radius analysis, time series forecasting, and analysis of supply and demand. As a result of applying the proposed methods, the effectiveness of several funeral facilities operating in New Taipei City, Taipei City, and Keelung City was evaluated. It was revealed that some of them operated in areas where there was a large surplus of demand for the funeral services, whereas others suffered from the lack of them.

Importantly, the research highlights the need for centralization and collaboration in efforts aimed at improving the management and distribution of funeral facilities. The research revealed that through analytics-based planning, organizations will be able to forecast future needs, enhance resource allocation, optimize operations, and enhance service accessibility [3]. This will ensure that organizations will be better placed to address future operational challenges using an informed approach.

A significant contribution of the Taiwan study relates to the focus on the use of data analytics and planning optimization in managing funeral services. The study highlights how data can be used by organizations to develop effective plans and make sound managerial decisions. Indeed, the findings of this study are important for the proposed capstone as the system to be developed will be aimed at facilitating data analytics and planning optimization.

Also, the study indicated that planning and forecasting were crucial in funeral management operations. Without adequate analysis of trends and future demands, many funeral organizations experience shortages, scheduling

problems, and ineffective delivery of services. It would be easier for organizations to plan and optimize resource allocation through analysis techniques and centralized systems. These results emphasize the need for the incorporation of planning and analysis functionality in the proposed system.

Another major result of the Taiwan research was the indication that funeral management not only entails operational tasks but also requires strategic planning and decision making. The above result is very important in the proposed capstone because the capstone will include features that will help with planning and analysis.

Another relevant study conducted abroad which may serve as a basis for conducting the current study is ―Development of Mobile App: Funeral Services System (FSS)‖ carried out in 2023 by researchers from Universiti Teknologi MARA. In this study, the focus was on the development of a mobile app that will help to facilitate the process of organizing funerals as well as increase access to the service. In their opinion, the organization of funerals becomes quite difficult emotionally and administratively because of the many activities and processes that need to be completed [4].

For the purpose of making the process easier, the researchers have developed a Funeral Services System (FSS), a mobile-based application, which would automate funeral arrangements and facilitate customer-funeral agent interaction [4]. Some of its functions include registration, funeral reservation, approval of bookings, automated allocation of funeral agents depending on the user‘s location, and centralized database management.

The use of automation and technology in operations made the process convenient for the stakeholders involved. With the use of the mobile application, the transactions concerning funerals became fast, efficient, and easy to coordinate [4]. Funeral assignments through the use of automation became easier. Also, the process of managing the database made it easy to record customer details and track their transactions.

Database security and information management is yet another area discussed by the researchers. It is important that there are measures put in place to protect funeral information such as authentication and passwords among others. Information that is stored within the database of any organization needs to be protected using effective database management techniques [4].

The outcomes found by the FSS study can serve as strong evidence to prove that the capstone proposal is valid since both studies have common points regarding the use of IT technologies, automation, and databases to optimize funeral management procedures. Like the proposed ―Data-Driven Funeral Management System for Planning and Optimization,‖ the FSS app was designed to make funeral planning and operation easier and more convenient.

In addition, the research proved that funeral management systems can help provide increased ease for the customers in terms of accessing services. This is because funeral management systems allow customers to organize their services via computerized applications, thereby reducing the levels of stress and ensuring effective communication between the customers and providers of funeral services. This clearly supports the capstone proposed, which seeks not only to improve the efficiency of funeral management but also to provide better services to customers.

From the reviewed foreign literature, it is clear that there are common areas within these studies that greatly help in developing the proposed capstone study. To begin with, it is evident that all the studies have established that the conventional methods of funeral management are characterized by manual activities and ineffective coordination. There are issues such as manual recording of data, scheduling problems, database disconnections, and ineffective communication.

Secondly, all reviewed articles stressed the significance of information systems and centralization in enhancing the efficiency of funeral organizations. By adopting automation and digitization, funeral organizations can manage records, optimize processes, minimize human mistakes, and serve customers effectively. The above findings justify the objectives of the proposed capstone by stressing the need for designing an integrated funeral management system that facilitates scheduling, managing clients, monitoring, and analyzing performance.

Thirdly, all reviewed articles pointed out the significance of analytics, forecasting, and optimization in improving the performance of funeral organizations. Forecasting, monitoring, and predictive analysis can help funeral organizations make better plans and optimize their performances. The above ideas relate strongly to the objectives of the proposed study concerning data analytics and operational optimization.

Moreover, from the discussed foreign studies, there was evidence of a general global trend towards digitalization of funeral service industries. This is because funeral businesses are gradually incorporating the use of mobile apps, automated management system and centralized data analysis technologies in their activities. From the above trend, it can be deduced that technology-driven funeral management systems will be increasingly relevant in modern funeral service businesses.

Finally, the studies provided strong evidence that adoption of technologies does not only result into increased efficiency in operations but also ensures organizational sustainability and client satisfaction. Automating processes decreases the administrative workload, increases accuracy and efficiency in planning, enhances communication processes and provides better accessibility of funeral services for the mourning families.

In conclusion, from what has been reviewed from the foreign literature sources, the international background for the research is well laid out in terms of theory and practice. It is clear from all the studies that information technology, automation, centralized data processing, analysis, forecasting, and optimization are indispensable when it comes to improving the management of funeral services and performance thereof. These studies justify the development of an integrated data management framework for managing funeral services effectively.

Another significant foreign study contributing to the proposed capstone research is the study titled ―Design, Development and Implementation of a Digital Burial Record-Keeping and Management System in Bangladesh‖ conducted by Md Hafizur Rahman et al. (2025). This study examined the development of digital burials recording systems in Bangladesh, which aim at resolving issues arising from manual burials recording procedures. According to the study, the use of manual burials recording systems results in missing files, inefficiency in data management, delays in reporting, and challenges in accessing information [22].

Observations, interviews, and consultations were made by the researchers for the purpose of assessing the disadvantages of the current system. From the results of the research, the development of the digital burial management system including the use of a website, mobile app, dashboard, and digital database system was made. The developed system also involved warnings and monitoring capabilities that would help enhance the management procedures at cemeteries.

The contribution of this study includes the importance attached to digitalization, especially in terms of using digital databases and monitors. Digitalization helped cemetery supervisors and administrators improve their level of productivity and efficiency in performing burial records, along with increasing information access and decision making abilities [22].

In addition, the study brought out how technology-based funeral and burial procedures could assist with operational planning and organizational sustainability even with issues such as lack of technological knowledge and adequate infrastructure. With proper implementation and training, the researchers noted that digital technology greatly increases the effectiveness of management.

The relevance of this study to the suggested capstone project lies on the validation of the significance of having a centralized database system, digitization of the records, monitoring of the processes, and information management in funeral operations. Like the suggested "Data-Driven Funeral Management System for Planning and Optimization," this study pointed out the importance of the use of digital technology in efficiency, planning, and management.

Another relevant study is titled ―Online Presence of the Funeral Industry: The Example of the Quebec Federation of Funeral Cooperatives,‖ authored by Elisabeth Beaunoyer, Alexandre Guitton, and Matthieu J. Guitton [23] (2022). The research was dedicated to the analysis of the impact of digital technology upon funeral establishments, including their websites, social networks, online mourning sites, and digital support for those who are mourning. As the authors claim, funeral companies become more dependent upon online platforms since these platforms help to communicate with clients and provide funeral planning services.

Another major finding from this research paper was the increasing significance of technology in the time of the coronavirus epidemic because funeral businesses had no other way but to find alternatives for interacting with the clients and providing the required services. As a result, funeral businesses began using technology for creating web pages, where the clients could receive necessary information about funeral arrangements.

The main conclusion drawn from this research was the importance of centralization of funeral processes through use of online tools and platforms that make it easier for users to arrange funeral services and obtain required information about them. For instance, families could use an online system called "Espace Famille" in order to communicate with the funeral homes, get information and plan funerals.

Further, the paper explains the role played by digital grief support systems and digital communication channels in expanding the duties of funeral establishments beyond operational duties. Funeral establishments were able to offer psychological assistance, memorial services, and ongoing communication through digital channels.

The research conducted aligns with the capstone project proposal since both studies emphasize the application of digital technology in managing funerals, communicating, making funeral services accessible, and offering customer service. This study clearly shows that centralized digital systems can improve efficiencies and customer engagements.

Another foreign study that is also very relevant to the current study is ―Operational System Exclusive Funeral Muslim in Indonesia _Darul Muqamah Memorial Park'‖ by Adi Hardono et al. (2021). This research was concerned with the operational system of a uniquely designed funeral and burial operation in Indonesia. According to the researchers, due to the rise of urban population growth, crowded cemeteries, and lack of funeral management processes, it became necessary to have a modern funeral management system [24].

To solve this problem, the researchers designed the Darul Muqamah Memorial Park operational system, which included registration, booking, logistics, burial process management, database development, and monitoring. With this new funeral operational system, customers would be able to book and register their funeral services using the website. This would enable the managers to make all funeral arrangements in advance [24].

The other significant impact of the study was the establishment of a database system used to record data regarding burials and locates the graves. The families could access this information quickly using the database system. Furthermore, the introduction of the CCTV systems and other security devices enhanced the operations in the cemeteries.

Furthermore, the study utilized the SIPOC operational model in the analysis of the funeral management processes. This model revealed the need for a process flow, coordination among funeral workers, and efficient operational management in enhancing effectiveness and customer satisfaction [24].

Moreover, the research highlighted the importance of planning, management of resources, and optimization of operations within funeral services. Planning of manpower, logistics, facilities, and cemetery resources played an important role in enhancing the efficiency of operations.

The results from the Indonesia study have great relevance to the capstone project being proposed as both projects aim at optimizing funeral service operations through the use of technology, data management, operational control, and planning. The research proves the fact that funeral management system technology can increase efficiency, enhance customer satisfaction, and provide sustainable operations.

2.4.2 Synthesis of Local Studies

All the discussed local studies show that the significance of the utilization of innovations in technology, digitalization, and the integration of information systems into funeral services management in the Philippines is steadily increasing. While the studies considered different aspects of funeral service management, in all cases, there was an emphasis on

the need to increase operational efficiency, enhance customer experience, coordinate efforts, and become more sustainable by implementing technology. The results of the research can serve as a rationale for developing the proposed ―Data-Driven Funeral Management System.‖

This study by Derogongan et al. (2019) showed the historical development of funeral homes in the Philippines and how the process of transformation from the traditional concept of ―funenaria‖ has taken place to the current era of technology-based funeral homes. According to the authors, the funeral home is now not only concerned with organizing funerals and aiding in burials but has evolved to become an institution offering its services through advanced technological means. It was found that the process of modernization led to increased requirements for such homes, and as a result, various technological means were adopted [25].

A key observation from the above research work relates to the fact that conventional methods of manual funeral management will be increasingly difficult to sustain within a highly competitive business environment. Some of the operational issues identified in the research include manual scheduling, coordination issues, ineffective communication with customers, and inadequate record keeping among others. This research provides strong motivation for conducting our capstone research project on funeral management.

In addition, it was emphasized that the integration of technology increases accessibility, enhances the customer experience, and promotes sustainability. The corporate funeral homes used digitalization to facilitate the arrangements process and to enhance the relationship between customers and the funeral home, which indicates that digital transformation can be extremely beneficial for a funeral business. This also confirms that the aim of the proposed project is justified since it focuses on analyzing different processes, scheduling events, creating plans, and managing operations.

One of the important contributions made in the Derogongan et al. research paper is associated with sustainability and competition in the funeral industry. According to the authors, traditional funeral homes have problems with remaining competitive since they use inefficient manual systems for their management and operations. The need for the implementation of an automated and analytical system becomes obvious as a result. As a matter of fact, the proposed project will address this problem through analytical tools for managers [25].

On the other hand, the research ―Everplan: A Multiplatform Funeral Arrangement and Management System for Funeral Services‖ carried out by Pangasinan State University researchers emphasized the relevance of digitization to the management of funeral services. The study highlighted the need for simplification of funeral planning procedures in terms of coordinating the activities of the funeral service firms and their customers [26]. Instead of engaging in complicated and lengthy physical interactions and negotiations like in typical funerals, the Everplan application was designed to facilitate the search and comparison of funeral services, booking of service requests, and transactions between customers and providers of funeral services through an online portal [26].

The results obtained in the study indicated that the use of digital systems in managing funeral services significantly simplified the process, ensuring convenience and efficiency in funeral planning. With such a system,

families did not have to engage in endless comparisons of different funeral packages or spend much time tracking funeral transactions. It is evident that this research finding relates directly to the proposed capstone project.

Moreover, Everplan highlighted the importance of automation and centralization of information management in avoiding problems related to communication and scheduling. Funeral providers found it easier to manage their transactions because of automation, notifications, and centralized requests [26]. It is consistent with the suggested study‘s goal of enhancing funeral operations through systematic scheduling and organization.

It was also found out in this study that the acceptance level of the technology was high among the users. The system was evaluated positively in its functionality, usability, reliability, maintainability, portability, and efficiency, all of which fall under the ISO 9126-1 Software Quality Standards. Thus, this proves that funeral management systems are indeed feasible and practicable in the Philippines. The high level of acceptance of Everplan shows that funeral providers and customers are ready to use funeral technologies provided that it provides them with ease [26].

Despite the fact that Everplan concentrated mainly on funeral arrangements management and customer interactions, it is also quite useful for our capstone proposal. In particular, the proposed Data-Driven Funeral Management System aims at not only managing transactions but also includes features of data analysis and optimization. The features like data analytics, predictive scheduling, inventory management, operational analysis, and planning optimization may help funeral homes in making strategic decisions [26].

Furthermore, the research conducted by the students from Davao Del Norte State College named "Information Systems Development Plan for Sta. Teresa Funeral Homes" contributed even more to the understanding of how vital information systems are for funeral businesses. According to the findings of the study, there were many issues associated with inefficient operation of funeral homes due to manual operations [7]. Specifically, attendance management, payroll processing, caskets production control, inventory management, and coordination were affected by manual systems.

In order to solve these challenges in operation, the researchers suggested the use of a Production Monitoring System (PMS) and a Biometric Time and Attendance System (BTAS). The two systems automate the process of operational monitoring, time attendance monitoring, payroll administration, and inventory management.
With automation, it was easier for management to manage resources, schedule productions, and track employees‘ performance [7]. This clearly shows how well the purpose of this proposed capstone relates to this research finding.

The most significant aspect from the Sta. Teresa study is the importance of real-time monitoring and management. According to the study, by automating the monitoring system, companies can easily compare expected results with real-time results, monitor resources used, and increase their efficiency in planning. All these points clearly match the aim of this proposed Data-Driven Funeral Management System.

Also, it is clear that using automated systems can help prevent any mistakes made by humans and lessen the burden of manual work related to record keeping [7]. Therefore, the findings of this study prove that moving towards automated systems can help funeral homes become more efficient in their services and administration. In this regard,

the proposed capstone project aims at eliminating any inefficiency arising out of manual processes by using automated systems for scheduling and managing funerals.

In addition, according to the findings of the Sta. Teresa study, it is important to note that technology will have an instrumental role in facilitating the development of a digital system for managing funerals [7]. In this regard, the researchers suggested the use of enterprise software, desktops, and biometric systems to support automatic operations. It clearly indicates that for funeral homes, digital transformation is possible through the proper use of software solutions and technology.

From the review of the local literature sources, several important recurring topics can be noted that will help to design the capstone project properly. Firstly, it should be emphasized that all sources acknowledged the fact that traditional funeral management methods become ineffective when it comes to managing the operations of modern funeral business companies. Such problems as manual scheduling, fragmentation in record keeping, ineffective coordination, inventory issues, and problems in communicating with customers continue existing.

Secondly, the authors confirmed that implementation of data management systems and various information technologies helps funeral houses to manage their businesses effectively and successfully. Using automation and centralization technologies, funeral services companies are capable of making their operations better, reducing possible mistakes and miscommunications and making their clients more satisfied. Therefore, information provided by the literature sources helps to confirm the validity of the chosen topic.

Lastly, the research shows the increasing importance of having efficient data management and operational monitoring practices within the field of funeral services. Either in terms of production monitoring methods, customer transaction systems, or even scheduling software, the studies shows that having the right data is crucial in improving efficiency and facilitating planning. This concept forms one of the key aspects around which the capstone study will be formulated and which will include data analytics and optimization techniques.

In addition, the findings of local literature show that Filipino funeral homes are progressively adapting themselves to modern technology. But despite the changes, there remains the issue of inefficiency in operations due to the use of manual systems and poor management practices. This is why the formulation of the ―Data-Driven Funeral Management System for Planning and Optimization‖ becomes necessary for achieving the set objectives.

In general, these local studies give ample justification for the significance, importance, and feasibility of conducting the suggested capstone study. The local literature shows that technological innovation, automation, centralized management, and data-based operation have become necessary aspects of funeral service management in the Philippines. Therefore, there is adequate local groundwork on which an efficient funeral management system can be developed.

2.4.3 Synthesis of Related Studies

The reviewed literature highlights the significant role of information systems, data-driven technologies, analytics, and database management systems in improving organizational operations and decision-making processes. The studies consistently emphasized that organizations benefit from the use of modern technological systems that can efficiently collect, process, store, and analyze data for planning, monitoring, and operational management. These concepts provide a strong theoretical and conceptual foundation for the development of the proposed study entitled ―A Data-Driven Funeral Management System for Planning and Optimization.‖

The literature on Management Information Systems (MIS) explained that organizations require accurate, timely, and organized information to support effective decision-making and operational efficiency. The study of ElEbiary et al. (2023) emphasized that properly designed information systems improve communication, coordination, reporting, and resource management within organizations. Similarly, the proposed funeral management system aims to improve funeral service operations through centralized record management, scheduling, service monitoring, and data organization. The literature supports the idea that integrating MIS into funeral service operations can reduce inefficiencies, improve productivity, and assist administrators in making informed decisions [8].

In addition, the literature on data-driven decision-making highlighted the importance of utilizing organized data and automated systems to support operational and strategic planning. Mokander et al. (2021) discussed how data-driven technologies and automated decision-making systems help organizations process large volumes of information efficiently while improving accuracy, reliability, and transparency. The study further emphasized the importance of analytics and governance in maintaining efficient operations. These concepts are closely related to the proposed study because the funeral management system intends to use data analysis and organized information to support planning, scheduling, inventory management, and operational optimization within funeral service organizations [28].

The reviewed literature also emphasized the growing importance of data analytics and visualization in improving organizational performance and planning. The study ―From Data to Decision: A Data Analytics Approach to Sales Optimization‖ explained that analytics and visualization tools help organizations identify trends, forecast outcomes, simplify complex information, and improve decision-making processes. Dashboards, charts, and graphical reports enable managers to better understand operational performance and respond effectively to organizational needs. In relation to the proposed system, these concepts support the integration of analytical and visualization features that may assist funeral home administrators in monitoring records, analyzing service trends, forecasting demand, and improving resource allocation and scheduling processes [6].

Furthermore, the literature on Database Management Systems (DBMS) emphasized the importance of efficient data storage, retrieval, consistency, and security in information systems. The reviewed study discussed how properly designed databases support reliable information processing and minimize issues such as redundancy, inconsistency, and poor data management. This literature is highly relevant to the proposed study because funeral service operations involve large amounts of client records, scheduling information, inventory data, and financial transactions that require proper organization and management. Through the implementation of a database management

system, the proposed funeral management system can ensure data accuracy, integrity, accessibility, and operational efficiency [29].

Overall, the reviewed literature demonstrates that the integration of management information systems, datadriven technologies, analytics, visualization tools, and database management systems can significantly improve organizational planning, operational efficiency, and decision-making processes. Although the reviewed studies focused on different organizational settings, their concepts and findings are directly applicable to the proposed ―Data-Driven Funeral Management System for Planning and Optimization.‖ The literature collectively supports the development of a centralized and data-oriented funeral management system capable of improving record management, operational monitoring, scheduling, inventory control, reporting, and strategic planning within funeral service organizations.

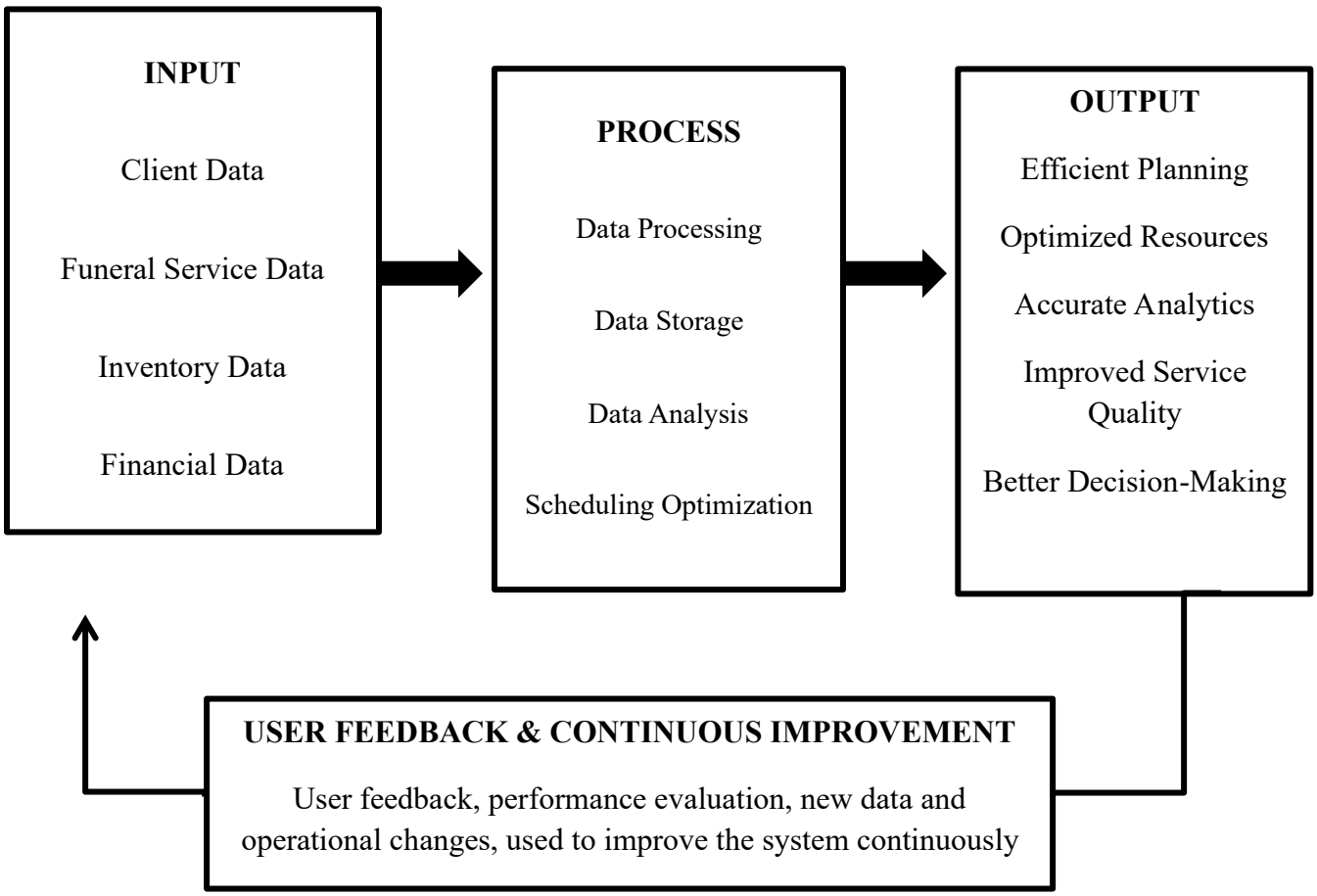
2.5 Conceptual Framework

The conceptual framework of this study illustrates the relationship between inputs, system processes, and outputs. The framework serves as a guide in understanding how data collected from funeral home operations can be transformed into a useful information that supports planning, decision-making, resource allocation, and service optimization.

The framework is based on the principle that effective funeral service management depends on accurate data collection, efficient information processing, and data-driven decision-making. By integrating funeral service records, client information, and financial transactions into a centralized database, the proposed system can generate meaningful insights that improve operational efficiency and service quality.

The framework follows the Input-Process-Output (IPO) Model, wherein various operational data serve as inputs, the proposed system acts as the processing mechanism, and optimized funeral service operations become the outputs. A feedback mechanism is also incorporated to ensure continuous improvement of the system and its services.

Diagram 1: Conceptual Framework – Data-Driven Funeral Management System



The diagram above presents the conceptual framework of the proposed system. The framework follows the **Input-Process-Output (IPO) Model**, which illustrates how data are collected, processed, and transformed into useful information that supports efficient funeral home operations.

The **Input** component consists of the essential data required by the system, namely client data, funeral service data, inventory data, and financial data. Client data include the personal information and service preferences of customers. Funeral service data contain details regarding funeral arrangements, schedules, and service packages. Inventory data refer to the available resources such as caskets, vehicles, equipment, and other funeral supplies, while financial data include payment records, service costs, and other related transactions. These data serve as the foundation for the operation of the proposed system.

The **Process** component represents the activities performed by the Data-Driven Funeral Management System. The collected data undergo data processing, where information is gathered, validated, and organized. The data are then stored in a centralized database through data storage for easy retrieval and management. Afterward, data analysis is conducted to identify trends, patterns, and operational insights that can support decision-making. Finally, scheduling optimization is performed to improve service coordination, minimize scheduling conflicts, and ensure the efficient allocation of resources and personnel.

The **Output** component represents the expected benefits of the proposed system. Through effective data processing and analysis, the system is expected to provide efficient planning, optimized resource utilization, accurate analytics, improved service quality, and better decision-making. These outputs contribute to more organized funeral operations, enhanced customer service, and improved overall management performance.

The framework also incorporates a **User Feedback & Continuous Improvement** mechanism. Feedback from users, performance evaluations, new operational data, and changes in business requirements are continuously collected and analyzed. This information is used to improve the system‘s functionality, accuracy, and effectiveness over time. The feedback loop ensures that the proposed system remains responsive to organizational needs and continuously supports the improvement of funeral management services.

2.6 Theoretical Framework

The study entitled **“A Data-Driven Funeral Management System for Planning and Optimization”** is grounded on three theories: **Management Information Systems (MIS) Theory**, **Data-Driven Decision-Making (DDDM) Theory**, and **Operations Management Theory**. These theories provide the foundation for understanding how technology, data, and efficient operational practices can be used to improve the management of funeral home services.

The **Management Information Systems (MIS) Theory** serves as the primary foundation of this study. This theory explains how information systems help organizations collect, process, store, and distribute information to support daily operations and managerial decision-making. According to Laudon & Laudon [2], management information systems enable organizations to improve operational efficiency, coordinate activities, and provide timely information for managerial decision-making. In funeral homes, managing client records, service schedules, inventory, and financial transactions through manual methods can lead to delays, errors, and difficulties in monitoring operations. By implementing a centralized information system, funeral home administrators can access accurate and organized information whenever needed. As a result, the organization can improve efficiency, reduce administrative workload, and provide better services to clients.

The study is also supported by the **Data-Driven Decision-Making (DDDM) Theory**, which emphasizes the use of factual data and analytical insights when making decisions. According to Provost and Fawcett [5], datadriven decision-making involves collecting and analyzing data to support evidence-based actions that improve organizational performance. Rather than relying solely on experience or intuition, managers use available data to identify trends, predict future needs, and evaluate organizational performance. In the proposed system, operational data such as funeral service records, inventory usage, scheduling information, and financial transactions are analyzed to generate useful insights. These insights help administrators make informed decisions regarding resource allocation, service planning, and operational improvements, ultimately leading to more effective management practices.

Another important theory that supports this study is the **Operations Management Theory**. This theory focuses on the efficient management of resources, processes, and workflows to achieve organizational goals. According to Heizer, Render, & Munson [12], operations management involves designing and managing processes

that transform resources into valuable services while maximizing efficiency and quality. Funeral homes handle various resources, including staff, vehicles, funeral equipment, chapels, and service schedules. Effective coordination of these resources is essential to ensure timely and quality service delivery. Through scheduling optimization and resource management features, the proposed system helps funeral home administrators maximize available resources, minimize scheduling conflicts, reduce operational inefficiencies, and improve overall service performance [17].

These three theories work together to support the development of the proposed Data-Driven Funeral Management System. MIS Theory explains the role of technology in managing information. Data-Driven DecisionMaking Theory highlights the importance of using data for planning and decision-making, while Operations Management Theory focuses on optimizing resources and processes. Collectively, these theories provide a strong foundation for achieving the study's goal of improving funeral home operations through data-driven planning and optimization.

2.7 Research Gap Analysis

A research gap refers to an area that has not been fully explored or addressed by existing studies. Identifying research gaps is important because it helps researchers determine how their study can contribute to the improvement of existing knowledge, practices, and technologies. Through the review of related literature and studies, the researchers observed that although several funeral management systems and related information systems have been developed, many of them primarily focus on digitizing records, automating transactions, and improving communication between funeral service providers and clients. While these systems have successfully enhanced operational efficiency, there are still areas that require further improvement, particularly in the use of data analytics, planning, resource optimization, and decision support.

The proposed study, ―A Data-driven Funeral Management System for Planning and Optimization‖, seeks to address these limitations by integrating data-driven functionalities into funeral operations. Unlike existing system that mainly focus on automation and record management, the proposed system utilizes operational data to support planning, improve resource allocation, optimize scheduling, and assist management in making informed decisions.

Table 1 Research Gap Analysis

Related Study	**Focus of the Study**	**Major Findings**	**Research Gap**
Biryukova (2023)	Information systems and automation in funeral companies	Improved efficiency through automation and integrated information systems	Limited use of analytics for planning and managerial decisionmaking

Lin, Peng, & Song (2019)	Strategic planning and resource management of funeral facilities	Data analysis improved resource allocation and demand forecasting	Focused on facility planning rather than an integrated funeral management system
Funeral Services System (2023)	Mobile-based funeral service management	Improved service accessibility, booking, and coordination	Lacked analytics, resource optimization, and performance monitoring features
Rahman, et al. (2025)	Digital burial record management system	Improved record accuracy and accessibility	Focused mainly on burial records and not overall funeral operations
Beaunoyer, et al. (2022)	Digital platforms in funeral services	Enhanced communication and online service delivery	Limited operational management and analytical capabilities
Derogongan et al. (2019)	Modernization of funeral homes in the Philippines	Highlighted the importance of adopting technology in funeral services	Did not provide a datadriven approach for planning and optimization
Everplan (2025)	Funeral arrangement and management platform	Improved customer interaction and funeral planning convenience	Limited support for analytics and strategic decision-making
Sta. Teresa Funeral Homes IS Development Plan (2021)	Inventory and attendance management systems	Improved monitoring and operational efficiency	Focused only on selected operations and lacked a centralized management approach
Proposed System	Data-driven Funeral Management System for Planning and Optimization	Integrates scheduling, inventory management, analytics, communication, and planning tools in one system	Addresses the identified gaps through data-driven decision support, planning, and resource optimization features

The studies reviewed demonstrate that information systems have become valuable tools in improving funeral service operations. Previous researchers emphasized the importance of automation, digital record keeping, online

booking, inventory management, and communication systems in increasing efficiency and improving customer service. These studies show that technology can help funeral homes streamline their operations and reduce the challenges associated with manual processes.

However, the review also revealed that many existing systems focus primarily on operational automation and transaction management. Only a few studies explored how operational data can be analyzed and utilized to support planning, forecasting, resource allocation, and decision-making. Furthermore, most existing systems do not provide comprehensive analytical tools that can help funeral home administrators monitor performance, identify trends, and optimize resources. To address these limitations, the proposed Data-driven Funeral Management System for Planning and Optimization integrates data analytics, scheduling optimization, inventory management/monitoring, and decision-support features into a centralized platform. By doing so, the study aims to provide a more comprehensive solution that not only automates funeral home operations but also support strategic planning and continuous operational improvement.

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CHAPTER 3: METHODOLOGY

Introduction

This chapter presents the methodology used in conducting the study and developing the proposed system. This chapter covers all the research methodologies used by the researchers in collecting data for the project as well as for the development of the system itself. The discussion includes the application of the research design, procedures, and methods involved in gathering data and in the development of the proposed computerized system.

The purpose of this chapter is to explain the steps and procedures undertaken to ensure that the study is conducted systematically and accurately. It serves as the blueprint that guided the researchers from the initial stage of identifying the problem up to the final stage of evaluating the developed system. The methodology chosen aligns with the objectives of the study, which is to analyze, design, develop, and evaluate a data-driven web-based system.

3.1 Research Design

Research design is the framework or plan for a study that guides the collection, analysis, interpretation, and presentation of data [30]. In system development studies, the research design helps researchers identify user requirements, analyze existing processes, develop appropriate solutions, and evaluate the effectiveness of the developed system.

For this study entitled ―Data-Driven Funeral Management System for Planning and Optimization‖, the researchers utilized a Descriptive-Developmental Research Design. This design combines the descriptive approach, which focuses on analyzing and understanding existing situations and problems, with the developmental approach, which involves the design, creation, implementation, and evaluation of a software system intended to address identified issues [31].

The descriptive component helped the researcher understand the current situation of the funeral home[30]. This includes identifying the common problems in funeral service such managing client‘s records, scheduling services, monitoring inventory, handling payments, etc.. To gather needed information, the researcher conducted interviews and discussion with the client.

The developmental component focused on designing and developing the system that address the problems identified during the data gathering process[31]. The proposed system aims to improve the efficiency and accuracy of the funeral home. It is also intended to help the funeral home manage data more effectively and support better planning and decision-making.

By combining descriptive and developmental research approaches, the study not only investigates existing operational problems but also proposes and evaluates a technological solution designed to improve organizational efficiency and decision-making.

3.1.1 Understanding the Research Design Used

The Descriptive-Developmental Research Design is commonly used in information technology and software engineering research because it allows researchers to examine existing conditions while simultaneously developing a solution to address identified problems [31].

The descriptive aspect of the design involves collecting factual information regarding the current processes being performed within the organization [31]. In this study, the researchers analyzed how funeral services are currently managed at Alfonso Somo Funeral Homes. Particular attention was given to the handling of client records, scheduling procedures, inventory management, financial transactions, reporting activities, and communication processes. Through this analysis, the researchers were able to identify inefficiencies, limitations, and areas that require technological intervention.

The information gathered serves as the foundation for identifying system requirements and determining the functionalities that must be included in the proposed solution.

Meanwhile, the developmental aspect focuses on creating a functional system based on the information gathered during descriptive phase. Developmental research involves transforming identified requirements into a working software system. It includes system analysis, system design, database development, interface creation, programming, testing, implementation, and evaluation [31].

In this study, the developmental process resulted in the creation of a web-based Data-Driven Funeral Management System that integrates resource monitoring, inventory management, customer record management, financial transaction recording, and data analytics. The system aims to support both operational management and strategic planning within the funeral home.

The combination of these two approaches ensures that the development of the proposed system directly addresses the identified problems in the funeral home. As a result, the proposed system is designed based on operational needs and user requirements.

3.1.2 Justification for Using the Research Design

The selection of the Descriptive-Developmental Research Design is appropriate for this study because the primary objective is not merely to investigate a problem but also to develop a practical solution to address it.

First, the study seeks to understand the current operational challenges experienced by Alfonso Somo Funeral Homes. These challenges include manual record keeping, poor resource tracking, fragmented information storage, and limited use of data for decision-making. A descriptive approach is necessary to thoroughly analyze these issues and identify their causes and effects on organizational performance.

Second, the study aims to design and implement a Data-Driven Funeral Management System that can improve operational efficiency and planning processes. Since the research involves the actual creation of a software system, a

developmental approach is required. Developmental research provides the framework needed to transform user requirements into a functional technological solution.

Furthermore, the chosen research design aligns with the objectives of the study, particularly those related to system analysis, design, development, implementation, and evaluation. The design allows researchers to gather sufficient information about existing operations before proceeding with system development, thereby ensuring that the proposed system addresses real-world needs.

The Descriptive-Developmental Research Design also supports continuous improvement throughout the development process. Feedback from users can be incorporated during testing and evaluation stages, enabling the researchers to refine the system and enhance its effectiveness before deployment [31].

Because the study focuses on both understanding organizational problems and developing solution, the Descriptive-Developmental Research Design provides the most suitable methodological framework.

3.2 System Development Methodology

This study used Agile Development Model for developing the proposed system that consists of the following:

- **Requirements Analysis**
  - In this phase, information, needs, and requirements was gathered from the client. Interviews and discussions are conducted to identify the current problem/situation the funeral home. The information collected serves as the foundation for the features and functions of the proposed system.
- **System Design**
  - In this phase, the creation of the design and structure of the system was created. This includes designing the interface, database, process flow, and system architecture. During this phase, researchers make sure that the design is simple, organized, and easy for users to use and understand.
- **Development**
  - In this phase, the actual creation of the system was carried out. The features of the system such as registration, login, client information management, funeral scheduling, payment recording, inventory monitoring, and analytics were developed and coded. The database was also integrated and ensures that all components of the system functions properly.

- **Testing**

  - In this phase, checking the performance and reliability of the system is the focus. Different tests are conducted to identify bugs, errors, and technical issues within the system. Feedback from the client and users was also gathered to determine if the system meets the required expectations and standards. Improvements and corrections was also made to enhance system‘s quality.

- **Deployment (Prototype)**

  - In this phase, the prototype of the system will be deployed online and presented to the client for evaluation. How the system works will be demonstrated and users are allowed to test its features and functions. Feedback will be gathered from the users will help determine whether the proposed system effectively improves funeral services and supports planning and optimization.

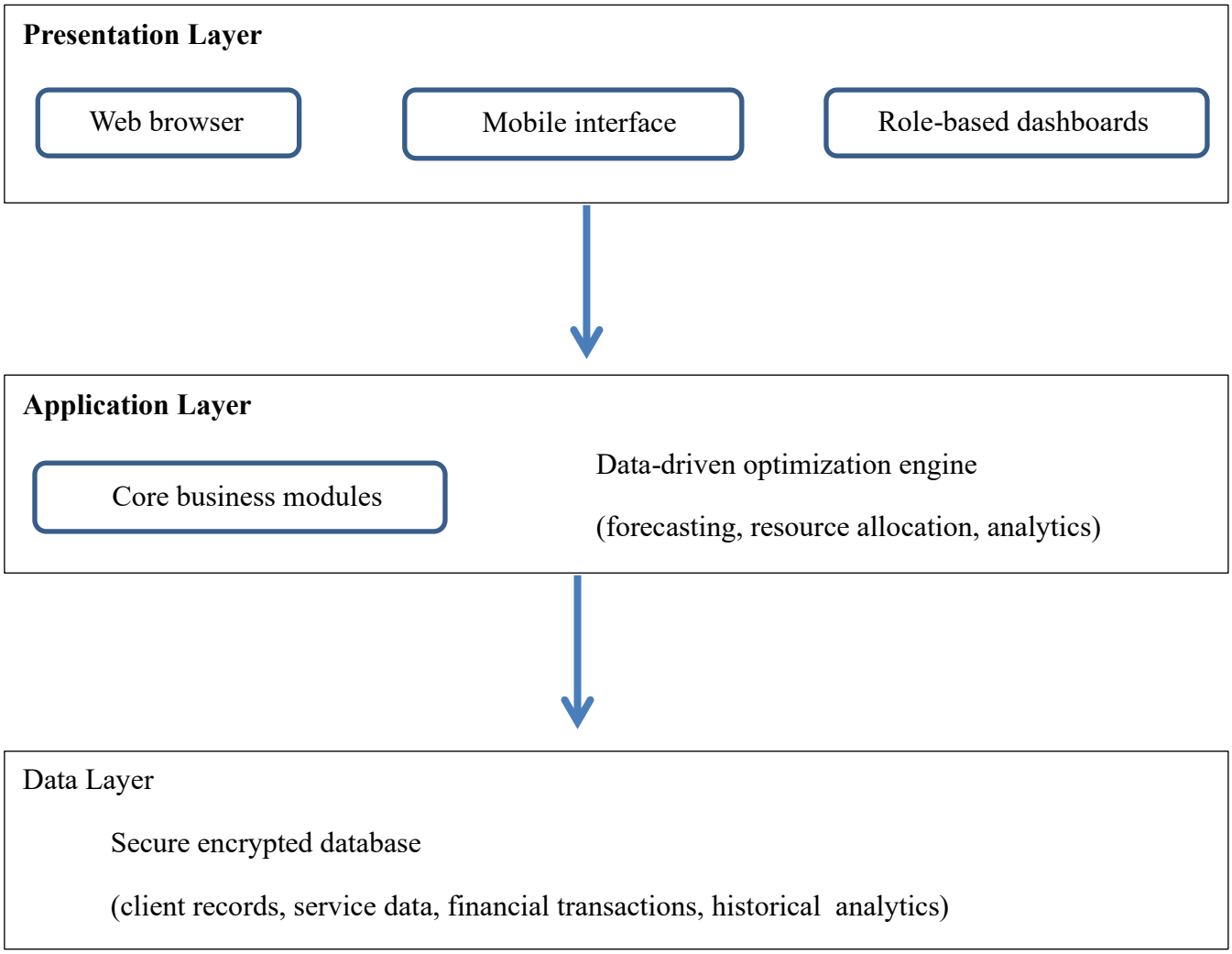
- **Maintenance**
  - In this phase, after collecting the respondents‘ provided feedback on system performance and usability, areas for future updates or improvements will be identified. Regular updates, bug fixes, and additional features will be integrated based on the evolving needs of users and the industry.

3.3 System Architecture

The proposed system adopts the three-tier client-server architecture – an enhanced extension of the basic client-server model – to separate presentation, business logic, and data management[32]. This structure ensures scalability, maintainability, data security, and performance efficiency, while supporting data-driven decision-making and optimization workflows.

The system is divided into three independent but interconnected layers:

1. Presentation Layer (Client Tier) – the user-facing interface for interaction.
2. Application/Business Logic Layer (Server Tier) – the core engine that processes requests, run business rules, and executes data analytics.
3. Data Layer (Database Tier) – secure storage and management of all system data.



3.3.1 User Interface Design / Wireframes

The User Interface (UI) Design or Wireframe is the design and structure of the intended funeral management system. As the name implies, it is the graphical representation or a blueprint that indicates how each page within the system will look and how the users will interact with different components of the software [33]. The wireframe entails the positioning of buttons, menus, forms, and all other components to make sure that they are easily accessible.

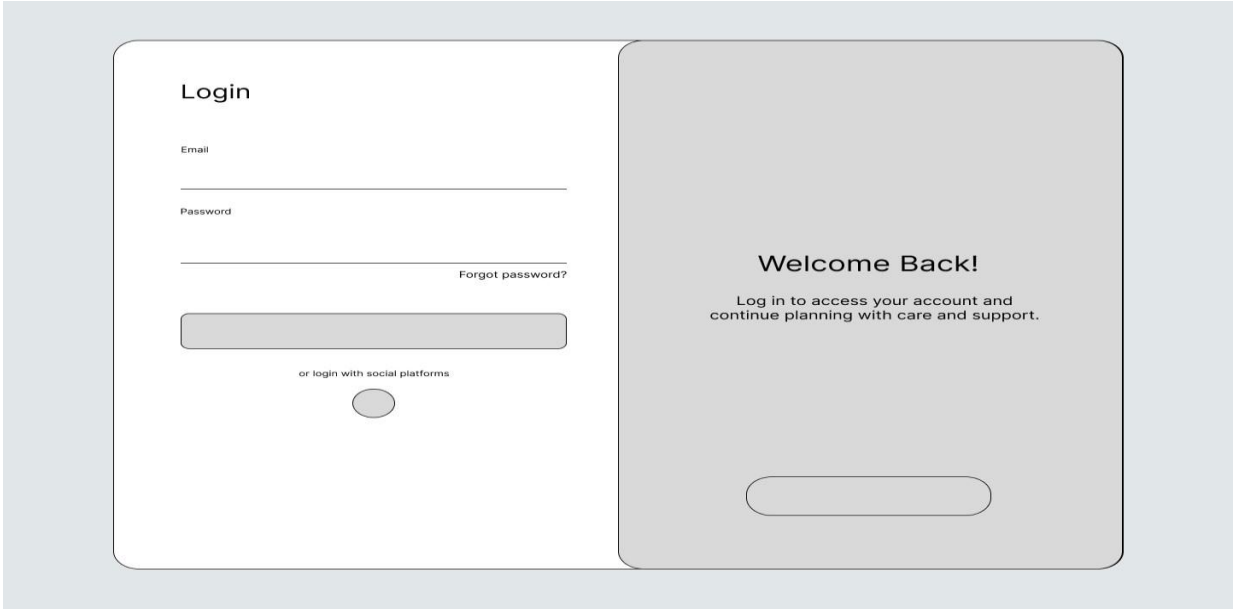
The main reason for having the UI wireframe for the intended application is to create an image of the future system interface in advance, even before its development process starts. The wireframe is essential for both developers and other stakeholders because it shows how the system will operate and be used by users. The wireframe provides insights into usability and allows the developers to improve layout and user navigation.

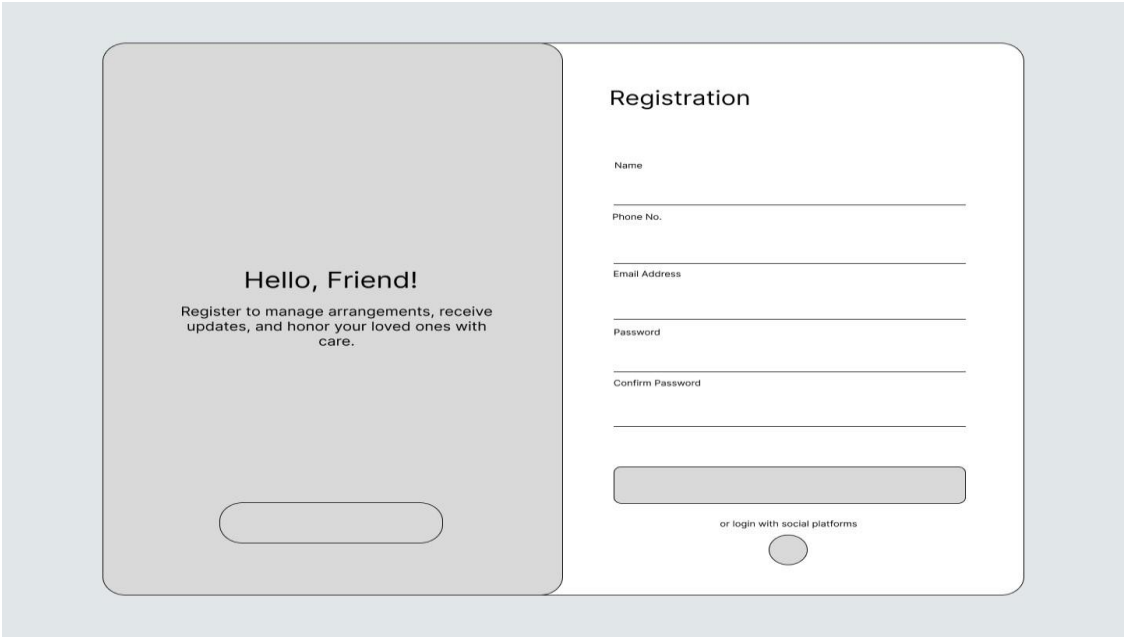
The wireframe of the proposed Funeral Management System includes some important interface designs like the login and registration interface, administrator interface, staffs interface, client/user interface, funeral package

selection interface, payment interface, and notification/messaging interface. The design of these interfaces will be in accordance with the function performed by each user.

Moreover, the design of the User Interface will play a significant part in increasing user satisfaction with respect to the proposed system. With a properly designed wireframe, the ease-of-use aspect of the system will be increased significantly.

Registration and Login Page





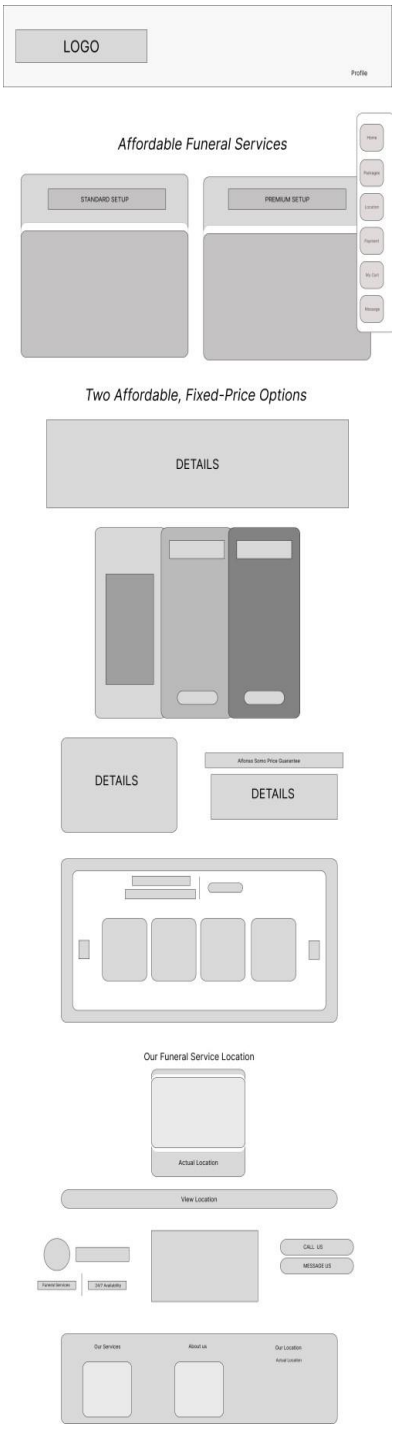
The registration and login page serves as the main entry point of the proposed Data-driven Funeral Management System. It is the first interface that users encounter when accessing the system and is responsible for providing secure access to authorized individuals. This page allows new users to create account while enabling existing users to login using their registered credentials.

The registration section contains input fields that allow users to provide their essential personal information such as their full name, contact number, email address, username and password. Validation mechanisms are incorporated to ensure that all required information is entered correctly before account creation. The system also verifies the uniqueness of email addresses and usernames to prevent duplicate registrations. These measures improve data accuracy and help maintain the integrity of the database.

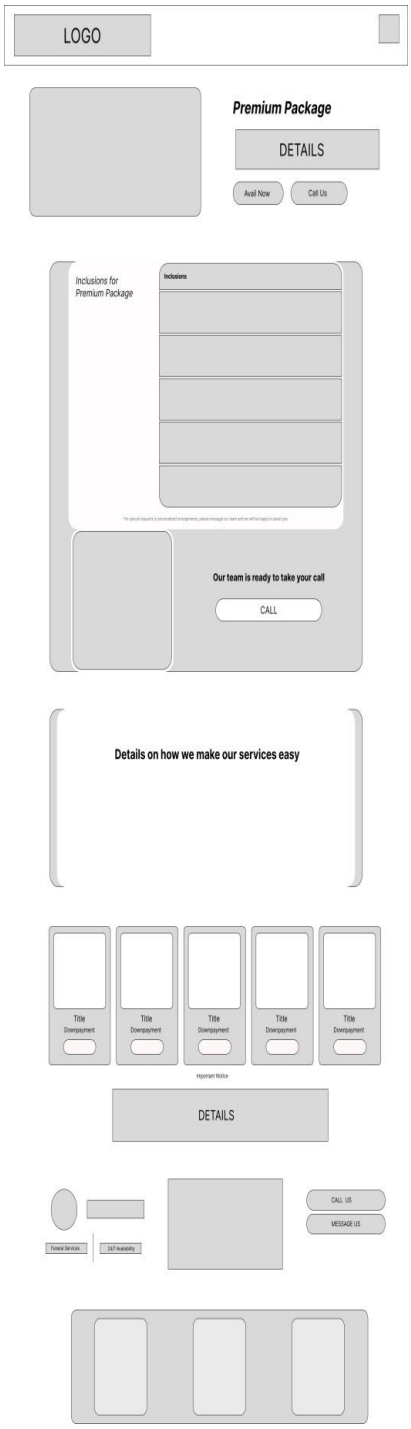
The login section allows registered users to access their accounts by entering their email address together with their password. Additional security features such as password encryption, account verification, and password recovery options may be integrated to further protect user information. Upon successful authentication, users are redirected to their respective dashboards based on their assigned roles, whether they are administrators, staff members, or clients.

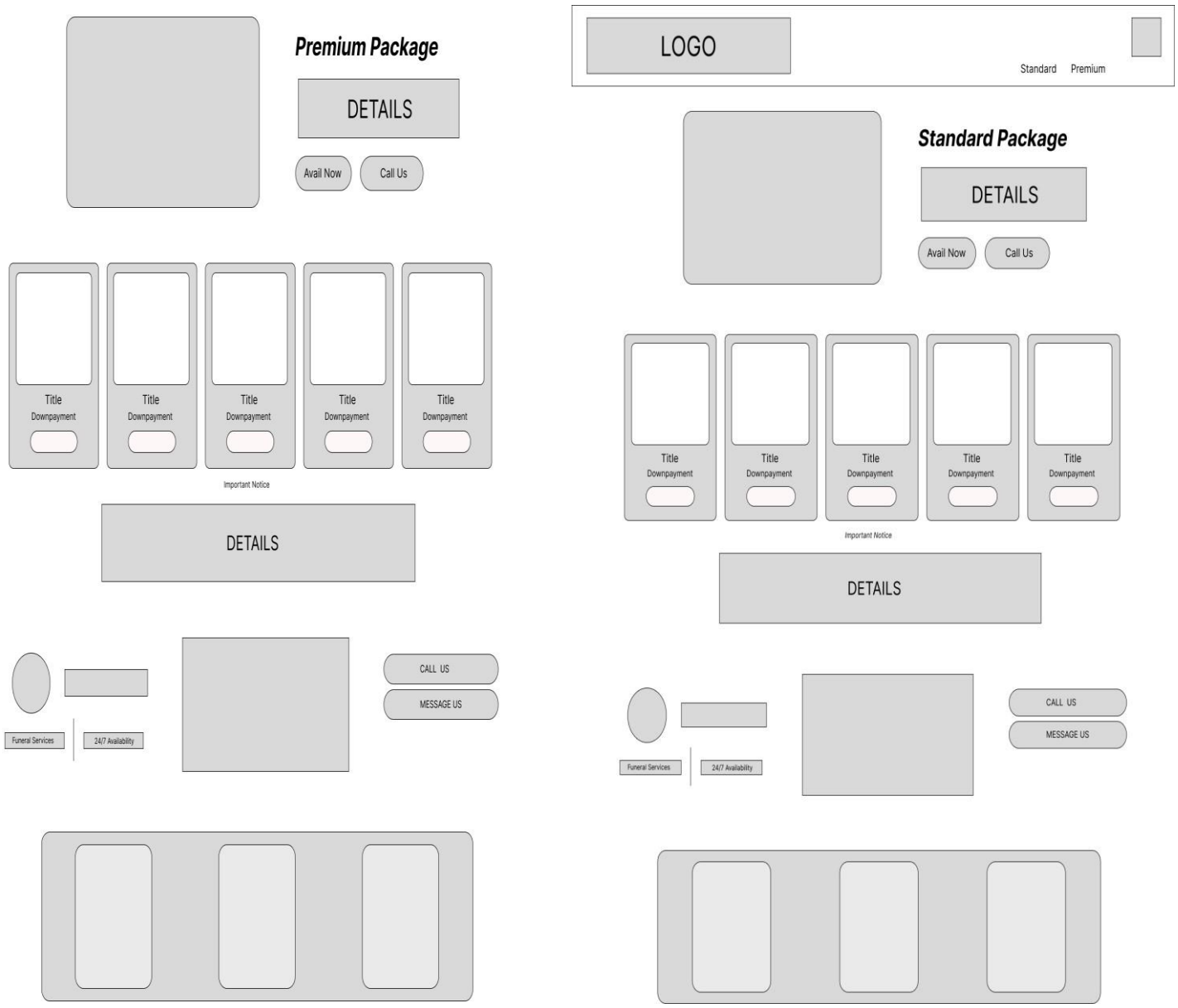
Furthermore, the page is designed with a simple and user-friendly layout to ensure accessibility even for users with limited technical knowledge. Clear instructions, properly labeled fields, and responsive design elements contribute to a positive user experience. By providing secure and convenient account management, the Registration and Login Page establishes the foundation for all interactions within the proposed Data-driven Funeral Management System.

Funeral Services and Packages Page (Client Dashboard)









The funeral services and packages page provides users with a comprehensive overview of the funeral services offered by the funeral home. This page serves as an online catalog where clients can browse, compare, and evaluate available funeral packages according to their preferences, requirements, and budget.

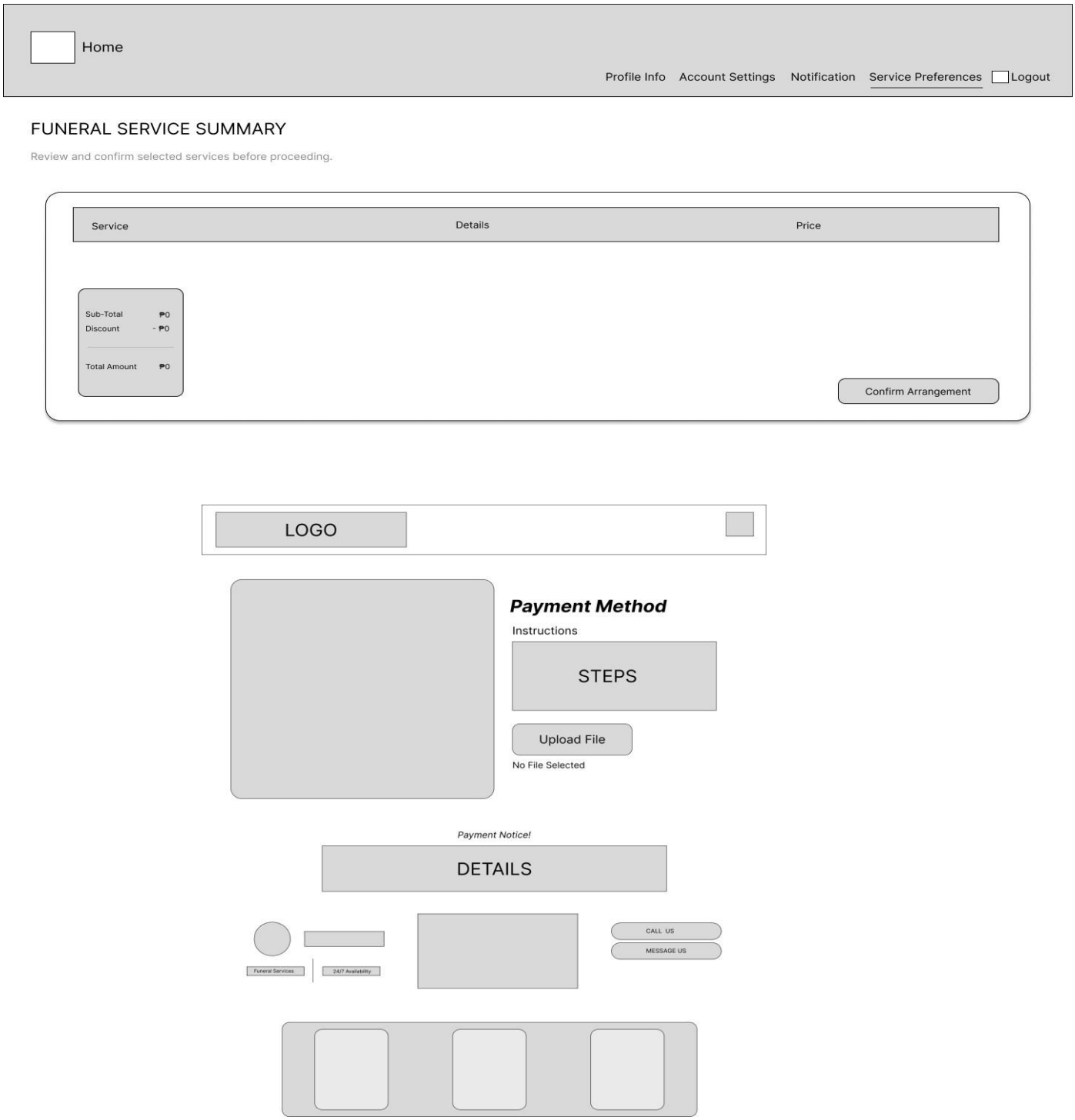
The page displays detailed information regarding each funeral package, including package names, descriptions, included services, pricing information, available amenities, transportation services, and other relevant inclusion. The information is presented in an organized manner to users easily understand the differences between available options.

Users can compare packages side-by-side to determine which service best suit their needs. Images, icons, and service highlights may also be incorporated to improve visual presentation and make the selection process easier.

Search and filtering functionalities can further assist users in narrowing down package options based on their preferred budget or service category.

The page also supports informed decision-making by providing transparent pricing and service details. This minimizes misunderstandings between clients and funeral service providers and allows families to make appropriate arrangements during emotionally challenging situations. Ultimately, the page contributes to customer satisfaction by simplifying funeral planning and improving service accessibility.

Funeral Service Summary and Payment Method Page



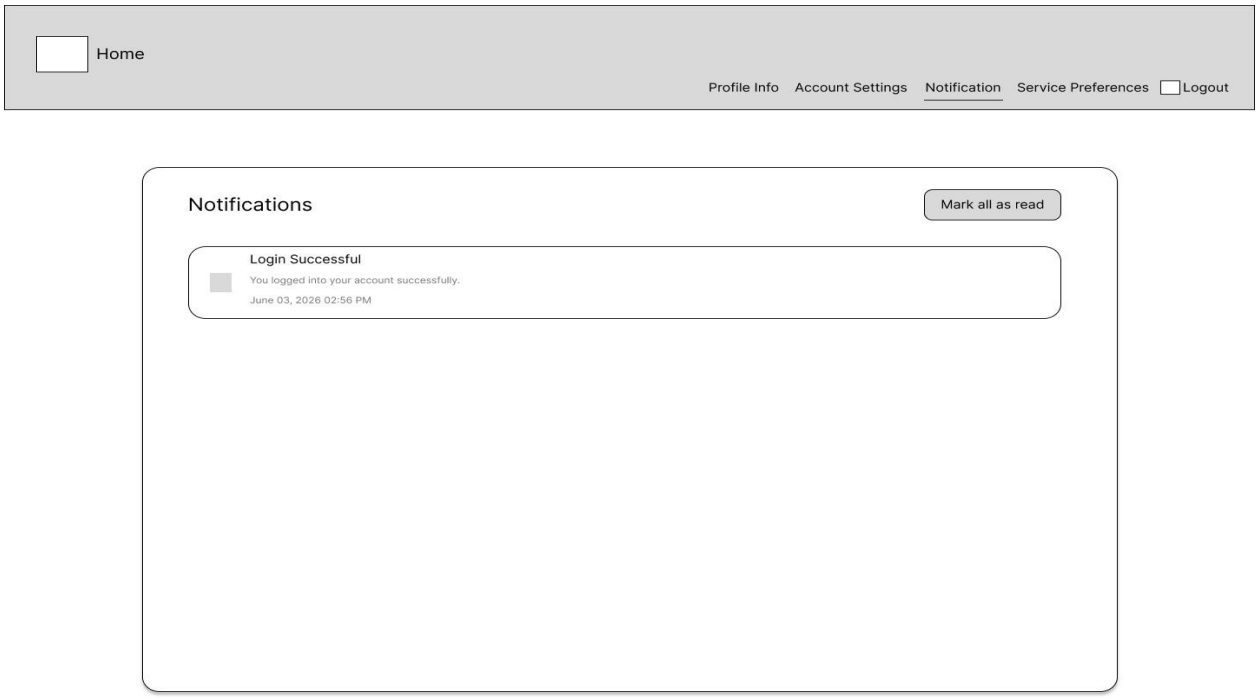
The funeral service summary and payment method page allows users to review all selected funeral services before finalizing their transaction. This page provides a complete summary of all selected services, ensuring that users can verify the accuracy of their choices before proceeding with payment.

The summary section displays important information such as selected funeral package, service preferences, additional requests, and the corresponding costs of each selected service. The system automatically calculates the total amount payable and presents a detailed breakdown of charges to promote transparency.

The payment section provides payment options that users may choose from including online payment or other supported payment methods. Users are guided through a secure payment process that protects sensitive financial information and reduces transaction risks.

In addition, the page includes confirmation features that allow users to review and modify their selections before final submission. This minimizes errors and ensures that all arrangements accurately reflect the client‘s intentions. By incorporating service verification and payment processing, the page improves transaction efficiency and enhances user experience.

Notification Page



The notification page functions as the system‘s communication center where users receive updates and alerts related to their funeral service transactions. This page ensures that users remain informed regarding the status of their requests and any relevant activities occurring within the system.

Notifications may include booking confirmations, payment verification updates, service status updates, funeral schedule reminders, and important announcements from the funeral service provider. Each notification is displayed chronologically, allowing users to track recent activities and remain updated.

The page may also provide notification categorization features that allow users to distinguish between payment notifications, service updates, reminders, and general announcements. Users can mark notifications as read or unread for future reference.

By providing real-time updates and reminders, the notification page improves communication efficiency and reduces the likelihood of missed appointments, delayed payments, or misunderstanding between clients and funeral service providers

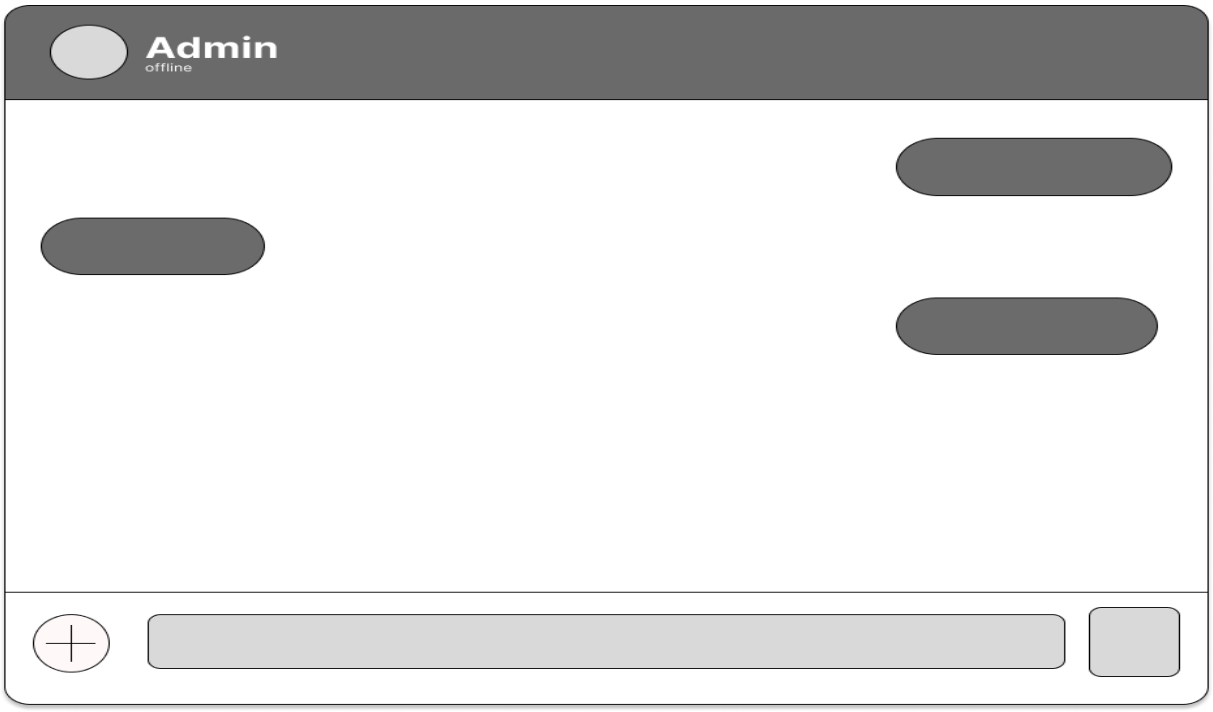
Messages Page

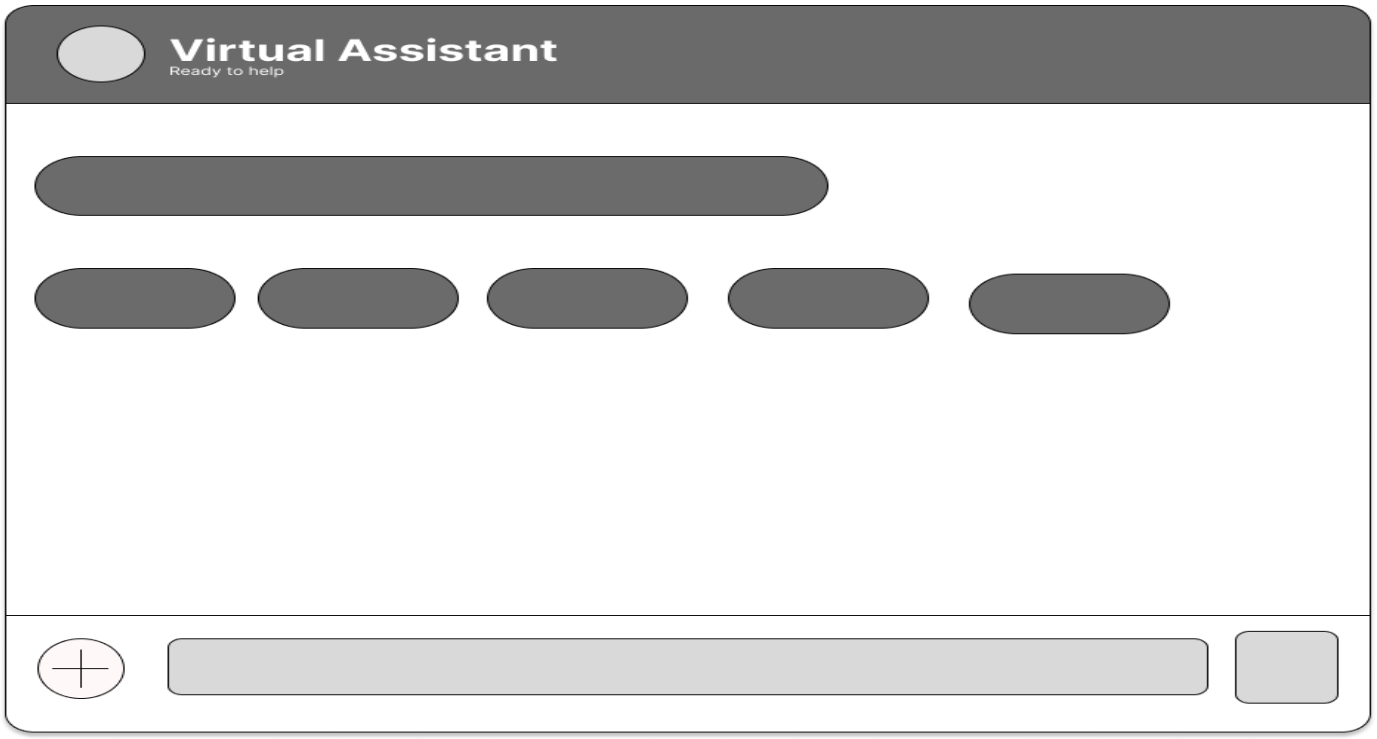


The messages page provides users with a centralized platform for communication that enables them to communicate directly with funeral home representative or system administrators. This page provides a convenient way for clients to ask questions, request assistance, and receive support without requiring physical visits or phone calls.

The interface displays conversation threads in an organized format, allowing users to view previous messages and maintain a complete communication history. Users can end inquiries regarding funeral packages, service schedules, payment concerns, and other funeral-related matters.

The messages page strengthens customer support by promoting direct communication and ensuring that important information can be exchanged efficiently between clients and funeral home personnel. **Live Admin and Virtual Assistant Chat Page**



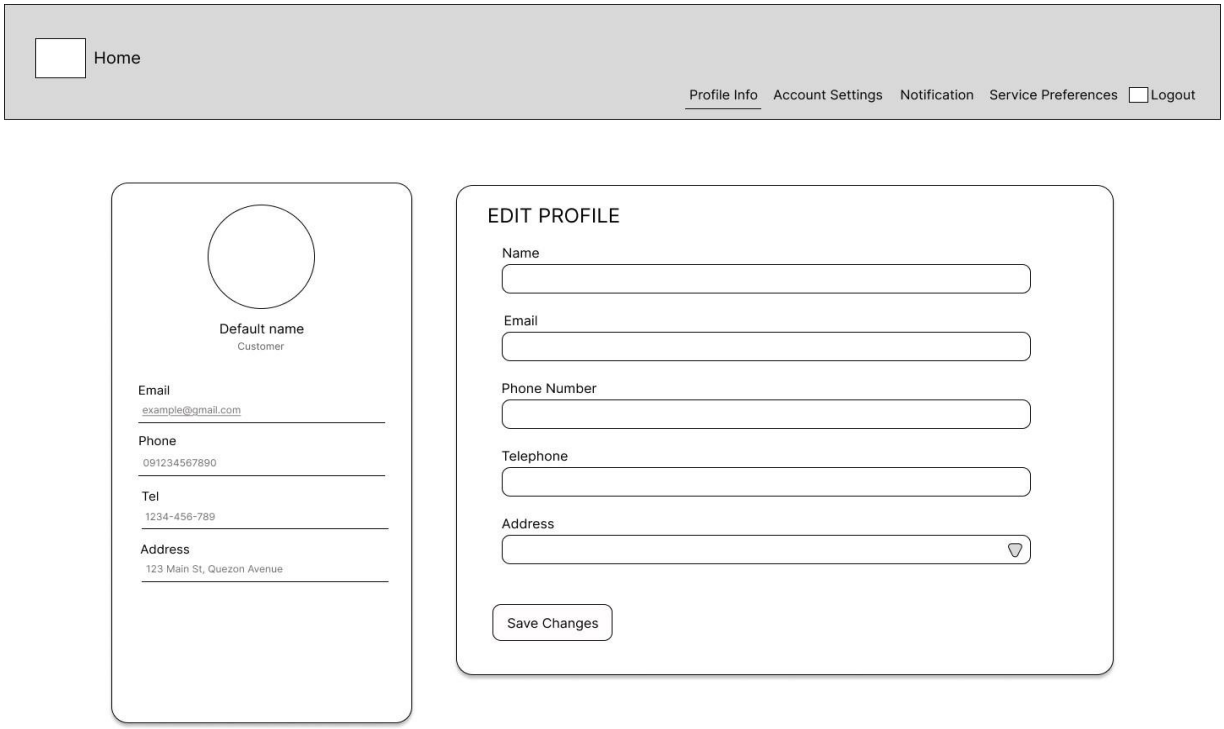


The live admin and virtual assistant chat page facilitates real-time communication between users and support personnel. This page is designed to improve customer service by ensuring that users receive prompt responses to their concerns and inquiries.

The virtual assistant handles frequently asked questions regarding funeral services, booking procedures, payment methods, service packages, and other related concerns. Through automated responses, users can obtain immediate assistance without waiting for administrator availability.

For more complex concerns, users may communicate directly with live administrators who can provide personalized guidance and support. The chat interface supports continuous conversations and maintains a history of previous interactions for future reference.

This functionality enhances accessibility, improves customer satisfaction, and ensures that users receive assistance whenever needed throughout the funeral planning process .**Profile Information Page**



The profile information page contains the personal information of the registered user, including contact information and account-related data within the system. It serves as the primary location where users can view and verify their account details to ensure accuracy and completeness.

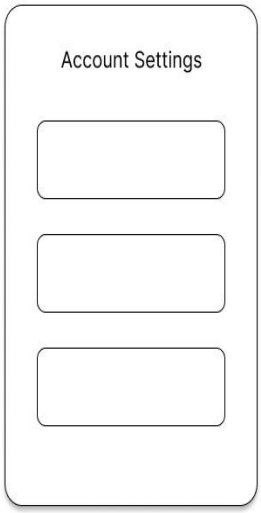
Information displayed on this page may include the user‘s full name, email address, mobile number, residential address, and registration details. Users can view their information and identify any inaccuracies that may require updating.

Maintaining accurate profile information is essential for effective communication, proper record management, and efficient service delivery. The page ensures that user records remain organized and accessible whenever needed.

Furthermore, the profile information page contributes to data accuracy and supports the overall reliability of the system‘s centralized database.

Account Setting Page





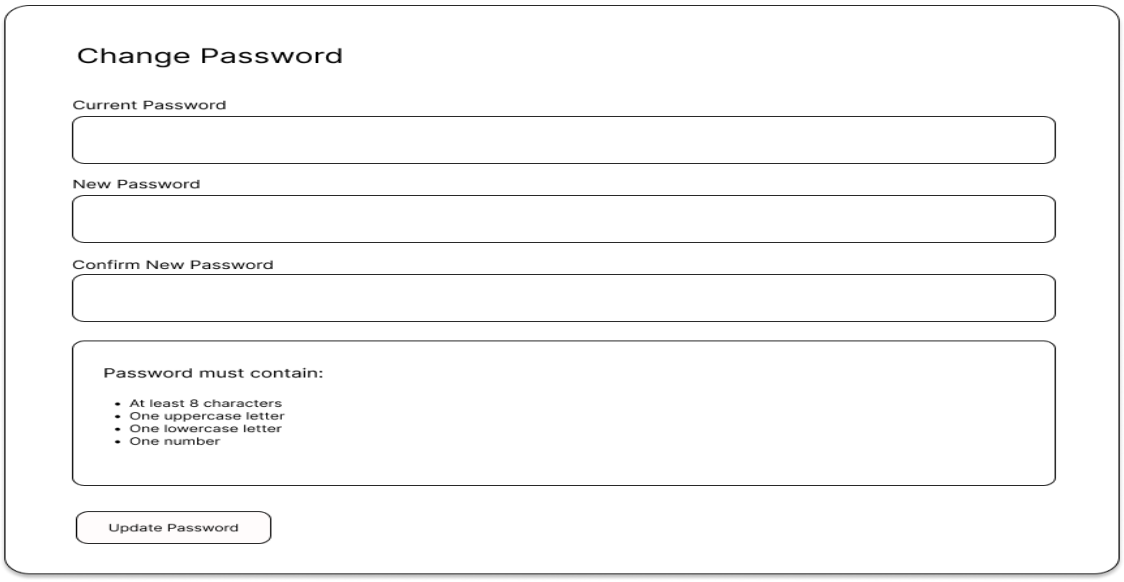
The account setting page allows users to manage various account preferences and customization options within the system. This page provides flexibility and control over how users interact with the platform.

Users may configure notification preferences, privacy controls, and account security settings. These customization options help create a personalized user experience that aligns with individual needs and preferences.

The page may also include settings related to accessibility features, allowing users to adjust display preferences for improved usability. Through centralized account management, users can conveniently control their interaction with the system without requiring administrative assistance.

Overall, the account setting page enhances convenience, flexibility, and user satisfaction. **Change**

Password Page



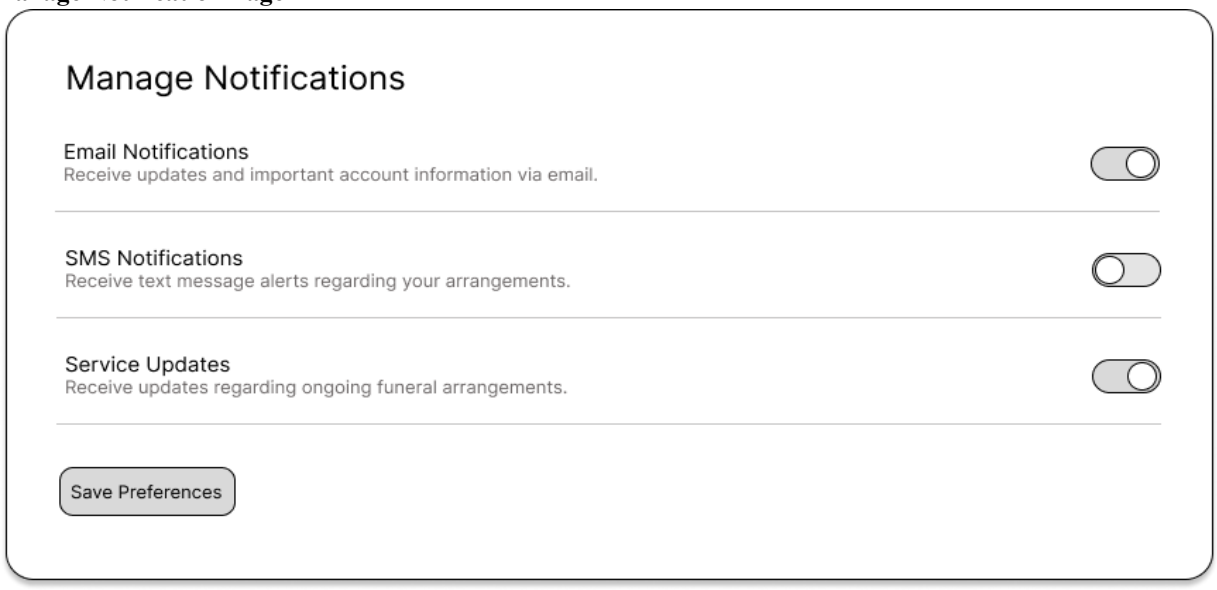
The change password page provides users with a secure mechanism for updating their account credentials. Since account security is a critical aspect of any information system, this page plays a significant role in protecting user information and preventing unauthorized access.

Users are required to enter their current password before creating a new one. Additional validation checks ensure that newly created passwords meets established security requirements, including minimum length and complexity standards.

The system may also provide password strength indicators and recommendations to encourage stronger password creation. Confirmation fields help reduce input errors and ensure successful password updates.

By allowing users to regularly update the credentials, the change password page strengthens system security and promotes responsible account management.

Manage Notification Page

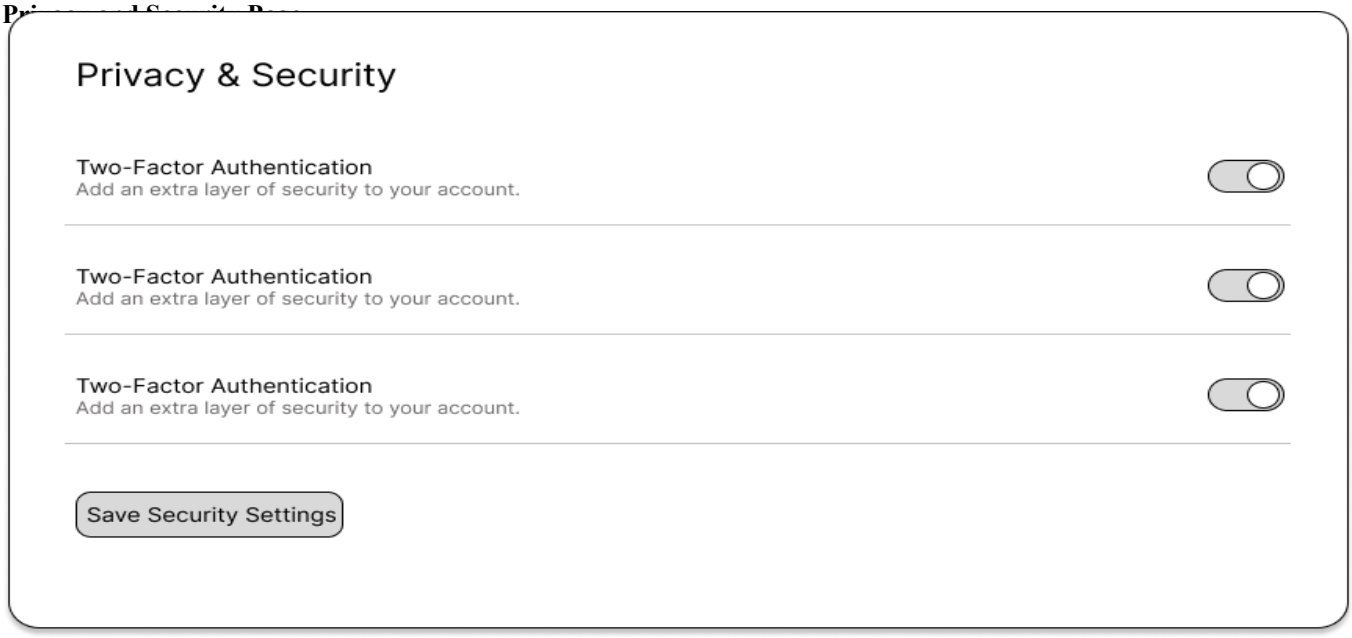


The manage notification page enables users to control how they receive important updates and information from the system. This page provides users with control over their communication preferences by enabling or disabling different notification channels.

The page enhances communication between the system and its users by ensuring that important updates, reminders, and service-related information are delivered through preferred channels. By allowing users to manage notification settings, the system promotes timely dissemination of information and helps users stay informed about relevant activities and arrangements.

Furthermore, manage notification page improves overall user experience by providing greater flexibility and control over account communications. It supports personalized interaction with the system, increases convenience, and reduces the likelihood of users missing important updates, thereby contributing to a more efficient and user-friendly platform.

Privacy and Security Page



The privacy and security page allows users to manage and enhance the security of their accounts within the system. It provides security options that users can easily enable or disable, helping protect sensitive personal data and transaction information.

By emphasizing privacy and security, the page helps build users trust and confidence in the system. **User's Edit Address Page**



The user's edit address page allows users to update and maintain their residential or contact address information within the system. Accurate address information is important for effective communication, service coordination, and record management.

This page provides editable fields where users can modify details such as house number, street name, barangay, municipality, province, and postal code. Validation mechanisms help ensure that address information is entered correctly and consistently.

Updated address information is immediately reflected in the centralized database, allowing funeral home personnel to access current client records whenever necessary.

This page contributes to operational efficiency by ensuring the location-related information remains accurate and up-to-date.

Funeral Address Page



The funeral address page provides users with information regarding the physical location of the funeral home where services are offered. This page is intended to improve accessibility and help clients easily locate the funeral establishment for consultations, service arrangements, and other transactions.

The page displays the complete address of the funeral home together with contact information, operating hours, and directions. Integration with digital maps may also be provided to assist users in navigating to the location efficiently.

Additional information regarding nearby landmarks may also be included to improve convenience for visitors. The page serves as a valuable reference for clients who may be unfamiliar with the funeral home‘s location.

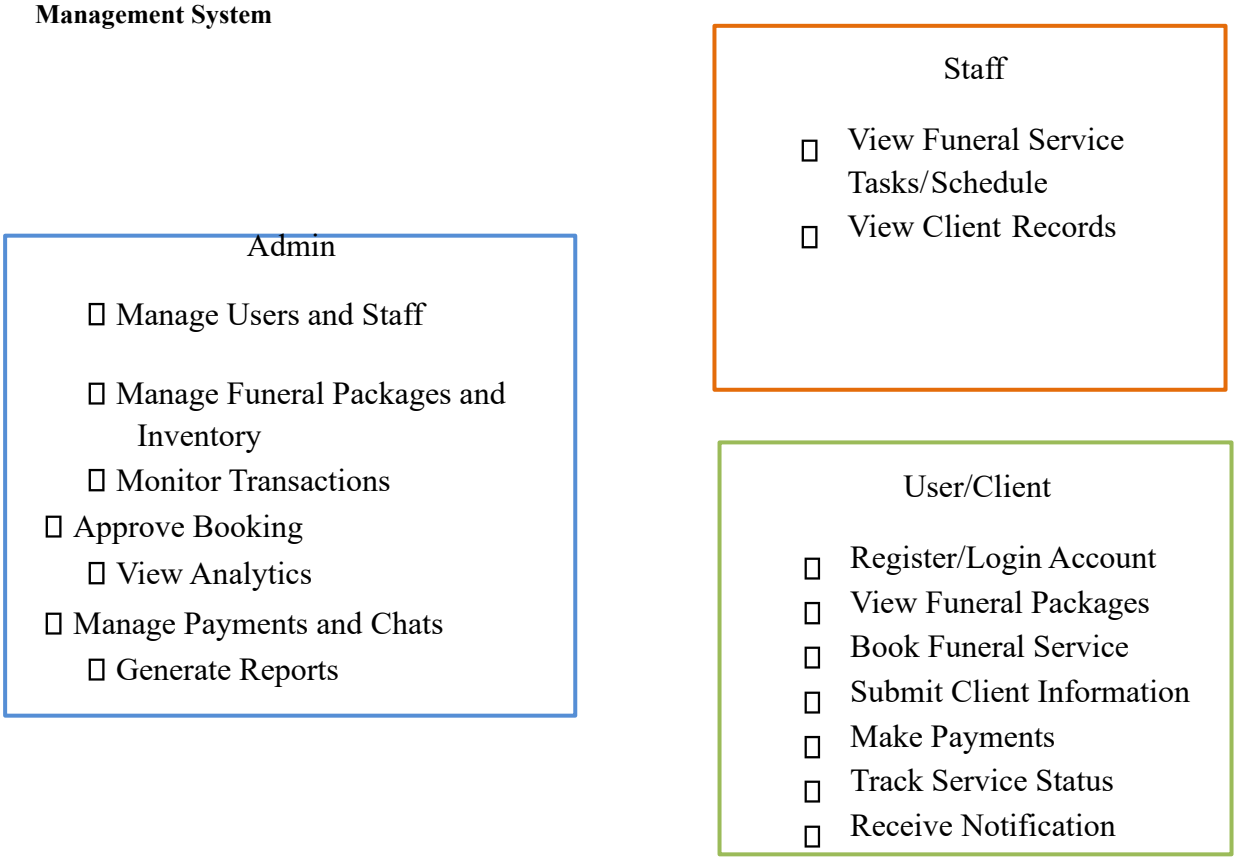
By providing accurate and accessible location information, the funeral address page enhances customer convenience, communication efficiency, and overall service accessibility.

3.3.2 Use Case Diagram

The Use Case Diagram gives a graphical depiction of the users interacting [34] with the proposed Funeral Management System. It shows the various actors in the process and their respective abilities. The diagram shows the functional requirements of the system and gives an idea on how the users interact with the system in order to execute certain tasks.

The proposed Funeral Management System is meant to enhance the efficiency of the management and execution of funeral services by using an efficient and automated system. The proposed system has three main actors which include the Admin, Staff and User/Client. All the three have their unique roles and functionalities in ensuring effective operation of the system. This makes the transaction of services, management of records, scheduling, booking, payment and notification among others much easier[34].

Use Case Diagram was helpful in giving a better picture of what the system entails. It is important in helping both the developers and researcher identify the processes that was implemented in the system development. **Funeral Management System**



The Use Case Diagram for Funeral Management System, consist of 3 primary actors: Admin, Staff, and User/Client. These actors interact with the system through certain responsibilities.

Admin

The admin has the most privileged access among others in the system and plays the role of overseeing and supervising the entire management and operation of the funerals' management services. Admin can manage and control user and staff account in order to allow access to the authorized persons. Apart from this, Admin has the capacity to oversee funeral packages and their inventories for easy management.

Additionally, the admin has the ability to monitor the transactions and approve booking request made by the clients. Admin also have access to the analytic tools in order to review performance and service trends. Additionally, Admin has the power to oversee payment transactions and chat between the client and the service. In the end, Admin is able to create reports, which can help for documentation, monitoring, and decision making.

Staff

Staff members play the part of assisting in the day-to-day activities of the funeral services. The staff members can access the funeral service schedule so as to know what needs to be done in the process of delivering the services as well as their respective assignments. In addition, the staff members can access the records of the clients for verification purposes.

The part played by the staff members assists in facilitating the operation process at the funeral home.

User/Client

User/Client refers to the customers who use the system in order to utilize funeral services. Users may register and login into their accounts and access the system securely. Users are entitled to viewing the available packages of funeral service and choose services that are appropriate for them.

Users can also place their funeral services order from the system by filling up the needed details about the clients. The payment can be made online as well through the system. Users can check the status of their orders or any other related updates or announcements concerning their transactions within the system.

In general, the Use Case Diagram showed that the Funeral Management System can effectively facilitate the communication and interactions among the administrators, employees, and the clients, leading to an effective management of funerals.

3.3.3 Data Flow Diagram

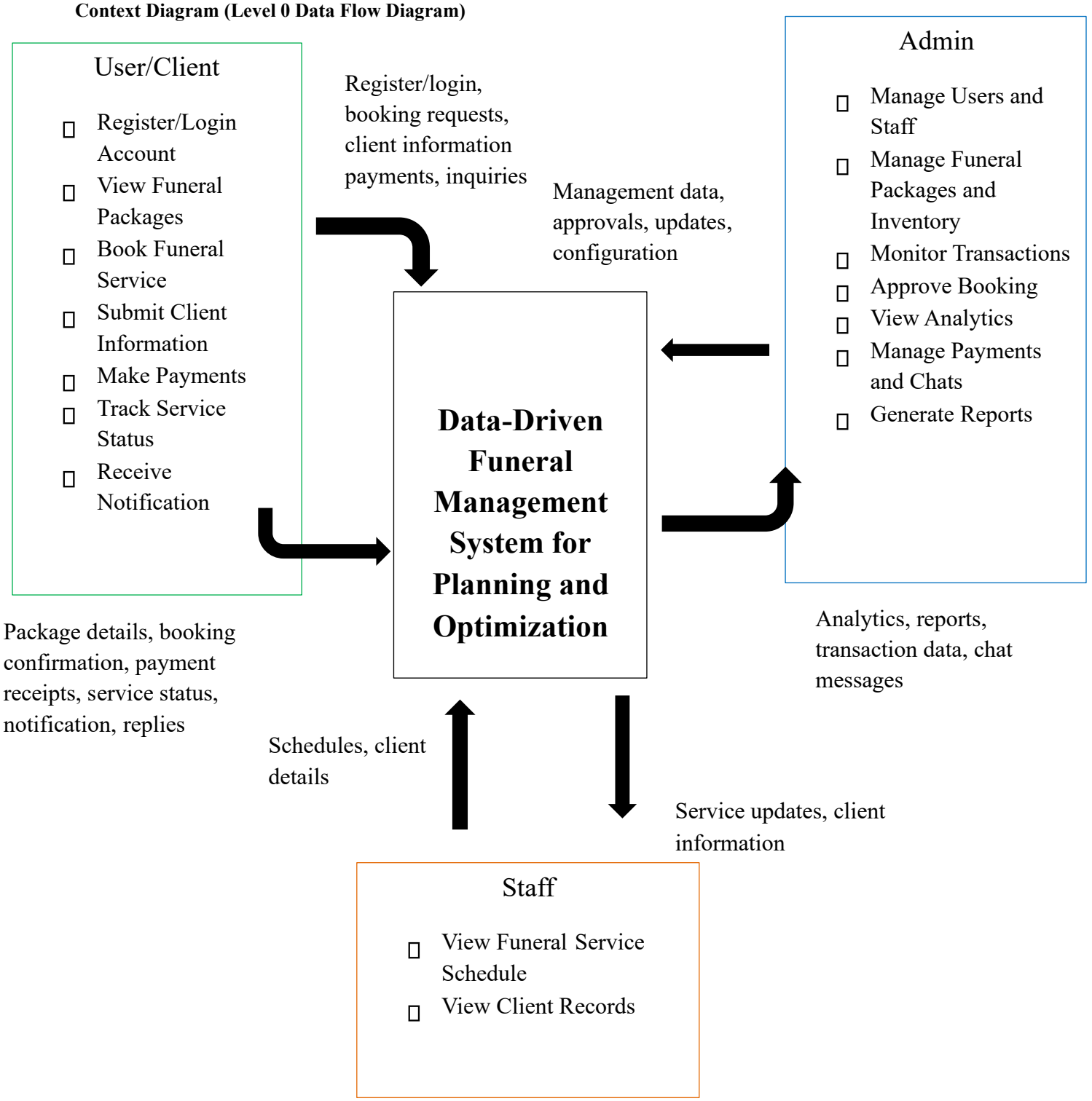
A Data Flow Diagram (DFD) is a visual presentation of the movement of data in the proposed Funeral Management System. This diagram shows the movement of data from users, processes, and data stores in the proposed system. With this diagram, we can see how the system receives, processes, stores, and sends out data.

This Data Flow Diagram is designed to visually represent the processes needed for managing funeral services in our application. The DFD describes the input and output process in our application, and the interaction process of the users such as the admin, staff, and user/client. With this, the researchers are able to analyze the processes of the system in order to make sure that the required processes are correctly arranged.

In the proposed Funeral Management System, the Data Flow Diagram indicates the process flow of data including clients‘ data, funerals booking, scheduling, billing, messaging, inventories and reporting. Additionally, it shows how the data is stored safely as well as how it can be accessed when required by any user who has permission to access this data. The diagram serves as one of the most important tools in analysis and design as it gives a clear picture of how the system works.

Moreover, through the use of Data Flow Diagram, the understanding of the system is much clearer since the connection between the processes and data sources are illustrated in a simpler way than before. This is expected to provide the proposed system with more efficiency regarding the management of funerals and transactions.

Context Diagram (Level 0 Data Flow Diagram)

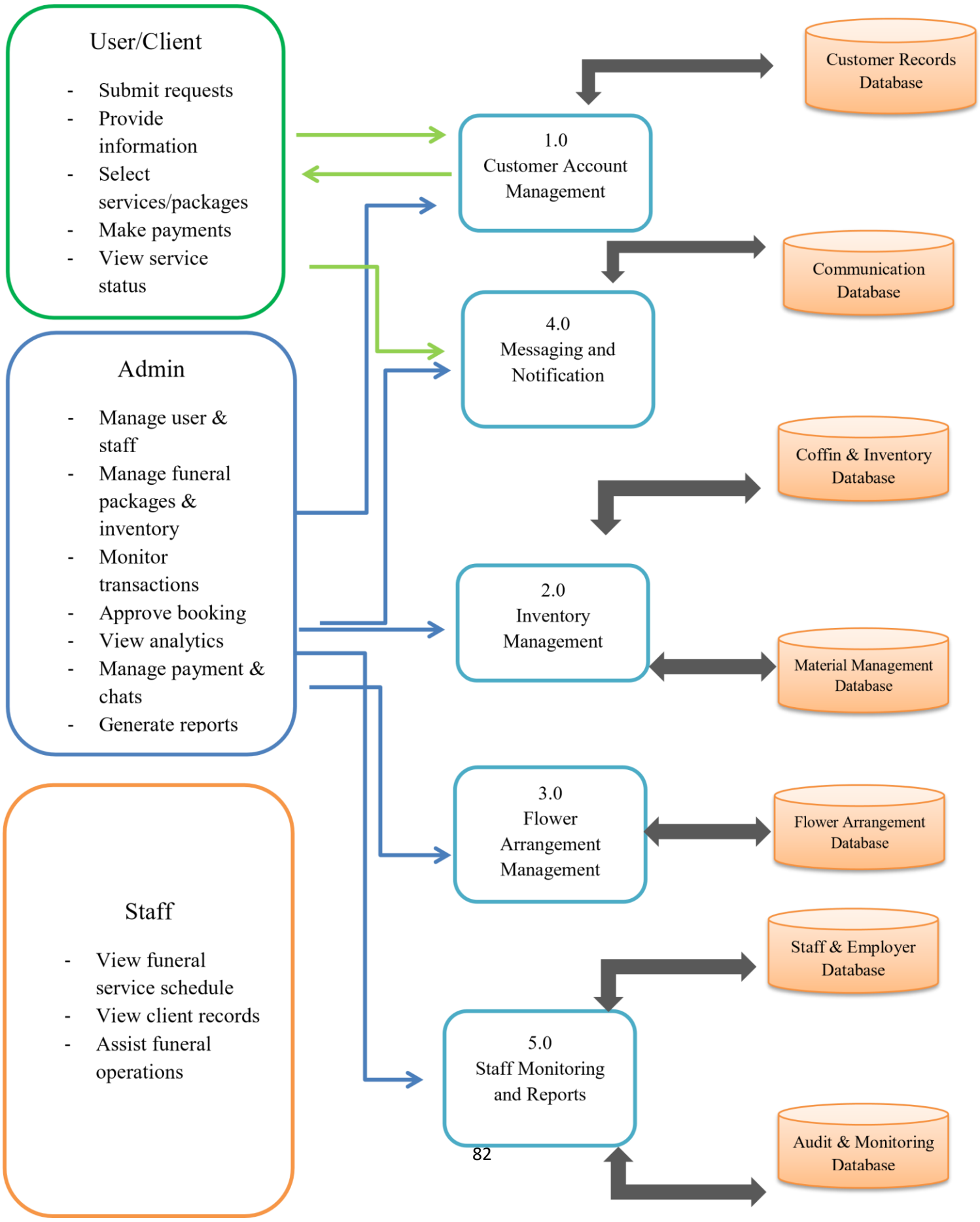


This Context Diagram or Level 0 DFD shown above presents the entire Data-Driven Funeral Management System for Planning and Optimization as one central process, outlining its boundaries and interactions with three key external entities: User/Client, Admin, and Staff. The user/client initiates data flow by sending registration details, booking requests, personal information, payments, and inquiries into the system; in return, the system provides package details, booking confirmations, payments confirmation, service status updates and notification. The admin manages the system by inputting updates, approvals, inventory adjustments, and configuration settings, while receiving analytics, reports,

transaction records, and chat messages to monitor and control all operations. Meanwhile, the staff receives essential data such as service schedules, client records, and service updates from the system, enabling them to perform their duties efficiently and support service delivery.

All information including client data, bookings, inventory records, financial transactions, and communications are processed, stored, and distributed through the central system, ensuring a smooth and organized flow. The diagram clearly demonstrates how the system streamlines funeral service planning, resource management, and record-keeping, ensuring transparency, efficiency and data-driven decision-making across all processes.

Level 1 Data Flow Diagram



The Level 1 Data Flow Diagram shows exactly how everything moves and connects in the Data-Driven Funeral Management System. There are three main groups using it: the clients or customers, the admin, and the staffs.

Clients initiate most transactions within the system by asking for services, submitting their details, selecting service packages, processing payment, and monitoring the stats of their transactions. Admin are responsible for managing the overall operations of the system. They manage everyone‘s accounts, update what services and items are available, keep an eye on all transactions, approve bookings, look over reports and numbers, handle payments and messages, and basically make sure everything works right. Staff members use the system to see their schedules, check client information, and help carry out the funeral service smoothly.

Inside the system, there are five main things happening. First the **Customer Account Management,** which handles everything related to the people using the service, including registration/login, account maintenance, service requests, and transaction records. Information provided by clients is securely stored in the ***Customer Records Database,*** and may also be accessed by admin for monitoring and management purpose.

Next is **Inventory Management,** which the admin handles – this focuses on monitoring and maintaining funeral-related materials and supplies such as coffins, equipment, and other service resources. Admin manage this process by updating stock levels, monitoring item availability, and maintaining inventory records. All inventoryrelated information is stored in the ***Material Management Database and Coffin & Inventory Database*** to ensure accurate tracking and resource availability.

Third is the **Flower Arrangement Management** which manages floral arrangement services, offered by Alfonso Somo Funeral Homes. This process allows client to select floral designs and arrangements according to their preferences while ensuring availability and proper coordination. All related information is maintained in the ***Flower Arrangement Database*** for efficient organization and retrieval**.**

Next is the **Messaging and Notifications**. This process serves as the communication center of the system by enabling interaction among clients, admin, and staff members. Through this module, users can exchange messages, receive updates, and obtain notifications regarding transactions, schedules, and service progress. Communication records and notification logs are stored in the ***Communication Database*** for documentation and reference purposes**.**

Finally, the **Staff Monitoring and Reports,** is responsible for tracking employee activities, schedules, and operational performance. Both admin and staff members contribute updates to this process, allowing the system to monitor workforce productivity and operational efficiency. All related records are stored in the ***Staff & Employer Database and Audit & Monitoring Database*** to maintain accountability, transparency, and accurate operational documentation**.**

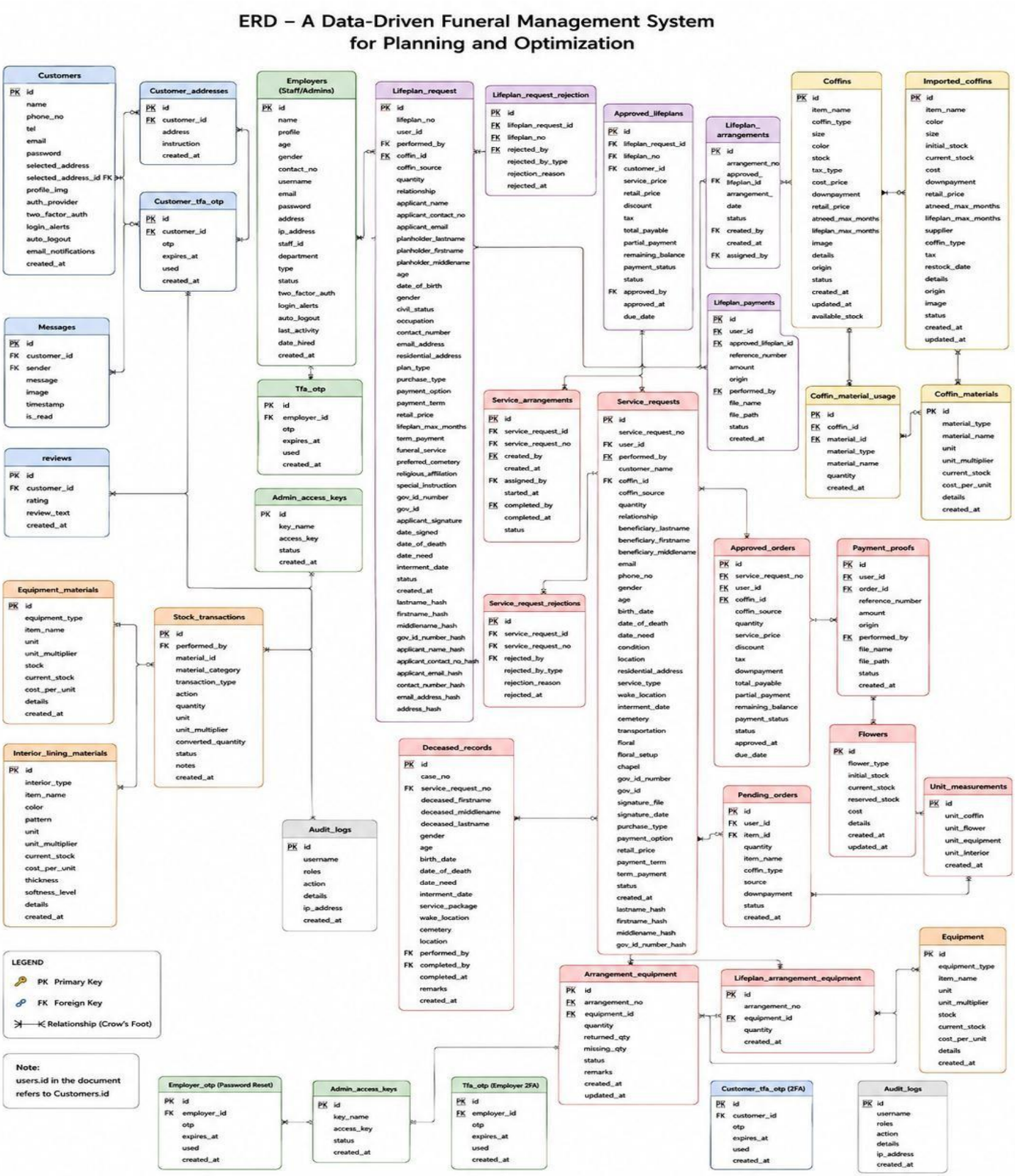
The arrows represented in the diagram indicate the movement and exchange of data between external entities, system processes, and databases. These data flows demonstrate how information is collected, processed, stored, and retrieved throughout the system. Overall, the Level 1 Data Flow Diagram provides a clear representation of how the proposed funeral service management system integrates customer management, inventory control, flower arrangement

services, communication processes, and staff monitoring into a centralized and organized platform that supports efficient service delivery and reliable record management.

3.3.4 Entity Relationship Diagram (ERD)

The Entity-Relationship Diagram (ERD) serves as the blueprint of the database structure of the proposed Data-Driven Funeral Management System. It illustrates the relationships among various entities involved in the operation of the system and provides a visual representation of how data are organized, stored, and connected [16].

The ERD was designed to ensure efficient data organization, minimize redundancy, and support the inventory management and customer service processes of the funeral home. Through the use of interconnected entities, the system can accurately track inventory, manage customer information, monitor transactions, and maintain operational transparency.


ERD – A Data-Driven Funeral Management System
for Planning and Optimization
Customers
id
name
phone_no
tel
email
password
selected_address
selected_address_id FK
profile_img
auth_provider
two_factor_auth
login_alerts
auto_logout
email_notifications
created_at
Customer_addresses
id
customer_id
address
instruction
created_at
Customer_tfa_otp
id
customer_id
otp
expires_at
used
created_at
Employers (Staff/Admins)
id
name
profile
age
gender
contact_no
username
email
password
address
ip_address
staff_id
department
type
status
two_factor_auth
login_alerts
auto_logout
last_activity
date_hired
created_at
Tfa_otp
id
employer_id
otp
expires_at
used
created_at
Admin_access_keys
id
key_name
access_key
status
created_at
Lifeplan_request
id
lifeplan_no
user_id
performed_by
coffin_id
coffin_source
quantity
relationship
applicant_name
applicant_contact_no
applicant_email
planholder_lastname
planholder_firstname
planholder_middlename
age
date_of_birth
gender
civil_status
occupation
contact_number
email_address
residential_address
plan_type
purchase_type
payment_option
payment_term
retail_price
lifeplan_max_months
term_payment
funeral_service
preferred_cemetery
religious_affiliation
special_instruction
gov_id_number
gov_id
applicant_signature
date_signed
date_of_death
date_need
interment_date
status
created_at
lastname_hash
firstname_hash
middlename_hash
gov_id_number_hash
applicant_name_hash
applicant_contact_no_hash
applicant_email_hash
contact_number_hash
email_address_hash
address_hash
Lifeplan_request_rejection
id
lifeplan_request_id
lifeplan_no
rejected_by
rejected_by_type
rejection_reason
rejected_at
Approved_lifeplans
id
lifeplan_request_id
lifeplan_no
customer_id
service_price
retail_price
discount
tax
total_payable
partial_payment
remaining_balance
payment_status
status
approved_by
approved_at
due_date
Lifeplan_ arrangements
id
arrangement_no
approved_ lifeplan_id
arrangement_ date
status
created_by
created_at
assigned_by
Lifeplan_payments
id
user_id
approved_lifeplan_id
reference_number
amount
origin
performed_by
file_name
file_path
status
created_at
Coffins
id
item_name
coffin_type
size
color
stock
tax_type
cost_price
downpayment
retail_price
atneed_max_months
lifeplan_max_months
image
details
origin
status
created_at
updated_at
available_stock
Imported_coffins
id
item_name
color
size
initial_stock
current_stock
cost
downpayment
retail_price
atneed_max_months
lifeplan_max_months
supplier
coffin_type
tax
restock_date
details
origin
image
status
created_at
updated_at
Coffin_material_usage
id
coffin_id
material_id
material_type
material_name
quantity
created_at
Coffin_materials
id
material_type
material_name
unit
unit_multiplier
current_stock
cost_per_unit
details
created_at
Messages
id
customer_id
sender
message
image
timestamp
is_read
reviews
id
customer_id
rating
review_text
created_at
Service_arrangements
id
service_request_id
service_request_no
created_by
created_at
assigned_by
started_at
completed_by
completed_at
status
Service_requests
id
service_request_no
user_id
performed_by
customer_name
coffin_id
coffin_source
quantity
relationship
beneficiary_lastname
beneficiary_firstname
beneficiary_middlename
email
phone_no
gender
age
birth_date
date_of_death
date_need
condition
location
residential_address
service_type
woke_location
interment_date
cemetery
transportation
floral
floral_setup
chapel
gov_id_number
gov_id
signature_file
signature_date
purchase_type
payment_option
retail_price
payment_term
term_payment
status
created_at
lastname_hash
firstname_hash
middlename_hash
gov_id_number_hash
Service_request_rejections
id
service_request_id
service_request_no
rejected_by
rejected_by_type
rejection_reason
rejected_at
Approved_orders
id
service_request_no
user_id
coffin_id
coffin_source
quantity
service_price
discount
tax
downpayment
total_payable
partial_payment
remaining_balance
payment_status
status
approved_at
due_date
Payment_proofs
id
user_id
order_id
reference_number
amount
origin
performed_by
file_name
file_path
status
created_at
Flowers
id
flower_type
initial_stock
current_stock
reserved_stock
cost
details
created_at
updated_at
Unit_measurements
id
unit_coffin
unit_flower
unit_equipment
unit_interior
created_at
Pending_orders
id
user_id
item_id
quantity
item_name
coffin_type
source
downpayment
status
created_at
Equipment_materials
id
equipment_type
item_name
unit
unit_multiplier
stock
current_stock
cost_per_unit
details
created_at
Stock_transactions
id
performed_by
material_id
material_category
transaction_type
action
quantity
unit
unit_multiplier
converted_quantity
status
notes
created_at
Interior_lining_materials
id
interior_type
item_name
color
pattern
unit
unit_multiplier
current_stock
cost_per_unit
thickness
softness_level
details
created_at
Audit_logs
id
username
roles
action
details
ip_address
created_at
Deceased_records
id
case_no
service_request_no
deceased_firstname
deceased_middlename
deceased_lastname
gender
age
birth_date
date_of_death
date_need
interment_date
service_package
wake_location
cemetery
location
performed_by
completed_by
completed_at
remarks
created_at
Arrangement_equipment
id
arrangement_no
equipment_id
quantity
returned_qty
missing_qty
status
remarks
created_at
updated_at
Lifeplan_arrangement_equipment
id
arrangement_no
equipment_id
quantity
created_at
Equipment
id
equipment_type
item_name
unit
unit_multiplier
stock
current_stock
cost_per_unit
details
created_at
LEGEND
PK Primary Key
FK Foreign Key
Relationship (Crow's Foot)
Note:
users.id in the document refers to Customers.id
Employer_otp (Password Reset)
id
employer_id
otp
expires_at
used
created_at
Admin_access_keys
id
key_name
access_key
status
created_at
Tfa_otp (Employer 2FA)
id
employer_id
otp
expires_at
used
created_at
Customer_tfa_otp (2FA)
id
customer_id
otp
expires_at
used
created_at
Audit_logs
id
username
roles
action
details
ip_address
created_at

Overall, the Entity Relationship Diagram serves as the foundation of the Data-Driven Funeral Management System. By integrating product management, inventory control, customer services, staff operations, and transaction tracking into a unified database structure, the ERD enables efficient information management across all aspects of funeral service operations. The relationships among entities ensure data integrity, reduce redundancy, and support real-time decision-making.

Most importantly, the ERD provides the data infrastructure necessary for planning and optimization. Through the collection and analysis of operational data, the system can forecast inventory demand, monitor resource utilization, identify service trends, and generate strategic insights. These capabilities allow funeral service providers to improve operational efficiency, reduce costs, enhance customer satisfaction, and deliver higher-quality funeral services. This comprehensive database design ultimately supports the system‘s objective of transforming traditional funeral management practices into a modern, data-driven operation.

3.4 Tools and Technologies Used

The process through which the creation of the proposed system was undertaken entailed the use of various technologies that ensured efficient, reliable, and usable application. The technologies used depended on their compatibility, availability, and efficacy in developing web information systems. The incorporation of the frontend, backend, database, and development technologies allowed the researchers to come up with a fully functional DataDriven Funeral Management System for Planning and Optimization.

Frontend: HTML, CSS, JavaScript

The frontend of the software was developed through the use of HTML, CSS, and JavaScript. The frontend refers to the technologies that were used in developing the interface of the system, which is directly operated by users.

HTML was responsible for providing the structure of the website and arranging its layout, including forms, tables, buttons, navigation menus, and other interface elements of the system. HTML formed the basic infrastructure for displaying data in the system.

CSS was employed to enhance the visual interface of the system. Through the use of CSS, the researchers were able to style the interface with the help of various styling attributes, such as colors, fonts, spacing, alignment, responsiveness, and other aspects.

JavaScript was used in adding interactive elements and functions to the system interface, such as validation of user inputs, interactive forms, modal dialogs, auto-calculations, updating of content, and other aspects.

Through the use of HTML, CSS, and JavaScript, the interface of the system was made functional and responsive to users' actions.

Backend: PHP

The backend technology used in developing the system was PHP (Personal Home Page). PHP is an efficient server-side programming language that was used due to its flexibility, support for MySQL database, and efficiency in performing server-side functions.

PHP handled all functions of the system including data processing, session management, and linking between the user interface and the database. The language supported the performance of functions in the system such as user authentication, data extraction, addition of data, modification of data, generation of reports, and management of funeral services.

Moreover, PHP made it possible for the system to handle large data volume effectively. This was achieved through server-side scripting, which made effective communication between the database and users possible.

Database: MySQL

The database management module of the system used MySQL, which is an open source RDBMS. MySQL was chosen because of its reliability, efficiency, speed, scalability, and interoperability with PHP-based web systems.

MySQL acted as the central database manager of the system. The database contained vital information such as customer details, funeral services information, inventory data, schedules, transaction records, reports, and users' accounts. MySQL was structured with tables and relations for efficient management of data.

Through the use of MySQL, the researchers were able to extract data effectively, store data securely, and process information quickly. MySQL also facilitated the data-oriented nature of the proposed system by providing means for analyzing and generating reports.

Moreover, MySQL facilitated data integrity and reduced data redundancy.

Development Tools: Visual Studio Code

The development tool mainly utilized in building the system was Visual Studio Code (VS Code). This is a source-code editor that provides lightweight and efficient features created by Microsoft.

The researchers relied on Visual Studio Code for coding, debugging, editing, and managing the system files. It offered several functionalities including code highlighting, code auto-completion, an integrated terminal, extensibility, and debugging tools among others that increased development efficiency.

The researchers made use of various extensions in Visual Studio Code to facilitate development of PHP, JavaScript, HTML, CSS, and MySQL. Additionally, it facilitated the organization of the projects and the version management process.

All in all, Visual Studio Code played a crucial role in the development of the system since it offered the necessary integrated development environment.

3.5 Data Gathering Techniques

In order to obtain insights necessary for the development of the proposed system, several techniques were used to collect information. These techniques assists in ensuring that the design of the system addresses the interests of various stakeholders in the funeral business especially funeral managers and other staff members in addition to tackling the issues faced by funeral service management today. Data obtained from these techniques forms the basis of determining the requirements of the system. These data collection techniques are as follows:

- Survey
    - The use of survey questionnaires among funeral management personnel, staff, and potential users are used to acquire quantitative data about their experience in funeral management processes. It enabled the researchers to collect data systematically from various participants. Some of the questions that were asked in the survey questionnaire include those concerning current management process practices, scheduling difficulties and record-keeping difficulties, communication processes, and importance of having a database-driven tool during decision making. The survey enables the researchers to know the general problems that users face and which features should be considered when developing the database.

- Interview ○ Interviews are conducted with the management of the funeral homes in order to have more extensive information about the workflow process of the business. The interview helped the researchers have direct knowledge on what is going on currently in terms of the manual procedures involved in their business, their management issues, customer services problems, and expectations regarding the proposed computerization. Interviews will also help clear up the information obtained from the survey and will provide expert opinions on how the system should work.

3.6 Testing Procedures

Testing is an important stage in the system development process to ensure that the system designed operates effectively and fulfill its purpose. The tests involved help detect any mistakes or faults within the system before its implementation. The test phases employed in this research will involve the following processes:

- **Unit Testing**

    Unit testing focuses on the individual components or modules of the system. Every function was tested independently to ascertain that everything operates properly. In this stage, the tester determines whether all the functions or modules like registration, log-in, scheduling, and data input operate properly.

All the errors identified were fixed before moving to the next stage.

- **System Testing**

  The purpose of system testing is to evaluate the whole integrated system itself. Once all of the modules have been integrated, the system itself needs to be tested in order to see whether it adheres to the overall requirements set out by the study. This includes testing the interrelation of different modules, system performance, and accuracy of the data provided by the system, the system's security aspects, and its user-friendliness. The Data-Driven Funeral Management System also undergo system testing to make sure all of its functions – like scheduling, data analysis, dashboard generation, and user management – work properly.

- **User Acceptance Testing**

  The primary purpose of this testing phase is to confirm that the developed system functions exactly according to the operational needs of Alfonso Somo and to validate that the entire system is fully stable and ready for official live deployment.

3.7 Evaluation Criteria

The following evaluation involved a number of software quality measures which ensures that the system is able to fulfill all the requirements necessary for such a system as a web-based funeral management system. This assessment process provides the basis for evaluating whether the system is effective, dependable, and suitable for implementation and use.

- **Functionality**

  This is where the system executes the necessary actions in accordance with its intended goals. This includes the ability of the system to properly execute user registration and log-in, schedule funerals and appointments, record transactions, produce reports, and analyze and visualize data. The system is described as being functional when all its features are working effectively in accordance with the specifications and there are no errors or missing outputs.

- **Usability**

  The usability factor assesses the simplicity and user-friendliness of the system by considering all sorts of users like administrative personnel, staff members, and clients. The criteria here consider the simplicity of the design of the interface, ease in navigation, access to features, and the overall experience of using the software. A well-usable system facilitates the user to perform operations easily and effortlessly without much technical understanding.

- **Reliability**

  This factor is about the capacity of the system to work consistently in regular operation for some period of time. This is the capability of the system to withstand continued usage without crashing, losing

information, or malfunctioning in any way. With reliability, we can be sure that our files are always kept safely and can be accessed at any time.

- **Security**

  The security aspect involves protecting the data and the assets from any unauthorized access or any form of malicious activity. This involves providing authentication services, role-based access controls, and security for the data that is stored in the application. The reason is that the data being processed by the application includes sensitive client and internal data.

- **Efficiency**

  This factor measures the effectiveness with which the system makes use of its available resources like processing speed, memory capacity, and network traffic when doing some work. The efficiency metric also looks at the speed of response by the system in processing requests and delivering pages and reports. An effective system must be able to process multiple users and processes without lagging.

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Chapter 4: SYSTEM DEVELOPMENT, IMPLEMENTATION, TESTING, EVALUATION, RESULTS, AND DISCUSSION

4.1 Introduction

This chapter presents the development, implementation, testing, evaluation, results, and discussion of the Data-Driven Funeral Management System for Planning and Optimization developed for Alfonso Somo Funeral Homes. It explains how the system was designed and developed based on the requirements gathered from the funeral home. The chapter also discusses the different system modules, software architectures, database design, hardware and software requirements, and the development methodology used throughout the project.

Furthermore, it describes how the system was implemented, tested, and evaluated to determine whether it successfully meet the objectives of the study. The findings presented in this chapter demonstrate how the developed system improves funeral service management by automating manual processes, centralizing information, enhancing resource monitoring, and providing data-driven insights that support planning and operational decision-making.

4.2 Overview of the Developed System

The Data-Driven Funeral Management System is a web-based information system developed to improve the daily operations of Alfonso Somo Funeral Homes. The system was designed to replace the manual record-keeping and fragmented processes with a centralized platform that efficiently manages client information, funeral service schedules, inventory, financial transactions, staff assignments, and operational reports. Through automation, the system minimizes paperwork, reduces human errors, and allows users to retrieve important information quickly and accurately.

Unlike the existing manual process, where records are stored in logbooks and/or paper files, the developed system stores all operational data in a centralized MySQL Database. This enables administrators and staff to access updated information in real time while maintaining data consistency and security. The system also incorporates analytical tools that transform historical operational data into meaningful reports and graphical dashboards. These reports helps the funeral management administrators identify service demand trends, monitor inventory utilization, evaluate business performance, and support future planning through historical trend analysis.

The developed system includes several integrated modules that work together to improve operational efficiency. These modules include user account management, client/deceased information management, funeral scheduling, inventory and resource monitoring, payment management, messaging, report generation, and an administrative dashboard for data visualization. Each module was designed to address the operational problems identified during the requirement analysis, particularly those related to manual processing, inefficient scheduling, poor inventory monitoring, and the lack of organized business records.

The system supports three categories of users: the administrator, staff, and clients. Each user is provided with a dashboard containing features appropriate to their assigned responsibilities. Administrators have full control over the system management, analytics, reports, inventory, and user accounts. Staff members

are responsible for managing their assigned tasks and schedules, viewing client records, and assisting clients. Clients can register accounts, view available funeral packages, submit funeral preferences, apply for life plan, monitor transaction status, communicate with the funeral home, and review payment information through the system.

Overall, the developed system serves as a complete management solution that improves coordination among personnel, enhances the accuracy of operational records, and supports data-driven decision-making. By integrating information management, scheduling, resource monitoring, and analytics into a single platform, the system enables Alfonso Somo Funeral Homes to provide more organized, efficient, and reliable funeral services while improving planning and optimization for future operations. **4.2.1 System Workflow Diagram**

Figure 4.2 illustrates the overall workflow of the developed system.

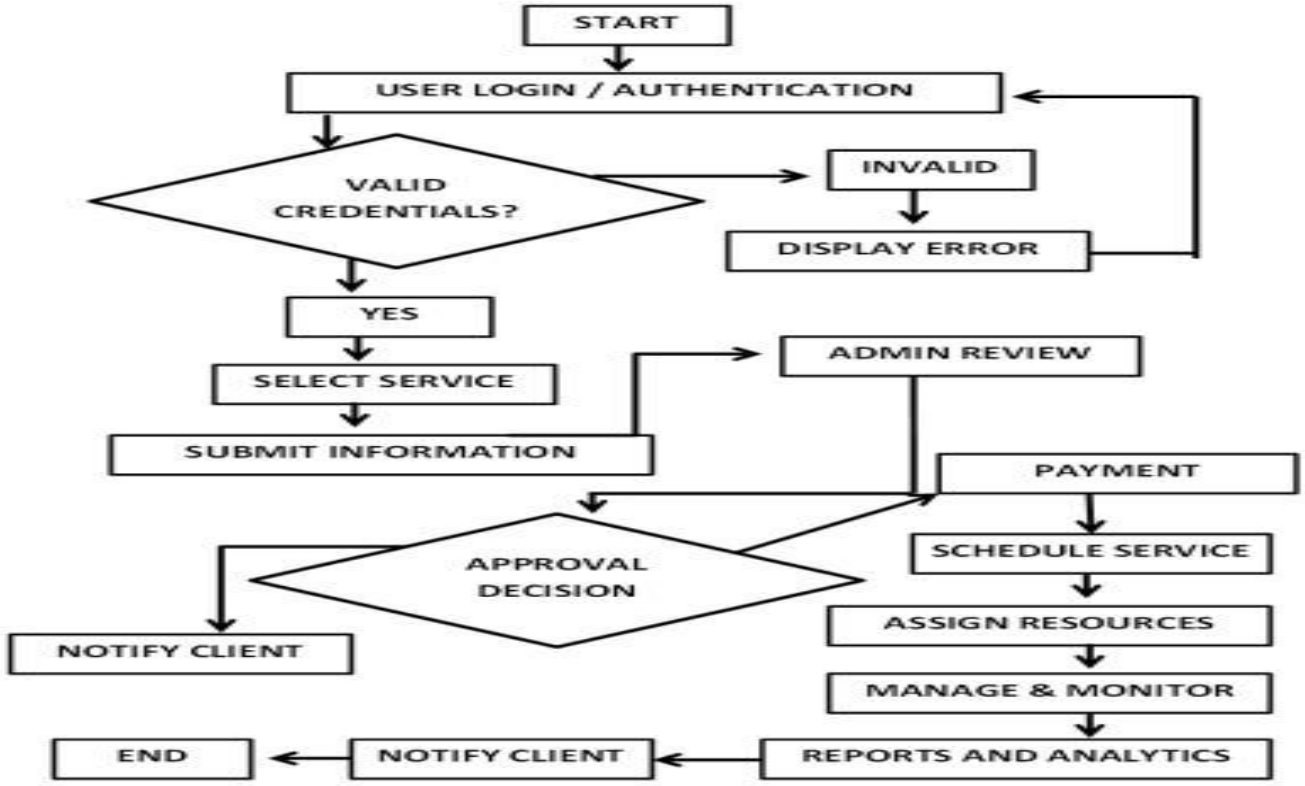


Figure 4.2 illustrates the overall workflow of the Data-Driven Funeral Management System, showing the sequence of processes from user authentication to the completion of funeral service management. The workflow begins when the user logs in to the system by entering their credentials. The system verifies the login information, and if credentials are invalid, an error message is displayed and it goes back to login page, prompting the user to try

again. If the credentials are valid, the user gains access to the system and proceeds to the service selection process/dashboard.

After successful logging in, and the user enters to the service selection dashboard, the user selects the desired funeral package/service and submits the required information. After submitting the required information, the administrator evaluates the submitted request to ensure that all required information have been completed before making an approval.

If the request is not approved, the system notifies the user/client regarding the status of application, and then the transaction ends. However, if the request is approved the client then proceed to payment, and the system schedules the service and assigns the necessary resources, including personnel, equipment, and other materials required for the service. Throughout the funeral operation, the system continuously manages and monitors schedules, resource utilization, and service progress to ensure efficient coordination and timely execution.

Upon completion of the funeral service, the system generates reports and analytics based on the collected operational data. These reports provide valuable insights that support monitoring, inventory management, and future planning. Finally, the client is notified of the completion and status of the service, marking the end of the workflow. This workflow demonstrates how Data-Driven Funeral Management System streamlines funeral service operations by integrating client management, scheduling, resource allocation, monitoring, and data-driven decision-making into a single, efficient process.

4.3 System Development Methodology

The development of the Data-Driven Funeral Management System followed the Agile Development Methodology, an iterative and collaborative approach the divides the project into manageable phases. Agile Methodology was selected because it allows continuous improvement of the system through iterative development, regular testing, and frequent feedback from the intended users. Since the operational requirements of the Alfonso Somo Funeral Homes evolved throughout the study, Agile methodology provided the flexibility needed to modify system features and improve functionalities without affecting the overall development process. Through incremental development, the researchers were able to complete each module individually, evaluate its performance, and integrate it with other system components after successful testing.

The Agile methodology also encouraged collaboration between the researchers and the personnel of Alfonso Somo Funeral Homes. Suggestions and observations gathered during each stage of development were carefully evaluated and incorporated into succeeding iterations of the system. This approach ensures that the final product closely reflected the actual operational needs of the funeral home while minimizing errors and unnecessary revisions.

The development process consisted of several major phases – the requirements gathering, system planning, system design, software development, testing, deployment, and maintenance. Each phase played an important role in ensuring the quality, reliability, and functionality of the developed system.

Requirements Gathering

The first phase involved collecting information from Alfonso Somo funeral homes through interviews. The researchers asked the funeral home management to determine the common challenges encountered in funeral home operations. The gathered information and requirements served as the basis for defining the functional and non-functional requirements of the system.

System Planning

After identifying the project requirements, the researchers established the project scope, objectives, operational constraints, and feasibility parameters. The planning stage also included development timelines to ensure the system remained practical and higly interpretable for the non-technical staff.

System Design

During this phase, the overall architecture of the system was designed, detailing structural process flows, wireframes, and module layouts. Use case diagrams, Data Flow Diagrams, and Entity-Relationship Diagrams were created to establish data dependencies. **Software Development**

The coding phase involved implementing the designed modules using the selected programming languages and frameworks. Individual system functionalities were developed incrementally, including user authentication, product management, transaction recording, customer profiling, analytics generation, and report visualization.

Testing

Each completed module underwent technical validation before full deployment. Integration testing evaluated data exchanges across modules, while system testing verified the overall performance of the application. User Acceptance Testing (UAT) was later conducted with selected respondents to validate whether the application met user expectations.

Deployment

Following successful testing, the developed system was deployed in a web-based environment. Database configurations and server initializations were executed to allow remote browser access for administrators, staff, and clients.

Maintenance

Ongoing performance tracking corrects software bugs, applies essential security patches, and updates operational interfaces based on continuous feedback from funeral home administrators.

4.4 System Architecture

The system utilizes a Three-Tier Client-Server Architecture which clearly separate the user interface, business logic processing, and database layers. This architectural approach improves system maintainability, scalability, and security by organizing software components according to their respective responsibilities. By separating each layer according to its specific responsibility, the system becomes easier to manage, update, and expand as the operational requirements of Alfonso Somo Funeral Homes continue to grow.

The architecture allows users to access the system through a web browser while all business processes and data management are handled by the application server and database server. This setup enables multiple authorized users to access the system simultaneously without affecting the consistency and integrity of the stored information. The three-tier architecture also ensures that sensitive operational data remains protected since users interact only with the system interface and do not have direct access to the database.

Presentation Layer

The presentation layer serves as the interface between the users and the developed system. It is responsible for displaying information, receiving user inputs, and presenting system outputs in an organized and user-friendly manner. This layer was developed using HTML, CSS, and JavaScript to provide responsive web pages that can be accessed through desktop computers, laptops, tablets, and smartphones with an internet connection.

The interface was designed to be simple and easy to understand to accommodate users with different levels of computer literacy. Menus, buttons, forms, tables, and graphical dashboards were arranged logically to reduce user confusion and improve navigation throughout the system. Each category of user is provided with an interface that corresponds to their assigned responsibilities.

For administrators, the interface includes modules for dashboard analytics, user management, client records, scheduling, inventory monitoring, payment management, and reports. Staff members are provided with interface for viewing client information and monitoring tasks and schedules. Clients are given access to funeral service packages, preferences requests, life plan application, payment information, service status, and messaging features.

By providing separate interfaces for each user role, the system maintains security while ensuring that users only access the information and functions necessary for their assigned responsibilities.

Application Layer

The application layer contains the business logic of the system. It serves as the bridge between the user interface and the database by processing user request, validating information, executing system operations, and generating the appropriate responses. This layer was developed using PHP, which handles server-side processing and ensures that system transactions are performed accurately.

Whenever a user submits information through the interface, the application layer verifies the entered data before storing it in the database. Validation procedures help prevent incomplete records, duplicate entries, and invalid information from being saved. This improves the reliability and accuracy of the system while reducing common errors associated with manual record management.

The application layer is responsible for executing the major functions of the developed system. These include user authentication, client registration, deceased information management, scheduling, inventory monitoring, payment processing, report generation, messaging, and data analytics. Each module follows predefined business rules to ensure consistency in funeral service operations.

Another important function of this layer is the generation of analytical reports and dashboard summaries. Historical records stored in the database are processed to produce chart, tables, and graphical reports that assist administrators in identifying service demand trends, monitoring inventory usage, evaluating sales performance, and planning future operations. These analytical capabilities support the primary objective of developing a data-driven management system for planning and optimization.

Data Layer

The data layer serves as the repository of all information managed by the system. It consists of a centralized MySQL database that stores operational records securely and systematically. This layer is responsible for organizing, retrieving, updating, and protecting all data generated during the operation of the funeral home.

The database contains information related to user accounts, clients, deceased records, staff assignments and service schedules, inventory, funeral packages, payments, messages, reports and audit logs. Relationships among these tables were established through primary keys to maintain data integrity and eliminate unnecessary duplication of records.

Because all information is stored within a centralized database, administrators can easily retrieve updated records whenever needed. The use of database constraints and validation mechanisms also helps maintain the accuracy and consistency of operational data throughout the system.

Regular database backup procedures are recommended to prevent data loss caused by hardware failure, accidental deletion, or unexpected system interruptions. Proper database maintenance also ensures the long-term reliability and security of the developed system.

System Interaction

The three layers work together to provide efficient and reliable funeral service management. When a user logs into the system using a web browser, the request is first received by the presentation layer. The entered information is then forwarded to the application layer, where it is validated and processed according to the business rules established by

the researchers. If the request requires retrieving or updating records, the application layer communicates with the MySQL database located in the data layer.

After processing is completed, the requested information is returned to the application layer, which prepares the appropriate response before displaying through the presentation layer. This process occurs continuously as users perform different activities such as updating inventory, booking funeral services, processing payments, viewing schedules, and monitoring operational performance.

The separation of these three layers allows the system to operate efficiently while simplifying future enhancements. Changes made to one layer can often be implemented without significantly affecting the other layers, making the system easier to maintain and upgrade.

Overall, the three-tier client-server architecture provides a stable and organized foundation for the DataDriven Funeral Management System. Its structured design enhances maintainability, improves data security, supports efficient processing of funeral service transactions, and enables the system to deliver reliable information that assist Alfonso Somo Funeral Homes in planning, resource management, and operational decision-making.

4.5 Hardware and Software Requirements

The successful implementation of the Data-Driven Funeral Management System requires appropriate hardware and software resources to ensure efficient operation and reliable performance. These requirements provide the necessary computing environment for processing client information, managing funeral service records, monitoring inventory utilization, generating reports, and performing data analysis. Selecting suitable hardware and software also helps maintain system stability, improve response time, supports multiple users accessing the system simultaneously.

The recommended specifications were determined based on the technologies used during the development of the system and the expected operational requirements of Alfonso Somo Funeral Homes. Since the system is web-based, users only need a computer or mobile devices with a stable internet connection and a modern web browser to access the system. The server computer should possess sufficient processing capability and storage capacity to manage the centralized database and support continuous system operation.

4.5.1 Hardware Requirements

The hardware requirements define the minimum and recommended specifications needed for the efficient operation of the developed system. These specifications ensure that users can access the system without experiencing significant delays while processing transactions or generating reports.

The administrator‘s computer should be equipped with at least an Intel Core i3 processor, 4gb of RAM, and 256gb SSD storage. However for smoother multitasking and faster system performance, the researchers recommend using an Intel Core i5 processor or higher, 8gb RAM, and 512gb SSD storage. The additional memory allows multiple browser tabs, reports, and system module to operate simultaneously without affecting system responsiveness.

A monitor with a minimum screen resolution of 1366 x 768 pixels is sufficient for viewing the system interface. However, a Full HD display is recommended because it provides a clearer view of dashboards, tables, graphs, and reports, allowing administrator to analyze operational data more effectively.

Since the system operates through the internet, a reliable network connection is necessary. A minimum internet speed of 10mbps is sufficient for normal operations, while 50mbps or higher is recommended to provide faster access when multiple users are connected simultaneously.

The system may also be accessed using smartphones or tablets for convenient monitoring of schedules, client information, and notifications. Android version 10 or higher is sufficient for normal operation, although newer versions provide better compatibility and improved security.

Overall, the recommended hardware configuration provides adequate processing power for handling funeral service transactions, database operations, inventory monitoring, and data visualization while ensuring a smooth user experience.

Table 4.1 presents the recommended hardware specifications

Component	Minimum Specification	Recommended Specification
Processor	Intel Core i3	Intel Core i5 or higher
Memory (RAM)	4 GB	8 GB or higher
Storage	256 GB SSD	521 GB SSD
Monitor	1366 x 768 Resolution	Full HD Display
Internet Connection	10 mbps	50 mbps or higher
Mobile Device	Android 10	Higher than Android 10

4.5.2 Software Requirements

The software requirements consist of the operating system, development tools, programming languages, database management system, and supporting application used in developing and operating the proposed system.

These software components were selected because of their reliability, compatibility, ease of use, and wide acceptance in modern web system development.

The developed system can operate on Microsoft Windows 10 or Windows 11, although other operating systems capable of supporting PHP and MySQL may also be used. During development, the researchers utilized Visual Studio Code as primary source code editor because it provides an organized development environment and supports various programming languages used in the project.

The system was developed using HTML, CSS, and JavaScript for designing the user interface and creating interactive web pages. These technologies enable users to interact with the system through responsive and visually organized interfaces.

The server-side functions of the system were implemented using PHP, which processes user requests, validates input data, performs business logic, and communicates with the database. PHP was selected because of its compatibility with MySQL, ease of deployment, and extensive support for web-based systems/applications.

The centralized database was developed using MySQL, which stores user accounts, client information, funeral service records, schedules, inventory data, payment transactions, reports, and analytical information. MySQL provides efficient data storage, fast retrieval of records, and strong support for relational database management.

To create a local development environment, the developer used XAMPP, which integrates the Apache web server, PHP interpreter, and MySQL database into a single software package. XAMPP simplified system development, testing, and debugging before deployment to the production server.

The system is accessed using modern web browsers such as Google Chrome. These browsers support the web technologies used in developing the system and ensure consistent performance across different devices.

Version control and source code management also supported using Git, allowing the developer to track system modifications, manage updates, and collaborate more effectively during future enhancements.

The combination of these software tools provided a stable development environment that enabled the researchers/developer to build a secure, reliable, and user-friendly funeral management system. Their compatibility with one another also simplified system maintenance and future expansion, ensuring that the system can continue to meet the operational needs of Alfonso Somo Funeral Homes.

Table 4.2 presents the software requirements

Software	**Purpose**
Windows 10 or 11	Operating System

Visual Studio Code	Source Code Editor
MySQL	Database Management System
PHP	Backend Development
HTML, CSS, JavaScript	Frontend Development
Google Chrome	Web Browser
Git	Version Control
XAMPP	Local Development Server

The selection of these software tools was based on their stability, community support, compatibility, and widespread adoption in modern web system development.

4.6 Database Design

The database design serves as the foundation of the Data-Driven Funeral Management System. It was carefully developed to ensure that all operational information of Alfonso Somo Funeral Homes is stored, organized, and managed efficiently. The database was designed using the relational database model implemented through MySQL, allowing the system to maintain accurate records while supporting fast retrieval, updating, and reporting of information. A well-structured database minimizes data redundancy, maintains data integrity, and improves the overall performance of the system.

Before the implementation of the database, the researchers analyzed the information requirements of the funeral home to identify the necessary entities, attributes, and relationships. This process ensured that all data related to funeral services, clients, inventory, payments, staff, and reports are properly organized within a centralized repository. The resulting database structure supports the major functions of the system while providing reliable storage for operational and historical records.

The Entity-Relationship Diagram (ERD) of the proposed system presented and discussed on Chapter 3, illustrates the database structure by showing the entities, their attributes, and the relationships among them, serving as the blueprint for the implementation of the database in the developed system.

The primary entities of the database include:

- Coffins
- Coffin Material Usage
- Flowers
- Flower Composition Materials
- Imported Coffins
- Equipment Materials
- Interior Lining Materials
- Unit Measurement
- Flower Materials
- Employer
- Audit Logs
- Stock Transactions
- Messages
- Customers
- Customer Addresses

- Coffin Materials

This database structure enables the system to efficiently manage funeral services, life plan applications, inventory monitoring, payment processing, communication, and administrative functions while ensuring data accuracy, security, and consistency across all system modules.

Chapter 5: Conclusion and Recommendations

5.1 Introduction

This chapter presents the summary of the findings, conclusions, and recommendations of the study entitled ―Data-Driven Funeral Management System for Planning and Optimization‖. It discusses the significant results obtained from the development, implementation, testing, and evaluation of the proposed system. The chapter also explains how the developed system addressed the problems identified in the existing manual operations of Alfonso Somo Funeral Homes, and is considered ready for deployment. Finally, recommendations are provided to guide the researchers and system developers in further improving the functionality and effectiveness of the proposed system.

5.2 Summary of Findings

The study successfully achieved its main objective of developing a Data-Driven Funeral Management System for Alfonso Somo Funeral Homes. The system was designed and developed to minimize the manual processes and integrate a computerized solution that manages client information, funeral scheduling, inventory monitoring, payments, messaging, and reports in a centralized database.

During the development process, the researchers completed the design, implementation, and testing of the system. The different modules were evaluated to verify that they functioned according to the specified requirements. After testing, the system performed its intended functions correctly and was able to process and organized data efficiently.

The developed system also provides dashboards and reports that allow administrators to monitor records and support planning and decision-making. Based on the testing and evaluation conducted by the researchers, it was found that the system is functional, reliable, and capable of supporting the operations of the funeral home.

Overall, the findings indicate that the objectives of the study were achieved and that the developed system is prepared for deployment.

5.3 Conclusion

Based on the findings of the study, the researchers conclude that the Data-Driven Funeral Management System for Planning and Optimization successfully met the objectives of the research. The system provides an organized and centralized platform for managing funeral home operations.

The results of the development confirmed that the system functions as intended. The features developed by the developer are capable of addressing the limitations of manual record management and improving the efficiency of administrative tasks.

Since the system successfully completed its development and researcher testing, it is concluded that it is ready for deployment at Alfonso Somo Funeral Homes.

5.4 Recommendations

To ensure the successful deployment and continuous improvement of the developed Data-Driven Funeral Management System, the following recommendations are proposed:

System Improvements

- Integrates SMS to automatically inform clients about booking confirmations, payment updates, schedule updates, and service reminders.
- Enhance the system‘s security and regular security updates to better protect confidential client and business information
- Develop additional report generation and export options, allowing reports to be downloaded in different formats.
- Develop a mobile application version to provide convenient access of users to the system.

Future Research Directions

- Evaluate the effectiveness of the system after deployment by conducting long-term satisfaction and performance studies to identify additional areas for improvement.
- Investigate the feasibility of implementing the system in multiple branches using cloud-based technology to support centralized management and real-time data synchronization.

Implementation Strategies

- Conduct user orientation and training to ensure that they are familiar with the system‘s features and daily operations.
- Establish a regular database backup and software updates to maintain system performance, protect data, and reduce risk of system failure.
- Encourage users to provide feedback and suggestions after deployment to help improve future version of the system.

5.5 Impact of the Study

The study demonstrates how information technology and analytics can be used to improve funeral management through automation and centralized information management. The developed system provides a practical solution that can help funeral homes organize records, monitor operations, and support planning through data-driven reports.

The successful development and evaluation of the system indicate that it has potential to improve operational efficiency once implemented. The study also serves as a reference for future researchers who plan to develop similar management information systems.

5.6 Summary

The Data-Driven Funeral Management System for Planning and Optimization was successfully designed, developed, and evaluated based on the objective of the study. It showed that the system meets the intended functional requirements and perform operations effectively.

Since the system successfully passed the development and testing phases, it is considered ready for deployment. Once approved and implemented in Alfonso Somo Funeral Home, the system is expected to provide more organized, efficient, and data-driven approach to funeral service management. Future enhancements may still be incorporated to further improve the system as the needs of the organization continue to grow.